

# The Causal Effects of Heterogeneous Expectation Formation in General Equilibrium

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## Abstract

This paper studies the causal effect of heterogeneously-formed expectations on aggregate economic outcomes. To do this, I develop a ‘temporary-information equilibrium’ approach that characterizes the causal impact of expectation changes in a broad class of macroeconomic models using simple cross-sectional statistics on how agents form expectations. This approach uncovers a novel ‘narrative heterogeneity channel’ through which heterogeneities in expectation formation can affect the transmission of both expectational and fundamental shocks. This channel arises in a wide range of commonly-used models of expectation formation. I apply this framework to data on firm and household expectations. First, I show how to use temporary-information equilibrium to interpret the results of a firm-expectations RCT, and find that the causal effect of inflation expectations on realized inflation is substantially larger than implied by standard benchmarks. Second, I calibrate a New Keynesian model to household survey data, and find that narrative heterogeneity accounts for a large but time-varying share of the causal effects of expectations on macroeconomic outcomes.

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# 1 Introduction

Expectations are fundamental to our understanding of macroeconomics. At least since [Lucas \(1972\)](#) they have been at the core of macroeconomic modeling, and more recently a vast empirical literature has demonstrated their importance for the decisions of firms, households, and investors.<sup>1</sup> Yet despite the seemingly obvious centrality of expectations, it is notoriously difficult to say precisely how a given change in expectations would affect aggregate outcomes.

The reason for this uncertainty is that in the equilibrium conditions that define the behavior of macroeconomic models, expectations are endogenous. Take, for example, the linearized New Keynesian Phillips Curve (NKPC), as presented in e.g. [Galí \(2008\)](#):

$$\pi_t = \beta E_t \pi_{t+1} + \kappa x_t \quad (1)$$

This shows that inflation  $\pi_t$  and expected future inflation  $E_t \pi_{t+1}$  are related to one another, but it *does not* reveal anything about the causal effect of inflation expectations. As emphasized recently by [Werning \(2022\)](#), the  $\beta$  coefficient combines the causal effect of  $E_t \pi_{t+1}$  on  $\pi_t$ , the causal effect of  $\pi_t$  on  $E_t \pi_{t+1}$ , and the fact that both  $E_t \pi_{t+1}$  and  $\pi_t$  react simultaneously to other shocks.

The best approach to date has been to study models in *temporary equilibrium* ([Grandmont, 1977](#); [Woodford, 2013](#); [Werning, 2022](#)). However, this approach necessarily abstracts from one of the central findings of recent empirical literature: that expectations are formed heterogeneously. Between otherwise similar decision-makers there are large differences in information, and in how people interpret any information they have.<sup>2</sup> How, if at all, does this heterogeneity affect the causal effects of aggregate expectation shifts, and macroeconomic dynamics?

In this paper, I develop a new ‘*temporary-information equilibrium*’ method that can reveal the causal effect of expectations in general equilibrium while allowing for heterogeneous expectation formation. Applying it to a general class of linear macroeconomic models, I uncover a novel ‘*narrative heterogeneity channel*’, through which heterogeneity influences both the causal effects of expectations and the transmission of fundamental shocks. I show that many popular models of expectation formation implicitly generate such a channel whenever sufficient heterogeneity is allowed for. I then demonstrate the use of my approach to studying expectations in general equilibrium models in two applications.

In the first application, I combine the Calvo model underpinning equation (1) with the information-treatment randomized controlled trial (RCT) results of [Baumann et al. \(2026\)](#) to measure the causal effects of firm inflation expectations on realized inflation. In doing so, I demonstrate how

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<sup>1</sup>See e.g. the extensive reviews in [Bachmann et al. \(2023\)](#), and the discussion of related literature below.

<sup>2</sup>See for example [Link et al. \(2023\)](#) for information, [Andre et al. \(2022\)](#) for subjective models used to interpret information, and [Pfajfar and Santoro \(2010\)](#), [Beutel and Weber \(2025\)](#), and [Macaulay and Moberly \(2026\)](#) for both.

temporary-information equilibrium can be used to map partial-equilibrium information-treatment RCT results into general equilibrium insights, offering a solution to the ‘missing intercept’ problem in these studies.

In the second application, I use unique features of a UK survey to provide a direct measurement of the narrative heterogeneity channel over time, in the context of household inflation expectations. I then combine this measurement with a New Keynesian model to measure the causal effects of household inflation expectations on output. I find that the narrative heterogeneity channel is substantial and time-varying.

**Causal effects of expectations.** The temporary-information equilibrium method is inspired by the information-provision experiments used to estimate causal effects of expectations on individual beliefs and actions (see [Haaland et al., 2023](#), for a review). In those experiments, researchers exogenously vary one part of the information available to participants, and use it as an instrument for the expectation of interest. They thus identify the causal effect of that expectation despite not controlling all aspects of expectation formation. I apply this same logic, developed for survey data, inside general equilibrium models.

To do this, I model each expectation as composed of information (a noisy signal) and a residual, which depends on expectations of other variables. With this setup, expectations are endogenous, just as in the rational-expectations NKPC in equation (1): expectations respond to each other (through what the experimental literature sometimes terms ‘cross-learning’), and to realized outcomes through the endogenous component of signals. However, while expectations are endogenous, the *noise in signals* is exogenous, and shocking this noise creates exogenous variation in a given expectation. Since all effects of the information shift come through the expectation of interest, we can use this exogenous variation to recover the causal effect of an expectation – just as information-provision experiments do in surveys.

This stands in contrast to the temporary-equilibrium approach, which fixes expectations as exogenous objects, solves for equilibrium as a function of those expectations, and then computes comparative statics. In other words, temporary equilibrium varies all agent expectations directly, while temporary-information equilibrium varies only part of information sets. In fact, temporary-equilibrium is nested as a special case of temporary-information equilibrium under particular parametric assumptions about agent information. Away from this special case, however, temporary-information equilibrium permits rich heterogeneity in expectation formation, and general-equilibrium feedback from endogenous variables back to expectations, all while preserving a causal interpretation of the results. Neither of these features can be considered in temporary equilibrium.

This temporary-information equilibrium approach is the key methodological contribution of the paper. The central qualitative insight to emerge from it, even before turning to applications, is a

previously overlooked channel of shock transmission, the ‘narrative heterogeneity channel.’ The channel arises whenever the precision of agents’ signals is correlated with the subjective models they use for cross-learning. For intuition, consider a Calvo price-setting model such as that behind equation (1), which will also be the leading example of Section 2. If the firms who receive precise signals on current inflation are also those who extrapolate most strongly from current to future inflation, then a given change in inflation will have a greater effect on aggregate pricing than if all firms had the same average information precision and cross-learning.<sup>3</sup> Aggregate dynamics therefore depend on the joint cross-sectional distribution of information and subjective models, not just on average beliefs. Indeed, I show that the same channel is also in operation in the transmission of fundamental shocks, whenever part of that transmission operates through agent expectations.

This narrative heterogeneity channel operates in the broad class of models to which I apply the temporary-information equilibrium method (Section 3). In the specific context of the Calvo model, I further show that the channel qualitatively alters features of the passthrough from inflation expectations to realized inflation derived in Werning (2022). Moreover, a wide range of common theories of expectation formation naturally imply narrative heterogeneity effects whenever heterogeneity is permitted in both information precision and the cross-learning used to interpret that information. This includes models of rational inattention (Maćkowiak et al., 2023), learning (Evans and McGough, 2020), level- $k$  thinking (Farhi and Werning, 2019) and others.

**Application 1: the effect of firm inflation expectations.** It is often difficult to formally map the estimates from information-provision RCTs into macroeconomic insights, because of a “missing intercept” problem: RCTs compare treated and untreated individuals, and therefore cannot reveal the general equilibrium effects of aggregate expectation changes. Moreover, since those general equilibrium effects likely also depend on the behavior of expectations, there are no available equivalence results allowing the intercept to be obtained from alternative time-series evidence, as is the case for fiscal transfers (Wolf, 2023).

Temporary-information equilibrium offers a solution. The conceptual link between the approach and information-provision RCTs means that standard IV estimates from those experiments directly identify the key statistics in the temporary-information equilibrium of standard macroeconomic models. I demonstrate this using the RCT on European firm inflation expectations in Baumann et al. (2026), showing how the temporary-information equilibrium of the Calvo model can map their estimates into general equilibrium insights for inflation dynamics.

I find that the Baumann et al. (2026) estimates imply that the general equilibrium effects of one-year-ahead inflation expectations are larger than would be implied by rational expectations, but

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<sup>3</sup>This echoes the redistribution channel in the broader heterogeneous-agent literature (Auclert, 2019), where shocks are amplified when exposure to shocks correlates with responsiveness to those shocks.

that the magnitudes depend strongly on the degree of price stickiness. If the average price duration is 5 quarters, the general equilibrium effect of inflation expectations is approximately 10% larger than that implied by rational expectations. If, however, the average price duration is 2 years, the difference rises to 40%.

In this application the total causal effects are inferred by mapping the estimated coefficients from the RCT into the relevant information and cross-learning statistics in the temporary-information equilibrium of the Calvo model. However, while the RCT estimates identify total causal effects, they do not allow the researcher to decompose that total effect into representative-agent and narrative-heterogeneity channels.<sup>4</sup> In the second application, I therefore turn to a context in which the narrative heterogeneity channel can be observed directly.

**Application 2: the effect of household inflation expectations.** The second application provides, to my knowledge, the first direct measurement of the narrative heterogeneity channel. Unique features of a UK household survey directly reveal properties of household inflation information and the subjective models used to interpret it. That is, we observe the two objects whose joint distribution drives the narrative heterogeneity channel.

I document two facts. First, information and subjective models are systematically correlated in the cross-section: households who believe inflation makes little difference to the strength of the economy use less information about inflation than households with other subjective models. Second, subjective models change with the state of the economy: in periods with high realized inflation, a greater proportion of households report that a further inflation increase would make the economy weaker. The first fact implies that the narrative heterogeneity channel operates, while the second implies that it varies systematically over time.

Interpreting these facts through the temporary-information equilibrium of a New Keynesian model, I find that increases in household inflation perceptions and expectations cause output to fall. When inflation expectations rise, the majority of households revise their expectations for future output down, and thus reduce consumption. However, this causal effect is time-varying. The model implies that the causal effects of household one-quarter-ahead inflation expectations fell by almost 90% between 2008 Q3 and 2015 Q1, when annualized inflation fell from 4.8% to 0.1%, as average cross-learning from inflation to output became correspondingly less negative. The narrative heterogeneity channel accounts for 45% of the causal effect in steady state, and amplifies its time-variation by 21%. The amplification of volatility occurs because, when inflation becomes high, the minority of households who believed inflation makes the economy stronger switch to more neutral subjective models, so the negative correlation between information and cross-learning becomes more pronounced.

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<sup>4</sup>I provide a discussion of how RCTs might be augmented in future to allow such decompositions in Section 4.2.

**Related literature.** Principally, this paper contributes to the vast literature on the role of expectations in macroeconomic dynamics. While many papers examine how particular models of expectation formation affect the transmission of fundamental shocks (Maćkowiak and Wiederholt, 2015; Eusepi and Preston, 2018; Bianchi et al., 2024, among many others), the causal effects of changes in expectations themselves are much less clear. Recent progress on this issue has involved abstracting from cross-sectional heterogeneities in expectations and expectation formation (e.g. Werning, 2022; Ascari et al., 2023; Chahrour and Jurado, 2025), despite their empirical prominence (e.g. Mankiw et al., 2004; Andrade et al., 2016; Link et al., 2025, among many others). I extend this literature by isolating the causal effects of *heterogeneously-formed* expectations on aggregate economic dynamics. Beyond allowing for heterogeneity, temporary-information equilibrium also imposes less structure on expectations than approaches studying belief shocks within particular parametric models of expectation formation (e.g. Ascari et al., 2023). This allows the method to connect directly to empirical estimates from information-provision experiments and surveys.

The approach uncovers a novel *narrative heterogeneity channel* through which heterogeneity in expectation formation affects aggregate dynamics, adding to the mechanisms identified in existing literature (e.g. Branch and Evans, 2006; Andrade et al., 2019, and many others). Conceptually, it relates to channels in the heterogeneous-household literature generated by correlations between shock exposure and the marginal propensity to consume (Auclert, 2019; Bilbiie, 2019).<sup>5</sup> I show that the relevant cross-sectional correlations for the narrative heterogeneity channel arise naturally in many common models of expectation formation, whenever heterogeneity is permitted in both information and subjective models. This complements empirical work documenting these correlations in survey data (Beutel and Weber, 2025; Macaulay and Moberly, 2026).

Methodologically, this paper is most closely related to Werning (2022), who studies the causal effect of inflation expectations in price-setting models in temporary equilibrium (Grandmont, 1977). My temporary-information equilibrium nests this, and extends it to settings with rich heterogeneity in expectation formation, as well as general equilibrium feedback from endogenous variables to expectations.<sup>6</sup> The method is also conceptually related to the large empirical literature on information-provision RCTs (reviewed in Fuster and Zafar, 2023; Haaland et al., 2023).

Translating the results of information-provision RCTs into insights for macroeconomic dynamics is complicated by the missing intercept problem that affects all cross-sectional empirical work in macroeconomics (Wolf, 2023; Donaldson, 2025; Matthes et al., 2025).<sup>7</sup> Because the expectation

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<sup>5</sup>Nord (2022) and Guerreiro (2023) study a further related channel driven by cross-sectional correlations between the sensitivity of expectations to a shock, and the consumption responses to expectations.

<sup>6</sup>Allowing for the reaction of expectations to endogenous variables also connects this method to the literature on learning from prices (Lucas, 1972; Grossman and Stiglitz, 1980; Gaballo, 2018; Chahrour and Gaballo, 2021).

<sup>7</sup>A small number of papers use such RCTs or similar survey experiments to discipline macroeconomic models (Roth et al., 2021; Hoffmann et al., 2022; Andre et al., 2025). Related work disciplines models with non-experimental moments of expectations data (e.g. Leombroni et al., 2020; Ludwig et al., 2024). While this avoids the missing intercept

statistics that discipline the temporary-information equilibrium are closely related to the objects estimated in these RCTs, the method provides a way to map RCT results into general-equilibrium insights, through the lens of standard macroeconomic models. I illustrate this way of addressing the missing intercept problem using the RCT on firm expectations in [Baumann et al. \(2026\)](#). The results speak to the growing literature on firm inflation expectations (see [Candia et al., 2022](#), for a review).

Finally, my second application relates to the broader literature on household inflation expectations (see reviews in [D’Acunto et al., 2023](#); [Milani, 2023](#)). My contribution is to assess the causal effect of household inflation expectations on aggregate output, uncovering substantial time variation, driven in large part by the narrative heterogeneity channel.

## 2 Calvo model

### 2.1 Setup

We start, as in [Werning \(2022\)](#), with the log-linearized first order condition of a price-setting firm:

$$p_{it}^* - P_{t-1} = \sum_{s=0}^{\infty} (\beta\lambda)^s E_{it} \pi_{t+s} + (1 - \beta\lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s E_{it} mc_{t+s} + \mu \quad (2)$$

where  $p_{it}^*$  is the desired reset price of firm  $i$  in period  $t$ ;  $P_{t-1}$  is the aggregate price level in period  $t - 1$ ;  $\pi_{t+s} \equiv P_{t+s} - P_{t+s-1}$  is inflation in the period to  $t + s$ ; and  $mc_{t+s}$  is real marginal costs in period  $t + s$ . The parameters  $\beta$  and  $\lambda$  are the discount factor and the probability a firm is unable to reset their price in a given period respectively, and both lie in  $[0, 1]$ .  $\mu$  is the desired steady state markup, and the operator  $E_{it}$  denotes the expectation of firm  $i$  in period  $t$ .

In New Keynesian models, the standard approach is to impose rational expectations, and then derive the familiar New Keynesian Phillips Curve, in which the coefficient on  $E_t \pi_{t+1}$  is  $\beta$  (e.g. [Galí, 2008](#)). However, as discussed extensively in [Werning \(2022\)](#), this coefficient does not reveal the causal effects of inflation expectations, because the use of rational expectations means that expected inflation is endogenous. High inflation expectations simultaneously cause and are caused by high current inflation, making it impossible to disentangle the strength of each individual mechanism.

**Temporary equilibrium.** Instead of imposing rational expectations, [Grandmont \(1977\)](#) and others recommend holding expectations fixed, then aggregating and solving for equilibrium in terms of

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problem, that data cannot reveal how expectations change in response to shocks ([Piazzesi and Schneider, 2016](#)).



expectations. In the Calvo model this is straightforward:<sup>8</sup>

$$\begin{aligned}\pi_t &= (1 - \lambda) \left( \int p_{it}^* di - P_{t-1} \right) \\ &= (1 - \lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s \bar{E}_t \pi_{t+s} + (1 - \lambda)(1 - \beta\lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s \bar{E}_t mc_{t+s} + (1 - \lambda)\mu\end{aligned}\quad (3)$$

where  $\bar{E}_t(\cdot) = \int_i E_{it}(\cdot) di$  denotes aggregate expectation across firms.

With this, we can compute comparative statics with respect to any aggregate expectation. Since we have not imposed rational expectations, expectations are exogenous in these equations, and the comparative statics can be interpreted as the causal effects of changes in expectations. Specifically,

$$\frac{d\pi_t}{d\bar{E}_t(\pi_t)} = 1 - \lambda \quad (4)$$

gives the causal effect of a change in aggregate expectations of current inflation, *holding all other expectations constant*. This is a direct consequence of removing all endogeneity from expectations: expectations of one variable cannot therefore react to expectations of another.

In many settings this is more extreme than is desirable. This Calvo model is a good example. Increasing  $\bar{E}_t(\pi_t)$  but leaving all longer-horizon  $\bar{E}_t(\pi_{t+s})$  unchanged, as in equation (4), entails a shift in expected current inflation with no pass-through whatsoever to expected inflation at any future horizon: the price level is expected to jump once and then resume its previous path. Most evidence on firm (and household) inflation expectations finds that shocks to short-run expected inflation do pass through, at least in part, to expectations at longer horizons (Weber et al., 2022), so this may not be the most informative way to study inflation expectations.

**Extended temporary equilibrium.** The simple way to avoid this problem is to impose some structure to link expectations that we think should co-move after solving for equilibrium (3), but before computing comparative statics. This is the approach taken in Werning (2022), who assumes

$$\bar{E}_t(\pi_{t+s}) = \pi^e, \quad (5)$$

i.e. that per-period expected inflation is constant across all horizons. While each expectation in equation (3) is no longer fully exogenous (as inflation expectations at each horizon are endogenous

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<sup>8</sup>Note we do not need to include a general equilibrium process determining  $mc_{t+s}$ , because it is *expected*  $mc_{t+s}$  that enters the firm first order condition, and these (like all other expectations) are held constant in temporary equilibrium.



to each other), the common factor  $\pi^e$  is still exogenous. The comparative static

$$\frac{d\pi_t}{d\pi^e} = \frac{1 - \lambda}{1 - \beta\lambda} \quad (6)$$

therefore gives the causal effect of a change in aggregate expectations of per-period inflation, assuming that per-period inflation is expected to be equal at all horizons. [Werning \(2022\)](#) gives an extensive analysis of this result. In principle, the researcher can also specify relationships between expected inflation and expected marginal costs, and recompute the causal effects.

This is a useful advance on studying the equilibrium New Keynesian Phillips Curve, and on the standard temporary equilibrium approach. However, it cannot address how heterogeneity in expectation formation affects shock transmission, how general equilibrium feedback alters the results, or what role expectations play in the transmission of fundamental shocks. To answer these questions, I propose an alternative approach that nests the results above as special cases.

**Temporary-information equilibrium.** When assessing the causal impact of a particular expectation, information-treatment RCTs do not vary that expectation directly. Rather, they exogenously vary the *information* participants have on that variable, and that information affects the expectation of interest. There are also commonly effects on expectations of other variables, known in that literature as ‘cross-learning’ ([Haaland et al., 2023](#)). Taking this procedure as our inspiration, we assume that the expectation of any variable  $z_j$  is composed of two parts: a signal  $s_{it}(z_j)$  and a residual  $R_{it}(z_j)$ .

$$E_{it}(z_j) = (1 - \tau_{ij})R_{it}(z_j) + \tau_{ij}s_{it}(z_j) \quad (7)$$

This follows the form of a standard signal extraction problem, including that used by [Maćkowiak and Wiederholt \(2024\)](#) to model information-treatment RCTs. Under that signal extraction interpretation,  $R_{it}(z_j)$  reflects the firm’s prior belief, and  $\tau_{ij}$  reflects the signal to noise ratio.

Rather than exogenously varying the expectation as in temporary equilibrium, we will analyze the effects of exogenous variation in the signal, allowing the residual component to (potentially) respond endogenously. Specifically, the residual is the expectation firm  $i$  would set if they received no information on that variable. The only way to form such an expectation is by using information and beliefs about *other* variables. Since the model here is linear, we impose

$$R_{it}(z_j) = \sum_{z_k \in \mathbf{z}: z_k \neq z_j} M_{ijk} E_{it}(z_k) \quad (8)$$

where  $\mathbf{z}$  is the vector of variables of interest in the model. Stacking equation (7) over expectations

of all  $z$ , substituting in equation (8), and rearranging gives:

$$E_{it}(z) = \chi_i \tau_i s_{it} \quad (9)$$

where  $\tau_i$  is a square diagonal matrix, with  $j$ th diagonal element equal to  $\tau_{ij}$ ;  $s_{it}$  is a column vector stacking  $s_{it}(z_j)$  over  $z_j$ , and the square matrix  $\chi_i$  is defined as

$$\chi_i \equiv (I - (I - \tau_i)M_i)^{-1} \quad (10)$$

where  $M_i$  has zeros on the diagonal, and  $(j, k)$ th element equal to  $M_{ijk}$  in equation (8).

We restrict attention to cases in which  $M_i$  is upper-triangular. Conceptually, this means we only consider cases where the subjective model of each agent can be expressed as a recursive system of equations, as discussed in the literature on Directed Acyclic Graphs (Spiegler, 2020).<sup>9</sup> This simplifies the algebra, but is not critical to the method. Appendix A shows how to relax it.

With this assumption,  $\chi_i$  is upper-triangular, with diagonal elements equal to 1. Each element of  $\tau_i$  therefore represents the degree to which a firm updates their expectations of a particular variable in response to direct information about that variable. In many applications this will be proportional to the precision of the agent's information. Off-diagonal elements of  $\chi_i$  capture cross-learning, or equivalently the links across expectations imposed in extended temporary equilibrium analysis.

The cross-learning matrix  $\chi_i$  captures both direct and indirect links between expectations of different variables: after a signal on short-run inflation, a firm may update expectations of long-run inflation partly because they believe inflation is persistent. They may also update their expectations of short-run marginal costs, which in turn affects their expectations of long-run marginal costs, which in turn affects their expectations of long-run inflation. It will be convenient from here to work directly with  $\chi_i$ , rather than considering each of these possible links individually.<sup>10</sup>

Substituting equation (9) into equation (3) yields

$$\pi_t = (1 - \lambda) \left( \sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i\pi_{t+s}} \tau_i s_{it} + (1 - \beta\lambda) \chi_{imc_{t+s}} \tau_i s_{it}) di + \mu \right) \quad (11)$$

where  $\chi_{i\pi_{t+s}}$  and  $\chi_{imc_{t+s}}$  are rows of  $\chi_i$  corresponding to  $E_{it}(\pi_{t+s})$  and  $E_{it}(mc_{t+s})$  respectively.

<sup>9</sup>As an example, this means that if the firm uses  $E_{it}\pi_t$  to update  $R_{it}mc_t$ , they cannot simultaneously use  $E_{it}mc_t$  to update  $R_{it}\pi_t$ .  $E_{it}mc_t$  may still affect  $R_{it}\pi_{t+s}$  for  $s > 0$ , as these are treated as separate elements of  $z$  from  $\pi_t$ .

<sup>10</sup>This has a parallel in the literature on production networks (Carvalho and Tahbaz-Salehi, 2019). The direct links in  $M_i$  are analogous to the elements of the input-output matrix, and  $\chi_i$  is the corresponding Leontief inverse.

## 2.2 The causal effect of inflation expectations

I now use the temporary-information equilibrium (11) to study the causal effect of changes in expected  $\pi_t$  and in expected  $\pi_{t+1}$ . The former is the effect studied in Werning (2022), while the latter more closely corresponds to the questions in the empirical information-treatment literature.

**Partial equilibrium.** To begin with, consider an exogenous shock to the signals firms receive about current inflation  $s_{it}(\pi_t)$ , holding constant the signals firms receive about all other variables. All proofs for this section and Section 3 are in Appendix A.

**Proposition 1** *The causal effect of a shock  $\varepsilon_{\pi_t}$  that increases  $s_{it}(\pi_t)$  equally for all firms is*

$$\frac{1}{\bar{\tau}_{\pi_t}} \frac{d\pi_t}{d\varepsilon_{\pi_t}} = (1 - \lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s \int_i \left( \chi_{i\pi_{t+s}\pi_t} \frac{\tau_{i\pi_t}}{\bar{\tau}_{\pi_t}} + (1 - \beta\lambda) \chi_{imc_{t+s}\pi_t} \frac{\tau_{i\pi_t}}{\bar{\tau}_{\pi_t}} \right) di \quad (12)$$

where  $\chi_{i\pi_{t+s}\pi_t}$  and  $\chi_{imc_{t+s}\pi_t}$  refer to the elements of  $\chi_i$  that correspond to the cross-learning from a signal about  $\pi_t$  to expectations of  $\pi_{t+s}$  and  $mc_{t+s}$  respectively. Scaling the derivative by the average response to  $\pi_t$  signals,  $\bar{\tau}_{\pi_t} \equiv \int_i \tau_{i\pi_t} di$ , implies that there is a unit increase in  $\bar{E}_t(\pi_t)$ .

As in temporary equilibrium analysis, this can be interpreted as the causal effect of inflation expectations. Firm signals about  $\pi_t$  have been varied exogenously, and the only way those signals affect the firm decision problem is through  $E_{it}(\pi_t)$ . This is therefore the theoretical equivalent of information-treatment RCTs using exogenous variation in information as an instrument for expectations, and thus determining the effects of expectations on other (individual-level) outcomes. In keeping with this IV-like interpretation, we scale the effect such that the result reflects the causal effect of a unit increase in the relevant aggregate expectation, which in this case is  $\bar{E}_t(\pi_t)$ .

This straightforwardly nests the two approaches above. Setting  $\chi_{iz_j\pi_t} = 0$  for all firms  $i$  and variables  $z_j \neq \pi_t$  recovers the temporary equilibrium case in (4), where one expectation is shocked and all others are held constant. Setting  $\chi_{imc_{t+s}\pi_t} = 0$  and  $\chi_{i\pi_{t+s}\pi_t} = 1$  recovers the extended temporary equilibrium in (6), in which a rise in expected  $\pi_t$  increases expected per-period inflation one-for-one at all horizons. By selecting different  $\tau_{i\pi_t}$  and  $\chi_i$ , we can also solve for the partial-equilibrium effect of  $\pi_t$  expectations under a range of common assumptions on expectation formation, which have different implications for responses to information and the linkages between expectations across horizons and variables. We return to this in Section 2.4.

Similarly, the causal effect of  $\pi_{t+1}$  expectations is given by Proposition 2.

**Proposition 2** *The causal effect of a shock  $\varepsilon_{\pi_{t+1}}$  that increases  $s_{it}(\pi_{t+1})$  equally for all firms is*

$$\frac{1}{\bar{\tau}_{\pi_{t+1}}} \frac{d\pi_t}{d\varepsilon_{\pi_{t+1}}} = (1 - \lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s \int_i \left( \chi_{i\pi_{t+s}\pi_{t+1}} \frac{\tau_{i\pi_{t+1}}}{\bar{\tau}_{\pi_{t+1}}} + (1 - \beta\lambda) \chi_{imc_{t+s}\pi_{t+1}} \frac{\tau_{i\pi_{t+1}}}{\bar{\tau}_{\pi_{t+1}}} \right) di \quad (13)$$

where the scaling implies there is a unit increase in  $\bar{E}_t \pi_{t+1}$ .

**The narrative heterogeneity channel.** A key novel insight from equations (12) and (13) concerns the impact of heterogeneity in expectation formation. There is substantial evidence of heterogeneity across households and firms in the precision of information (e.g. [Link et al., 2023](#)), and more directly in the responsiveness to signals ([Balla-Elliott, 2025](#)). There is also substantial heterogeneity in how agents update from one expectation to another (e.g. [Andre et al., 2022](#); [Macaulay and Moberly, 2026](#)). Holding signals, rather than expectations, fixed allows us to study how this heterogeneity affects the causal effect of expectations. Using the definition of a covariance, we can rewrite (12) as

$$\begin{aligned} \frac{1}{\bar{\tau}_{\pi_t}} \frac{d\pi_t}{d\varepsilon_{\pi_t}} = & (1 - \lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s \left( \bar{\chi}_{\pi_t+s\pi_t} + (1 - \beta\lambda) \bar{\chi}_{mc_t+s\pi_t} \right) \\ & + (1 - \lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s \left( \frac{1}{\bar{\tau}_{\pi_t}} \text{Cov}_I[\chi_{i\pi_t+s\pi_t}, \tau_{i\pi_t}] + \frac{1 - \beta\lambda}{\bar{\tau}_{\pi_t}} \text{Cov}_I[\chi_{imc_t+s\pi_t}, \tau_{i\pi_t}] \right) \end{aligned} \quad (14)$$

where  $\bar{\chi}_{z\pi_t}$  is the average of  $\chi_{iz\pi_t}$  across firms for each variable  $z$ , and  $\text{Cov}_I[\cdot, \cdot]$  denotes a cross-sectional covariance. Equation (13) admits an equivalent decomposition for the effects of  $\varepsilon_{\pi_{t+1}}$ .

The first summation captures representative-agent effects: the causal effect of expected current inflation depends on the average cross-learning to expectations of marginal costs and longer-run inflation across firms. The second summation captures the effects of heterogeneity. If the firms with the most precise information on  $\pi_t$  (i.e. the highest  $\tau_{i\pi_t}$ ) are also the firms who extrapolate most strongly from  $\pi_t$  to other variables (high  $\chi_{i\pi_t+s\pi_t}$  and  $\chi_{imc_t+s\pi_t}$ ), information is concentrated among those who will extrapolate and react most strongly to it. That amplifies the effect of an aggregate signal relative to what would be observed if all firms formed expectations in the same way.

We will refer to this broad category of effects as the ‘narrative heterogeneity channel,’ reflecting the idea that a narrative combines information with a subjective model to interpret it ([Gibbons and Prusak, 2020](#)). These effects were not visible in temporary equilibrium analysis, because such exercises implicitly remove all heterogeneity in expectations. Once we allow for heterogeneity, mathematically the narrative heterogeneity terms are a trivial consequence of the expectation of a product. In Section 2.4 we will argue that *economically* they are not trivial at all. A wide range of popular models of expectation formation imply potentially large and time-varying narrative heterogeneity effects, with important qualitative and quantitative consequences.

**General equilibrium.** All of the exercises so far have studied the causal effect of a given expectation rise, holding signals on all other variables constant. However, holding signals constant means abstracting from second-round effects: when expectations of  $\pi_{t+1}$  rise, that affects current realized  $\pi_t$  (equation (13)), which in turn should affect the information firms receive about  $\pi_t$ , which

has its own effects. To assess the full passthrough from inflation expectations to realized inflation in general equilibrium, this feedback should be taken into account.

The temporary-information approach can easily accommodate this. As is standard in models with noisy information (e.g. [Coibion and Gorodnichenko, 2015](#)), suppose that firm signals are equal to the variable of interest plus additively separable noise, which I allow to consist of a public and an idiosyncratic component:

$$s_{it}(\pi_{t+s}) = \pi_{t+s} + \varepsilon_{\pi t+s} + \tilde{\varepsilon}_{i\pi t+s} \quad (15)$$

where  $\varepsilon_{\pi t+s}$  and  $\tilde{\varepsilon}_{i\pi t+s}$  are both mean zero, and  $\tilde{\varepsilon}_{i\pi t+s}$  is additionally i.i.d. across firms.

This functional form assumption preserves the causal interpretation of the exercises above: exogenous variation in  $\varepsilon_{\pi t+s}$  only affects firm decisions through  $E_{it}(\pi_{t+s})$ , so can be used to assess the causal effects of that expectation on aggregate outcomes. However, that effect now includes an endogenous response of signals to the shock, through the dependence of the signal on realized  $\pi_{t+s}$ .

For the causal effect of  $\bar{E}_t(\pi_t)$ , this general equilibrium effect turns out to be irrelevant. The partial equilibrium effect in equation (12) increases  $\pi_t$ , which causes a second round of  $\bar{E}_t(\pi_t)$  updates, with associated consequences for realized inflation. However, since that general equilibrium effect operates through  $\bar{E}_t(\pi_t)$ , when we scale to ensure a unit impact on aggregate expectations we get back exactly to the partial equilibrium effect. I show this formally in [Appendix A](#).

However, for the causal effects of  $\bar{E}_t(\pi_{t+1})$ , the general equilibrium feedback is not irrelevant. From here, it will be useful to introduce the following notation: let  $\Omega^{PE}(\pi_{t+s})$  denote the partial-equilibrium effect of a change in  $s_{it}(\pi_{t+s})$ , scaled to ensure a unit impact on  $\bar{E}_t(\pi_{t+s})$  as in [Propositions 1 and 2](#). Let  $\Omega^{GE}(\pi_{t+s})$  denote the equivalent when we allow for general equilibrium feedback through the endogenous part of signals.

**Proposition 3** *The causal effect of  $\bar{E}_t(\pi_{t+1})$  on realized  $\pi_t$ , after accounting for general equilibrium effects through current inflation signals, is given by*

$$\Omega^{GE}(\pi_{t+1}) = K \cdot \frac{d\pi_t}{d\varepsilon_{\pi t+1}} = \frac{\Omega^{PE}(\pi_{t+1})}{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t) + \Omega^{PE}(\pi_{t+1}) \cdot (\bar{\chi}_{\pi_{t+1}\pi_t} \bar{\tau}_{\pi_t} + Cov_I[\chi_{i\pi_{t+1}\pi_t}, \tau_{i\pi_t}])} \quad (16)$$

where  $K$  is a scaling coefficient defined in [Appendix A](#) which ensures the impact on  $\bar{E}_t(\pi_{t+1})$  in general equilibrium is equal to 1.

This reveals two general-equilibrium forces at work after a shock to  $\bar{E}_t(\pi_{t+1})$ . First, the partial-equilibrium effect of the shock on realized  $\pi_t$  feeds back into signals  $s_{it}(\pi_t)$ . This change in signals in turn affects  $\pi_t$ , with strength governed by  $\Omega^{PE}(\pi_t)$ . Second, the resulting increase in  $E_{it}(\pi_t)$  causes firms to further update  $E_{it}(\pi_{t+1})$  through cross-learning. Since we scale to ensure a unit

increase in  $\bar{E}_t(\pi_{t+1})$ , this dampens the aggregate effect. Overall, general equilibrium effects amplify the partial-equilibrium effect of the shock if the first force dominates:

$$\Omega^{PE}(\pi_t) > \Omega^{PE}(\pi_{t+1}) \cdot \left( \bar{\chi}_{\pi_{t+1}\pi_t} + \frac{1}{\bar{\tau}_{\pi_t}} \text{Cov}_I[\chi_{i\pi_{t+1}\pi_t}, \tau_{i\pi_t}] \right) \quad (17)$$

Note that even if the desired model experiment has all firms receiving an identical signal about  $\pi_{t+1}$ , and so there is no narrative heterogeneity term in the direct partial-equilibrium effects of the shock, heterogeneity will still affect dynamics through these general equilibrium feedback channels. If realized  $\pi_t$  also affects real marginal costs, which have not been modeled here, that would further alter the dynamics. The temporary-information equilibrium can be used in the same way.

### 2.3 The role of expectations in the transmission of fundamental shocks

So far, we have been considering the causal effects of a shock or change in expectations. However, the temporary-information equilibrium apparatus is also useful in studying the related question of how expectations affect the transmission of *fundamental* shocks.

To see why, go back to equation (11), and allow for  $mc_t$  to be perfectly observed, with no cross-learning from it to any other expectations. In this setting we have

$$\pi_t = (1 - \lambda) \left( \left[ \sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i,\pi_{t+s},\pi_t} + (1 - \beta\lambda)\chi_{i,mc_{t+s},\pi_t}) \tau_{i,\pi_t} di \right] (\pi_t + \varepsilon_{\pi_t}) \right) + (1 - \lambda)(1 - \beta\lambda)mc_t + a \quad (18)$$

where  $a$  collects the steady-state markup  $\mu$  and all terms involving signals of  $\pi_{t+s}$  and  $mc_{t+s}$  for  $s \geq 1$ . Consider an i.i.d. shock to  $mc_t$ . Since the Calvo model is purely forward-looking, all variables return to steady state in period  $t + 1$ , and we can treat  $a$  as fixed. The aggregate effects of an i.i.d.  $mc_t$  shock are therefore given by

$$\frac{d\pi_t}{dmc_t} = (1 - \lambda) \left( \left[ \sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i,\pi_{t+s},\pi_t} + (1 - \beta\lambda)\chi_{i,mc_{t+s},\pi_t}) \tau_{i,\pi_t} di \right] \frac{d\pi_t}{dmc_t} + 1 - \beta\lambda \right) \quad (19)$$

which rearranges to

$$\frac{d\pi_t}{dmc_t} = \frac{(1 - \lambda)(1 - \beta\lambda)}{1 - \bar{\tau}_{\pi_t}\Omega^{PE}(\pi_t)} \quad (20)$$

This helps us to see the role of endogenous expectations in the transmission of the fundamental  $mc_t$  shock, as all expectational effects are contained in the  $\bar{\tau}_{\pi_t}\Omega^{PE}(\pi_t)$  term in the denominator.

Formally, we can compute a counterfactual shock transmission in which there is no response of expectations by setting  $\Omega^{PE}(\pi_t) = 0$ . We can then compare this counterfactual to equation (20) to compute the share of marginal cost shock transmission which can be attributed to expectations.

**Proposition 4** *The fraction of the transmission of an i.i.d.  $mc_t$  shock to realized  $\pi_t$  that is due to inflation expectations is*

$$\frac{\frac{d\pi_t}{dmc_t} - \frac{d\pi_t}{dmc_t} \Big|_{\sim E}}{\frac{d\pi_t}{dmc_t}} = \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t) \quad (21)$$

where  $\frac{d\pi_t}{dmc_t} \Big|_{\sim E}$  denotes the transmission in a counterfactual with  $\Omega^{PE}(\pi_t) = 0$ .

That is, the share of the transmission of a fundamental marginal cost shock that is due to the behavior of inflation expectations is equal to the partial equilibrium effect of an expectation shock, multiplied by the average responsiveness to inflation signals. The channels determining the causal effects of expectations also determine the role of expectations in the transmission of other shocks.

This is a particularly simple fundamental shock to analyze. However, the approach continues to work for more complicated interactions with multiple variables, or for alternative counterfactuals.<sup>11</sup> The key point is that studying the shock in the temporary-information equilibrium, rather than in a rational-expectations equilibrium or after imposing any other expectation-formation assumptions, allows us to separate the consequences of expectational shifts from their causes, without abstracting from heterogeneity in expectation formation.

## 2.4 The narrative heterogeneity channel

The analysis above identifies a covariance between firms' direct response to signals, and their cross-learning, as a potential factor in the effects of expectations in general equilibrium. We now consider this 'narrative heterogeneity channel' in more detail.

**An example.** We begin with a simple example to give intuition for how the narrative heterogeneity channel operates, and how it can change quantitative and qualitative conclusions about the effects of expectations. Suppose that there is no cross-learning from inflation to marginal costs ( $\chi_{imc_{t+s}\pi_t} = 0$ ), and, for horizons  $s \geq 1$ ,  $\chi_{i\pi_{t+s}\pi_t}$  and  $\tau_{i\pi_t}$  are each split equally between two values

$$\chi_{i\pi_{t+s}\pi_t} = \{0, 2\} \text{ for } s \geq 1, \quad \tau_{i\pi_t} = \{0, 1\} \quad (22)$$

These have been chosen such that  $\bar{\chi}_{\pi_{t+s}\pi_t} = 1$  for all  $s$  (recall that  $\chi_{i\pi_t\pi_t} = 1$  for all firms by construction). If  $\chi_{i\pi_{t+s}\pi_t}$  and  $\tau_{i\pi_t}$  are independent, the narrative heterogeneity terms equal 0, and

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<sup>11</sup>For example, if firms observe current  $\pi_t$ , we could allow  $E_{it}(\pi_t)$  to adjust in the counterfactual, but remove transmission to expected future  $\pi_{t+s}$ ,  $mc_{t+s}$  by modifying the counterfactual  $\chi_i, \tau_i$  and thus  $\Omega^{PE}(\pi_t)$  accordingly.



the causal effect of current inflation expectations in partial equilibrium is as in equation (6) and in Werning (2022) Proposition 1. However, if they are correlated, this causal effect can change dramatically. Proposition 5 considers the limit as  $\beta \rightarrow 1$ , which is a useful benchmark to gain intuition (and which is also used in Werning, 2022).

**Proposition 5** *If  $\tau_{i\pi_t}$  and  $\chi_{i\pi_t+s\pi_t}$  are split equally between the values in equation (22), the causal effect of expected inflation on realized inflation as  $\beta \rightarrow 1$  is given by*

$$\frac{1}{\bar{\tau}_{\pi_t}} \frac{d\pi_t}{d\varepsilon_{\pi_t}} = 1 + \lambda(2q - 1) \quad (23)$$

where  $q \equiv \Pr(\chi_{i\pi_t+s\pi_t} = 2 | \tau_{i\pi_t} = 1)$ .

Werning (2022) Proposition 1 states that the transmission from expectations to inflation is 1 under no-discounting, irrespective of price stickiness. Equation (23) shows this is no longer true when expectations are formed heterogeneously. Unless information precision is independent of subjective models (i.e.  $q = \frac{1}{2}$ ), transmission depends on  $\lambda$ , even though the representative-agent terms do not.

The intuition is that  $\chi_{i\pi_t+s\pi_t} = 1$  is a special case that balances two opposing forces. As prices get stickier, each price-setting firm expects to wait longer before they can change prices again, which increases their responsiveness to expectations of inflation in future periods. At the same time, stickier prices mean fewer firms are resetting prices. When  $\chi_{i\pi_t+s\pi_t} = 1$  for all firms, these forces exactly cancel out, leaving the causal effect of expectations independent of  $\lambda$ .

Heterogeneity breaks this balance, by affecting the first of these forces. When price stickiness increases, firms with high  $\chi_{i\pi_t+s\pi_t}$  increase their responsiveness to expected inflation a great deal, because the extra length of a typical fixed-price spell encompasses more periods over which they expect inflation to remain elevated. By the same logic in reverse, firms with a low  $\chi_{i\pi_t+s\pi_t}$  change their behavior less after an increase in stickiness. If information is concentrated among the former group, they dominate the passthrough from aggregate expectations to inflation, so that passthrough increases in stickiness.<sup>12</sup> If information is instead concentrated among the latter group (with low  $\chi_{i\pi_t+s\pi_t}$ ), that group dominates, and the passthrough decreases in stickiness.

**Common models of expectations generate narrative heterogeneity.** Previous literature has typically relaxed either full information or rational expectations, but not both simultaneously (see related literature discussion above). That allows for heterogeneity in either  $\chi_i$  or  $\tau_i$ , and not both,

<sup>12</sup>Macaulay and Moberly (2026) find evidence that among households greater Kalman gains on inflation signals are correlated with greater perceived inflation persistence, consistent with a positive narrative heterogeneity effect here and thus an amplified causal effect of current inflation expectations.

which forces the narrative heterogeneity terms to be 0. However, if heterogeneity is allowed in both components of expectation formation, many standard models of expectations naturally generate narrative heterogeneity effects.

In Appendix B, I consider the simple case in which the only possible signals are on  $\pi_t$ , where there is no public noise in those signals, and where both  $\pi_t$  and  $\tilde{\varepsilon}_{i\pi_t}$  are normally distributed.<sup>13</sup> I then introduce a range of common assumptions on expectation formation, and show that they each imply systematic correlations between direct responses to signals ( $\tau_{i\pi_t}$ ) and cross-learning ( $\chi_{i\pi_t+s\pi_t}, \chi_{imc_t+s\pi_t}$ ), whenever heterogeneity is allowed in both. Since these models of expectation formation are well-known, I leave the details and derivations to Appendix B, and focus here on the intuition for why the narrative heterogeneity channel arises under each set of assumptions.

*Rational inattention* (Maćkowiak et al., 2023): if firms can pay a cost to increase the precision of their signals, then  $\tau_{i\pi_t}$  is endogenous to the firm's expected use of information. Firms who believe there is a stronger pass-through from current inflation to future inflation and marginal costs find information on  $\pi_t$  to be more valuable, and so process more of it. If there is heterogeneity in cross-learning, high  $\tau_{i\pi_t}$  is therefore correlated with large magnitudes of  $\chi_{i\pi_t+s\pi_t}$  and  $\chi_{imc_t+s\pi_t}$ .

For further intuition, suppose all firms have  $\chi_{i\pi_t+1\pi_t} \geq 0$ . Rational inattention implies  $Cov_I[\chi_{i\pi_t+1\pi_t}, \tau_{i\pi_t}] > 0$ , because firms with larger  $\chi_{i\pi_t+1\pi_t}$  condition prices more strongly on signals of  $\pi_t$ , so they pay for more precision in those signals. This amplifies the causal effect of aggregate inflation expectations, as information is concentrated in those who respond most strongly to it (Proposition 5). This in turn amplifies the transmission of fundamental shocks (equation (20)).

*Learning* (Evans and McGough, 2020): if firms update their perceived laws of motion for inflation and marginal costs after observing their signals, then  $\chi_i$  is endogenous to information precision. Firms who observe more precise signals update their subjective models of the economy more strongly after each observation than those with less precise information. If there is heterogeneity in information precision, this therefore leads to systematic correlations between elements of  $\tau_i$  and  $\chi_i$ . Notably, the narrative heterogeneity channel generated in this case may vary over time as firms continually update their perceived laws of motion as they receive new signals.

*Sticky information* (Mankiw and Reis, 2002): if firms can pay a cost to observe information more frequently (as in e.g. Reis, 2006, and others), then subjective models influence this information choice, with a similar logic to the rational inattention model.

*Level- $k$  thinking* (Farhi and Werning, 2019): again, a narrative heterogeneity channel arises

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<sup>13</sup>The exception is level- $k$  thinking, where for tractability I allow for signals on  $mc_t$  rather than  $\pi_t$ .

whenever firms can pay a cost to engage in a further round of reasoning, as in [Alaoui and Penta \(2016, 2022\)](#). In this case, firms with more precise information have more to gain from engaging in costly reasoning about how to use that information. Reasoning causes firms to update their cross-learning parameters, so the elements of  $\chi_i$  differ between firms with different  $\tau_i$ .

*Information delegation to media outlets* ([Nimark and Pitschner, 2019](#)): firms with larger  $\chi_{i\pi_t+s\pi_t}$ ,  $\chi_{imc_t+s\pi_t}$  place more value on  $\pi_t$  information. Competitive media outlets specialize, and firms with large values of these cross-learning parameters select those which report  $\pi_t$  more often.

In this case, the resulting narrative heterogeneity channel is state-dependent. When inflation is close to steady state, almost no outlets find inflation sufficiently newsworthy for them to report it, and most firms have  $\tau_{i\pi_t} = 0$ . Similarly, when inflation is very far from steady state, it is highly newsworthy, and almost all outlets do report it, leading most firms to have  $\tau_{i\pi_t} = 1$ . It is only in the intermediate region where some outlets report inflation, and others don't, that the precision of firm information becomes heterogeneous, generating a narrative heterogeneity channel.

*Selective memory recall* ([Bordalo et al., 2023](#)): if signals on current economic conditions act as cues for selective recall (as in [Gennaioli et al., 2024](#)), firms with different information recall different past inflation experiences. Firms with precise signals (high  $\tau_{i\pi_t}$ ) recall experiences clustered around the realized  $\pi_t$ , while those with less precise signals draw from a wider set of experiences. If recalled experiences inform cross-learning, e.g. through perceptions of inflation persistence, this implies a correlation between  $\tau_i$  and  $\chi_i$ , and thus a narrative heterogeneity channel.

*Data generated through production* ([Farboodi and Veldkamp, 2025](#)): firm subjective models affect their response to a given shock. That in turn affects how much data each firm has to learn from in subsequent periods, again generating a correlation between subjective models and information.

*Heterogeneous price indices*: for households, inflation rates vary substantially across the population ([Kaplan and Schulhofer-Wohl, 2017](#); [Kiss and Strasser, 2024](#)). If households observe their own idiosyncratic price levels (heterogeneous information), and form expectations using the dynamics of those idiosyncratic price levels (heterogeneous cross-learning), then we could see a narrative heterogeneity effect even if all agents have rational expectations. I take this idea to the Calvo model by assuming firms belong to sectors which differ in the persistence of their inflation processes. Greater inflation persistence implies a weaker response to inflation signals but a stronger extrapolation from short to long-horizon inflation expectations, generating a negative covariance between  $\tau_{i\pi_t}$  and  $\chi_{i\pi_t+s\pi_t}$ , even though all cross-learning is consistent with rational expectations.

Overall, the narrative heterogeneity channel is a robust feature of many models of expectation formation, either because subjective models influence information acquisition, or because information influences the formation of subjective models. The final case (heterogeneous price indices) shows that the mechanism can arise even under rational expectations, if there is heterogeneity in the object to be forecast. Allowing for heterogeneity in both components of expectation formation therefore generates dynamics that are not present if heterogeneity is restricted to either information or subjective models alone. Moreover, in cases with such two-sided heterogeneity, the correlation between information and subjective models provides a novel way to test competing models of expectation formation. As an example, Empirical Result 1 in Section 5 documents a correlation that is consistent with rational inattention, but not with several other models in the list above.

### 3 General model

Section 2 used a Calvo model for concreteness, but nothing in the logic is specific to that setting. I now show how to apply the temporary-information equilibrium approach to a general class of linear models in discrete time, and show that the narrative heterogeneity channel is a general feature of the causal effects of expectations.

**Environment.** Consider a class of models in which the policy function of agent  $i$  can be written as

$$\mathbf{x}_{it} = A_i \mathbf{x}_{it-1} + \sum_{s=0}^{\infty} (B_{iws} E_{it} \mathbf{w}_{t+s} + B_{ixs} E_{it} \mathbf{x}_{t+s}) \quad (24)$$

and where general equilibrium conditions can be collectively written as

$$C \mathbf{x}_t + D \mathbf{w}_t + F \boldsymbol{\xi}_t = 0. \quad (25)$$

In these expressions,  $\mathbf{x}_{it}$  is an  $N_x \times 1$  vector of choice variables for agent  $i$  in period  $t$ ;  $\mathbf{x}_t \equiv \int_i \mathbf{x}_{it} di$  denotes aggregate actions;  $\mathbf{w}_t$  is a  $N_w \times 1$  vector of state variables; and  $\boldsymbol{\xi}_t$  is a  $N_\xi \times 1$  vector of exogenous shocks.  $A_i$ ,  $B_{iws}$ ,  $B_{ixs}$ ,  $C$ ,  $D$ , and  $F$  are matrices of coefficients, with the former three potentially varying across agents  $i$ .

The Calvo model of Section 2 is one example of a model in this class: equation (2) is straightforwardly of the form of (24). Moreover, the only state variable in  $\mathbf{w}_t$  is  $mc_t$ , and since this was treated as a pure shock variable, the only general equilibrium condition is  $mc_t + \xi_{mc,t} = 0$ , which is of the form of (25). However, a broad range of other models also fit into this class: (24) could describe the policy function of firms, households, investors, or other agents, and (25)

could capture technological constraints, market clearing conditions, or the decision rules of other categories of agents whose expectations are not the focus of the analysis.

As in Section 2.1, define  $z_t$  as the vector of all variables about which the agent will form expectations. Similarly, let  $B_i$  be the matrix that collects all  $B_{iws}, B_{ixs}$  coefficients.

$$z_t \equiv [x_t, w_t, x_{t+1}, w_{t+1}, \dots]', \quad B_i \equiv [B_{ix0}, B_{iw0}, B_{ix1}, B_{iw1}, \dots] \quad (26)$$

Following the same steps as Section 2.1, we can express expectations of  $z_t$  as in (9). Substituting this into (24) yields

$$x_{it} = A_i x_{it-1} + B_i \chi_i \tau_i s_{it} \quad (27)$$

**Temporary-information equilibrium.** For given distributions of  $\tau_i, \chi_i$ , and given exogenous shocks  $\xi_t$ , the temporary-information equilibrium is a sequence of actions  $x_{it}$ , endogenous states  $w_t$ , and signals  $s_{it}$  such that

1.  $x_{it}$  is chosen according to the policy function (27), and aggregate actions are  $x_t \equiv \int_i x_{it} di$ ;
2.  $w_t$  satisfies the general equilibrium condition (25);
3. a specified process links signals  $s_{it}$  with the corresponding states and aggregate actions.

In principle, many different processes could be specified for item (3). For simplicity, however, I restrict attention here to the form used in Section 2.2:

$$s_{it}(z_{jt}) = z_{jt} + \varepsilon_{jt} + \tilde{\varepsilon}_{ijt} \quad (28)$$

where  $\varepsilon_{jt}, \tilde{\varepsilon}_{ijt}$  are mean-zero white noise. The public noise  $\varepsilon_{jt}$  is common across agents  $i$ ; the idiosyncratic noise  $\tilde{\varepsilon}_{ijt}$  is independent across agents. From here, all results follow as in Section 2.

**Causal effects of expectations in partial equilibrium.** The causal effect of  $\bar{E}_t(z_{jt})$  on an element of the aggregate choice vector  $x_t$  in partial equilibrium (i.e. holding all signals aside from  $s_{it}(z_{jt})$  constant) is given by Proposition 6, which generalizes Propositions 1 and 2 in the Calvo model.

**Proposition 6** *The causal effect of  $\bar{E}_t(z_{jt})$  on aggregate action  $x_{kt}$  in partial equilibrium is*

$$\Omega_k^{PE}(z_{jt}) = \frac{1}{\bar{\tau}_j} \cdot \frac{dx_{kt}}{d\varepsilon_{jt}} = \sum_{m=0}^{N_z} \left( \bar{B}_{km} \bar{\chi}_{mj} + Cov_I \left[ B_{ikm}, \chi_{imj} \frac{\tau_{ij}}{\bar{\tau}_j} \right] + \bar{B}_{km} Cov_I \left[ \chi_{imj}, \frac{\tau_{ij}}{\bar{\tau}_j} \right] \right) \quad (29)$$

where  $B_{ikm}$  denotes the  $k, m$ th element of  $B_i$ , and  $\bar{B}_{km} \equiv \int_i B_{ikm} di$ . The scaling by  $\bar{\tau}_j^{-1}$  ensures that the information shock is such that  $\bar{E}_t(z_{jt})$  increases by 1.

This consists of three sets of effects. The first term inside the summation is the *representative agent channel*. This summarizes all shock transmission channels in models with a representative agent, and indeed in many models with heterogeneity, where average expectation formation is sufficient to capture shock responses to first order.<sup>14</sup> The third term captures the *narrative heterogeneity channel*, explored in detail for the Calvo model in Section 2.4.

The second term, the *response heterogeneity channel*, was not present in Section 2. Since  $\chi_{imj}\tau_{ij}$  gives the response of  $E_{it}(z_{mt})$  to a signal on  $z_{jt}$ , this reflects that shocks will be amplified if the agents whose expectations react the most to the shock are the agents whose actions are most sensitive to those expectations (i.e. who have the largest  $B_{ikm}$ ). This channel is behind the dynamics in Macaulay (2021), Guerreiro (2023), Macaulay and Moberly (2026), and Broer et al. (2026). Importantly, the existence of such response heterogeneity does not interfere with the narrative heterogeneity channel, the main novel effect introduced in this paper.

As in Section 2, Appendix A shows that choosing  $\tau_{ij}$  and  $\chi_{imj}$  appropriately in equation (29) recovers the causal effects computed from temporary equilibrium and extended temporary equilibrium analyses.

**Causal effects of expectations in general equilibrium.** The causal effect of  $\bar{E}_t(z_{jt})$  on an element of the aggregate choice vector  $x_t$  in general equilibrium is given by Proposition 7.

**Proposition 7** *The causal effect of  $\bar{E}_t(z_{jt})$  on aggregate action  $x_{kt}$  is*

$$\Omega_k^{GE}(z_{jt}) = \frac{[\Lambda^{GE}\Omega^{PE}]_{k,j}}{1 + \Gamma_j} \quad (30)$$

where  $\Omega^{PE}$  is an  $N_x \times N_z$  matrix collecting the partial-equilibrium effects of each expectation  $\bar{E}_t(z_{jt})$  on each aggregate action  $x_{kt}$ , and  $\Lambda^{GE}$  is an  $N_x \times N_x$  matrix capturing general-equilibrium feedback through aggregate actions and states. The denominator scales the effect such that  $\bar{E}_t(z_{jt})$  rises by 1. Both  $\Lambda^{GE}$  and  $\Gamma_j$  are defined in Appendix A.

Importantly, the GE feedback matrix  $\Lambda^{GE}$  will typically not be diagonal. This means that when computing the effects of  $\bar{E}_t(z_{jt})$  on a specific aggregate action  $x_{kt}$ , the effects of that expectation on *all* aggregate actions and state variables will affect the result, through learning from the endogenous component of signals. This generalizes the result in Proposition 3, where shocks to  $\bar{E}_t(\pi_{t+1})$  in part propagated by influencing  $\pi_t$ , which in turn affected  $\bar{E}_t(\pi_t)$ . Note in particular that in this general

<sup>14</sup>In Andrade et al. (2019), for example, households differ in their interpretation of forward guidance announcements, but the aggregate effects of an announcement depend only on the average over this mix of beliefs. Heterogeneity might affect the average information or cross-learning behavior sustained in equilibrium, but unless one of the other two terms in equation (29) is non-zero, those averages alone drive aggregate shock transmission.

class, there may be an additional round of such learning from endogenous signals: if a change in  $x_t$  also affects the states  $w_t$  through the equilibrium condition (25), that  $w_t$  change *also* affects agent information (see Corollary 1 in Appendix A). This was not present in Section 2 because we abstracted there from all responses of the state (marginal costs) to firm actions.

The other key insight from this is that all three channels identified in partial equilibrium also affect the strength of general equilibrium feedback. When exogenous variation in a given expectation affects aggregate actions and states, this affects the information agents receive about those variables, changing their expectations again. The effects of that second round of expectation updating also depend on the representative-agent, response heterogeneity, and narrative heterogeneity channels of Proposition 6. This is the core of the missing intercept problem discussed in Section 4.

Indeed, these same forces also determine the share of fundamental shock transmission due to expectation updating. Proposition 10 in Appendix A shows the analysis of Section 2.3 straightforwardly generalizes to this broad class of models, and general fundamental shocks.

Overall, then, this section demonstrates that temporary-information equilibrium can be applied beyond the Calvo model of Section 2. To apply it to a different context, a researcher must simply cast their preferred model in the form of equations (24)-(25), then select a way to discipline the key statistics on  $\tau$  and  $\chi$ . This could come from a particular theoretical model of expectation formation (as in Section 2.4), or from data, as I now go on to demonstrate across two applications.

## 4 Mapping RCT estimates into general equilibrium

Since the temporary-information equilibrium method is conceptually linked to information-treatment RCTs, the expectation statistics in the model correspond closely to parameters estimated in those experiments. Temporary-information equilibrium can therefore be used to translate RCT estimates into general equilibrium insights, overcoming the typical missing intercept problem in the empirical literature. This helps us uncover whether the departures from full-information rational-expectations observed in a given RCT are large or small, in terms of their macroeconomic implications.

I apply the method to the RCT conducted on European firms by Baumann et al. (2026), as this lends itself naturally to interpretation through the Calvo model already studied in Section 2. Specifically, I compute the causal effects of one-year-ahead inflation expectations as in Propositions 2 and 3. However, the methods Baumann et al. (2026) employ are common to a broad range of survey experiments, so the same procedure could be applied in other experiments, using the temporary-information equilibrium of other models as appropriate to the context.

**The missing intercept in information-treatment RCTs.** RCTs like Baumann et al. (2026) use information treatments to exogenously vary the expectations of some survey participants, while



keeping others as a control group. This cross-sectional variation enables researchers to identify the causal effect of expectations on individual beliefs and actions. While this has advanced our understanding of the micro-level effects of expectations, it is often hard to determine whether the results are of a magnitude that is meaningful for macroeconomic outcomes. This is because the individual-level effect cannot simply be aggregated to give a total macroeconomic effect of a given expectation change. If *all* inflation expectations change, for example, then aggregate realized inflation may change, along with interest rates, wages, and other variables, each of which may feed back into firm expectations and decisions. These general equilibrium effects form the missing intercept between the estimates from an information-provision RCT and macroeconomic insights.

This issue is present in all empirical work using cross-sectional variation to identify partial-equilibrium elasticities (see e.g. discussions in [Nakamura and Steinsson, 2018](#); [Guren et al., 2020](#); [Wolf, 2023](#), among many others). In information-provision experiments, however, the problem is particularly acute, for two reasons. First, the experiment typically occurs only once, or a small number of times. There is therefore limited scope for time-series approaches that could complement the cross-sectional identification (as in e.g. [Donaldson, 2025](#); [Matthes et al., 2025](#)).

Second, the features of expectations that drive the partial-equilibrium RCT results also affect the missing general equilibrium feedback. In studies of fiscal policy, researchers can use estimates from other literatures to quantify the general-equilibrium intercept, thanks to equivalence results from [Wolf \(2023\)](#). This approach cannot be taken for expectations experiments, because the features of expectations revealed in the partial-equilibrium RCT estimates *also* affect the intercept. If, for example, firms over-extrapolate from current to future inflation, that will affect both the RCT estimates and firm responses to subsequent general equilibrium changes in aggregate variables.

The temporary-information equilibrium approach developed here provides a way through this issue: the results in Section 3 can be applied to interpret the results from information-treatment RCTs in any model fitting the form introduced there.

## 4.1 Extracting key expectation statistics

[Baumann et al. \(2026\)](#) elicit prior beliefs about inflation in 1, 3, and 5 years' time. They then treat a random subsample of participants with information about inflation. They study two treatments: one treatment group is shown information on current inflation, and another is shown information on one-year-ahead inflation (from SPF forecasts).

**RCT regressions.** As standard in the RCT literature, [Baumann et al. \(2026\)](#) first estimate:

$$E_{it}^{post} \pi_{t+h} = a_0 + a_1 \cdot E_{it}^{prior} \pi_{t+h} + \sum_j b_j \cdot E_{it}^{prior} \pi_{t+h} \cdot D_{ij} + \sum_j c_j \cdot D_{ij} + \Gamma_{it} + \text{error}_i \quad (31)$$

where  $D_{ij}$  is an indicator = 1 if firm  $i$  is in treatment group  $j$ , and  $\Gamma_{it}$  is a vector of controls. They then estimate:

$$E_{it}^{post} Y_i = \gamma_0 + \gamma_1 \cdot E_{it}^{post} \pi_{t+h} + \gamma_2 \cdot E_{it}^{prior} \pi_{t+h} + \Gamma_{it} + \text{error}_i \quad (32)$$

for a variety of expected dependent variables  $E_{it}^{post} Y_i$ .  $E_{it}^{post} \pi_{t+h}$  is instrumented with variation from the information treatment.

The estimated coefficients in both of these regressions can now be mapped into the parameters in the temporary-information equilibrium framework developed above. Letting equation (7) describe the prior belief of agent  $i$  about  $\pi_{t+h}$ , the posterior after observing a further  $\pi_{t+h}$  signal  $s^{RCT} = \pi_{t+h} + \varepsilon^{RCT}$  in the experiment is:

$$E_{it}^{post} \pi_{t+h} = \tau_{i\pi_{t+h}}^{RCT} s^{RCT} + (1 - \tau_{i\pi_{t+h}}^{RCT}) \cdot \underbrace{\left[ \tau_{i\pi_{t+h}} s_{it}(\pi_{t+h}) + (1 - \tau_{i\pi_{t+h}}) \sum_{z_k \neq \pi_{t+h}} M_{i\pi_{t+h}k} E_{it}^{prior}(z_k) \right]}_{E_{it}^{prior} \pi_{t+h}} \quad (33)$$

That is, posteriors are a weighted average of priors and the new signal. As with equation (7), this follows standard signal extraction models, though we do not impose that  $\tau_{i\pi_{t+h}}^{RCT}$  must be the optimal one derived from Bayes' rule. Given this, the coefficient  $b_j$  in equation (31) is  $-1 \times$  the average degree of updating from the observed signal in treatment group  $j$  (Coibion et al., 2022; Haaland et al., 2023). For the group shown a signal about  $\pi_{t+1}$ , the  $b_j$  estimate when  $E_{it} \pi_{t+1}$  is the dependent variable is therefore equal to  $-\bar{\tau}_{\pi_{t+1}}^{RCT}$  in our model.<sup>15</sup>

Going further, the parameter  $\gamma_1$  in equation (32) can also be mapped into the model, in the following way.

**Proposition 8** *Under the expectation formation structure in equations (9) and (33), random assignment of  $D_{ij}$ , and the assumption that the treated variable is ordered last in  $M_i$ :*

(i) *Using  $D_{ij}$  as the only excluded instrument in equation (32), the IV coefficient identifies*

$$\gamma_{1,Y}^{Wald} = \int_i \chi_{iY\pi_{t+h}} \frac{\tau_{i\pi_{t+h}}^{RCT}}{\bar{\tau}_{\pi_{t+h}}^{RCT}} di = \bar{\chi}_{Y\pi_{t+h}} + \frac{1}{\bar{\tau}_{\pi_{t+h}}^{RCT}} Cov_I[\chi_{iY\pi_{t+h}}, \tau_{i\pi_{t+h}}^{RCT}] \quad (34)$$

(ii) *Using both  $D_{ij}$  and  $D_{ij} \times E_{it}^{prior} \pi_{t+h}$  as excluded instruments (i.e. using (31) as a first stage),*

<sup>15</sup>For other combinations of treatment and dependent variable, the signal is no longer precisely about the dependent variable. The coefficient therefore represents the average updating of the dependent variable ( $E_{it} \pi_{t+h}$ ) in response to the treatment signal (on  $\pi_{t+s}$  for some  $s$ ), which depends on both the direct response to information  $\tau_{i\pi_{t+s}}^{RCT}$  and the cross-learning  $\chi_{i\pi_{t+h}\pi_{t+s}}$  from the treated variable to the dependent variable.

the 2SLS estimand  $\gamma_{1,Y}^{2SLS}$  is a convex combination of  $\gamma_{1,Y}^{Wald}$  and a nuisance term specified in Appendix C.1.

**Proof.** Appendix C.1. ■

The core intuition here is that  $\gamma_1$ , as the estimate from an IV regression, identifies a *local* average treatment effect. In this context, that means it identifies a weighted average of how expectations of  $\pi_{t+1}$  affect expectations of  $Y_i$ , i.e. of the cross-learning parameter  $\chi_{i,Y,\pi_{t+1}}$ . Crucially, the weights are determined by how responsive each individual firm’s expectations of  $\pi_{t+h}$  are to information on  $\pi_{t+h}$ , which is precisely how  $\tau_{i,\pi_{t+h}}^{RCT}$  is defined in equation (33). In most applications of IV, this complier weighting is a nuisance. Here it is a benefit, because the causal effects of expectations in temporary-information equilibrium (Proposition 2) are themselves responsiveness-weighted averages of cross-learning. Thus,  $\gamma_1$  identifies the expectations statistic required by the theory.

**Discussion.** Beyond the simple intuition just stated, there are several points about Proposition 8 that merit further discussion.

First, it is standard practice in the RCT literature to use both the treatment indicator  $D_{ij}$  and its interaction with priors as instruments for  $E_{it}^{post} \pi_{t+h}$  in equation (32). Using the interaction as a second instrument has the advantage of increasing the power of the first-stage regression. However, Proposition 8 shows that this comes at the cost of achieving a clean mapping to the structural parameters of interest in temporary-information equilibrium.

Because only the 2SLS estimates are reported in Baumann et al. (2026), I proceed below under the assumption that  $\gamma_{1,Y}^{2SLS} \approx \gamma_{1,Y}^{Wald}$ . Appendix C.1 derives the relationship between the two and shows that, given the large gap between average prior expectations and the treatment signal in this experiment (5.83% vs. 2.8%), the 2SLS estimates are dominated by the treatment-control comparison, making  $\gamma_{1,Y}^{2SLS} \approx \gamma_{1,Y}^{Wald}$  a highly accurate approximation in this context.

Second, the experiment identifies cross-learning weighted by  $\tau_{i\pi_{t+h}}^{RCT}$ , which may in principle differ from the responsiveness to signals in everyday life  $\tau_{i\pi_{t+h}}$ . This is intuitive: we can only identify the effects of the type of information shock produced in the experiment. See Maćkowiak and Wiederholt (2024) for how  $\tau_{i\pi_{t+h}}$  and  $\tau_{i\pi_{t+h}}^{RCT}$  may be related if agents are rationally inattentive.

Third, the proof of Proposition 8 requires that the treated variable is ordered last in  $M_i$ . Given that  $M_i$  is upper-triangular, this ensures we are allowing for cross-learning from the treatment variable to all other variables in the system. Since this is the cross-learning we are interested in here (as opposed to cross-learning to the treated variable from other variables), this is the natural ordering to consider.

To compute the causal effects of expectations of one-year-ahead inflation in partial equilibrium,  $\gamma_1$  from equation (32) is therefore sufficient. For some exercises we will also make use of  $\bar{\tau}_{\pi_{t+1}}$  estimated from equation (31).

Finally, note that the experiment in [Baumann et al. \(2026\)](#) does not provide an estimate of  $\bar{\tau}_{\pi_t}$ , the average  $E_{it}(\pi_t)$  response to signals about  $\pi_t$ . This affects the evaluation of general equilibrium effects (see Proposition 3 and the associated discussion). In all calibrations below I set  $\bar{\tau}_{\pi_t} = 1$ , as if all firms are perfectly well-informed about current inflation. This is conservative, in that it imposes no further departure of the RCT calibration from the rational expectations and perfect foresight benchmarks I consider.<sup>16</sup> Future RCTs intending to use this method could, however, easily measure this parameter in the same way as [Baumann et al. \(2026\)](#) measure  $\bar{\tau}_{\pi_{t+1}}$ .

## 4.2 The causal effect of expected future inflation on current inflation

I now compute the causal effect of one-year-ahead inflation expectations on realized current inflation, under two sets of assumptions. In the first, I compute only the effects that operate through the updating of inflation expectations at various horizons, abstracting from any cross-learning from  $\pi_{t+1}$  to marginal cost expectations (i.e. setting  $\chi_{imc_{t+s}\pi_t} = \chi_{imc_{t+s}\pi_{t+1}} = 0$ ). This requires no extra structure beyond that described in Section 2. In the second, I impose a simple production function over labor and other inputs, that allows me to translate the estimated effects of the information treatment on wages and non-wage costs into effects on marginal costs. This therefore incorporates marginal cost cross-learning, at the expense of adding greater structure to the model.

In both exercises, I calibrate the relevant statistics on  $\chi_i$  and  $\tau_i$  using the estimated coefficients in equations (31) and (32), assuming one period is one year. Full details are reported in Appendix C.2. Of particular note is that the estimates of  $\gamma_{1,Y}$  are insignificantly different from 0 for every  $Y$  considered when using the  $\pi_t$  treatment as the instrument, while those using the  $\pi_{t+1}$  treatment are significant and larger in magnitude. I therefore set all cross-learning from  $\pi_t$  to 0. This gives a conservative estimate of the degree of learning from prices, but does not affect the partial-equilibrium effects of expected  $\pi_{t+1}$ , as these rely on cross-learning from  $\pi_{t+1}$  only. I relax this in Appendix C.6. Finally, I set  $\beta$  to 0.96, and plot the results for varying values of price stickiness  $\lambda$ . Pointwise 90% confidence intervals reflect the influence of sampling uncertainty in the RCT estimates, holding  $\beta$  and  $\lambda$  fixed; details are in Appendix C.3.

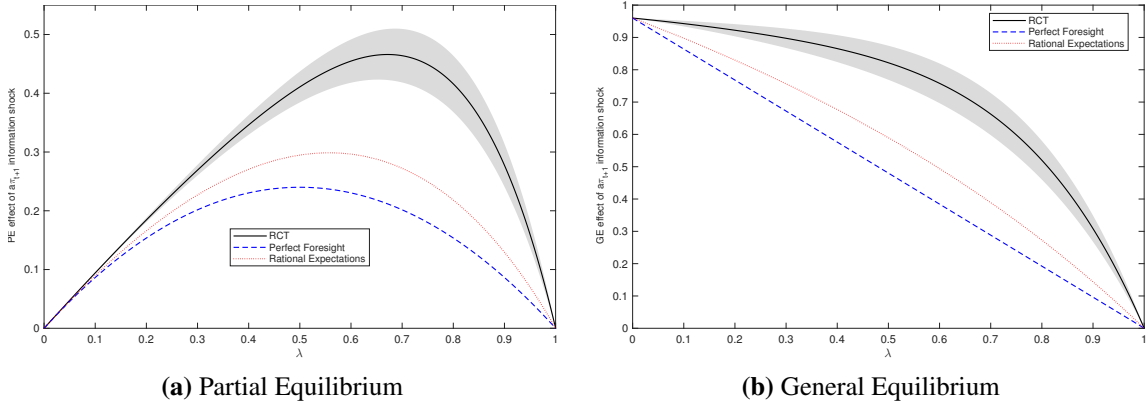
**Benchmarks.** I compare the RCT-implied causal effects of inflation expectations to two benchmarks: perfect foresight and an approximation to limited-information rational expectations. For perfect foresight, we straightforwardly set  $\tau_{ij} = 1$  for all variables  $z_j$ , which in turn implies  $\chi_i$  becomes the identity matrix (equation (10)).

<sup>16</sup>This assumption also has a convenient implication for the distinction between  $\tau_{i\pi_t}^{RCT}$  and  $\tau_{i\pi_t}$  discussed above. The general equilibrium feedback in Proposition 3 operates through  $\pi_t \rightarrow s_{it}(\pi_t) \rightarrow E_{it}(\pi_t)$ , which depends on how agents process ‘everyday’ signals about  $\pi_t$ , not experimental ones. With  $\bar{\tau}_{\pi_t} = 1$ , however, all agents observe  $\pi_t$  perfectly, and the two weighting schemes coincide.

Rational expectations is harder to specify, because we do not know the true joint dynamics of inflation and marginal costs in the Euro Area. For the model without marginal cost cross-learning, I provide a simple approximation by regressing annual Euro Area HICP inflation on a 12-month lag of itself. I then set  $\bar{\tau}_{\pi_{t+1}}$  to the value in the RCT-based calibration, but set cross-learning so that all firms extrapolate to future inflation using the degree of persistence implied by that AR(1) regression. For the case with cross-learning to future marginal costs, we cannot write down an equivalent rational-expectations benchmark without specifying a full model for the processes determining wages and costs. For that exercise I therefore compare the results to the perfect foresight benchmark, and to the RCT results without marginal cost cross-learning. Full details are given in Appendix C.2.

**The total causal effect of expectations without marginal cost cross-learning.** Figure 1 plots the causal effect of one-year-ahead inflation expectations on current inflation as price stickiness  $\lambda$  varies, for the case with no marginal cost cross-learning. Panel (a) plots the partial equilibrium effect  $\Omega^{PE}(\pi_{t+1})$ , while panel (b) plots the general equilibrium effect  $\Omega^{GE}(\pi_{t+1})$ , incorporating the endogenous response of  $\pi_t$  signals. The perfect-foresight benchmark in the partial equilibrium case is equivalent to the causal effect calculated in a standard temporary equilibrium analysis. The first key message from these figures is therefore that the extensions to this classic approach allowed for by temporary-information equilibrium are quantitatively and qualitatively important.

**Figure 1:** Partial and general equilibrium effects of a shock to  $s_{\pi_{t+1}}$ , comparing RCT estimates to benchmarks



*Note:* Figure plots the partial- and general-equilibrium causal effects of an increase in  $\bar{E}_t \pi_{t+1}$  in the Calvo model, following the formulae in Propositions 2 and 3, and a calibration derived from the estimates in Baumann et al. (2026) (detail in Appendix C.2). Gray shaded bands show 90% pointwise confidence intervals (detail in Appendix C.3).

The second message is that the macroeconomic significance of the RCT results depends critically on the degree of price stickiness. With an average price duration of 5 quarters ( $\lambda = 0.2$  since the calibration is annual), the general equilibrium effect implied by the RCT estimates is 11% larger than under rational expectations. However, if prices are sufficiently sticky that the average price

duration is two years ( $\lambda = 0.5$ ),<sup>17</sup> the RCT estimates imply general equilibrium causal effects of expectations that are 39% larger than rational expectations. The differences to the perfect foresight benchmark are universally larger, but the same strong dependence on price stickiness is present. With  $\lambda = 0.2$ , the RCT implies a 20% larger general equilibrium effect. With  $\lambda = 0.5$ , this figure jumps to 71%.

The reason for this dependence is that the RCT estimates depart most strongly from rational expectations at long horizons, where cross-learning from  $\pi_t$  and  $\pi_{t+1}$  information is found to be much stronger than the benchmark would predict.<sup>18</sup> This means that a given change in one-period-ahead expectations is associated with greater updating of long-horizon expectations, and thus a larger inflation response. This pattern of stronger departures from the rational benchmark at longer horizons is consistent with direct evidence that expectations overreact more strongly at long horizons than short ones (d'Arienzo, 2020; Bonaglia et al., 2025). The difference is even more pronounced when comparing to perfect foresight, where there is no such cross-learning at all.

At very flexible prices (small  $\lambda$ ), this does not have a large quantitative effect, because price durations are expected to be short and so long-horizon expectations are relatively unimportant for price-setting. As prices get stickier, long-horizon expectations become more important to current price-setting, and the differences between the RCT-implied model and the benchmarks grow.

Finally, note that even at small  $\lambda$ , the rational-expectations and perfect-foresight causal effects lie outside the 90% confidence interval around the RCT-implied effects. Estimation uncertainty in the RCT is therefore rather small for this macroeconomic question, relative to the uncertainty around the appropriate calibration of  $\lambda$ .

**Decomposition.** The mapping described in Section 4.1, and implemented here, shows how to combine the point estimates from an RCT with temporary-information equilibrium to obtain the total causal effect of expectations, including both representative-agent and narrative-heterogeneity effects. It cannot, however, separate the two, as in the decompositions in equation (14) and Proposition 6, without further information. One natural approach would be to directly elicit subjective models within the survey, for example using hypothetical scenarios (Andre et al., 2022) or open-ended questions (Haaland et al., 2025). The cross-sectional mean of such responses identifies  $\bar{\chi}_Y \pi_{t+1}$ , and the residual  $\gamma_{1,Y} - \bar{\chi}_Y \pi_{t+1}$  could then be used to identify the narrative heterogeneity term.

For example, if firms' average perceived persistence of inflation is close to the coefficient from an AR(1) regression on recent data, as Macaulay and Moberly (2026) document for households, then the representative-agent effect equals the rational-expectations benchmark already computed. The red line in Figure 1 then reflects the representative-agent component of the total causal effect,

<sup>17</sup>This is the calibration used in Werning (2022) footnote 8.

<sup>18</sup>For the rational expectations benchmark to deliver the same response-weighted cross-learning from  $\pi_{t+1}$  to  $\pi_{t+5}$  as the RCT, the annual persistence of inflation would need to be  $\approx 0.95$ , more than double its estimated value of 0.39.

and the gap between the red and black lines measures the narrative heterogeneity channel. Without independent measurement of  $\bar{\chi}_{\pi_t+s\pi_{t+1}}$  within the specific sample of firms in the RCT, this is necessarily a crude approximation, but it nevertheless suggests that the narrative heterogeneity channel is large and positive. That is, the firms who extrapolate most strongly from short to long-run expected inflation are also those whose short-run expectations react most strongly to signals. This is consistent with models of rational inattention, in which greater perceived persistence makes short-run inflation signals more valuable.

**Including marginal cost cross-learning.** Figure 2 repeats Figure 1, including cross-learning from  $\pi_{t+1}$  to expectations of  $mc_{t+1}$ . The estimates in Baumann et al. (2026) imply that greater expected  $\pi_{t+1}$  passes through to greater expected nominal marginal costs. However, this pass-through is less than one-for-one, so real marginal cost expectations fall. This therefore reduces the causal effect of expected inflation on realized inflation, especially at low values of  $\lambda$ , when the compensating effect of strong extrapolation to long-run inflation expectations is less powerful.

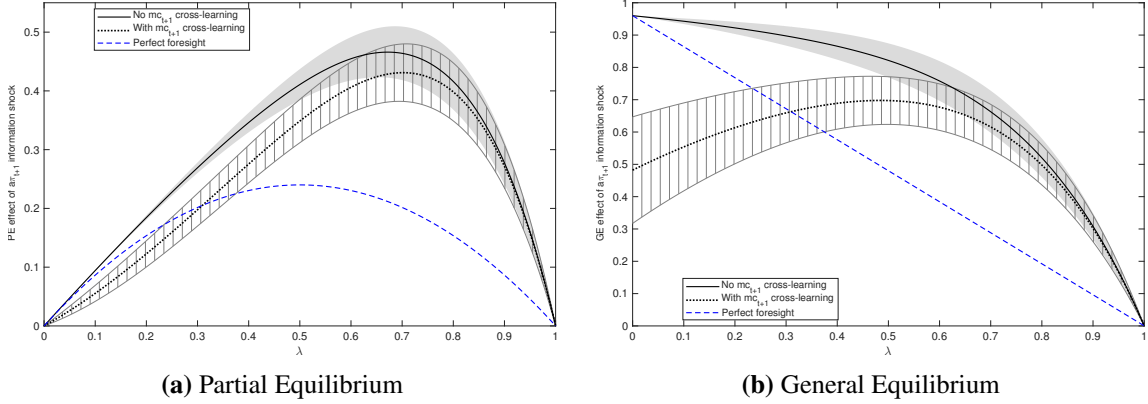
One implication of this is that, at  $\lambda \approx 0.3$ , the perfect-foresight results are a close approximation to the RCT-implied causal effects with marginal cost cross-learning. In this comparison, however, it is important to note that the perfect foresight benchmark only corresponds to standard temporary equilibrium analysis in the partial equilibrium figure. The temporary-information equilibrium approach is still required to compute the general equilibrium result, allowing for learning from the realized  $\pi_t$  change. Standard temporary equilibrium calculations therefore still miss an important component of the causal effect of expectations, even if in partial equilibrium they approximate the true causal effect in this case.

Appendix C also contains a number of other results extending this analysis. In particular, I apply the methods described in Section 2.3 to study the share of transmission of a marginal-cost news shock that is due to inflation expectations. I also compute alternative benchmarks using two popular models that depart from rational expectations: diagnostic expectations (Bordalo et al., 2020) and inattentiveness (Gabaix, 2020). In each case, adjusting the benchmark simply involves replacing the  $\chi_i$  and  $\tau_i$  parameters with those implied by the researcher’s chosen theory of expectations. The GE causal effect of expectations under diagnostic expectations, in particular, is found to be explosive for certain values of  $\lambda$ , as the over-sensitivity to news in such models creates very strong feedback effects through perceived inflation.

**Guidance for future experiments.** The methods employed here extend naturally to other contexts, to analyze the implications of other information-provision experiments. Indeed, if researchers have a particular expectation whose causal effects they would like to analyze, finding the temporary-information equilibrium of the appropriate model can provide guidance for the design of the relevant



**Figure 2:** Partial and general equilibrium effects of a shock to  $s_{\pi_{t+1}}$ , comparing RCT estimates with and without cross-learning to  $mc_{t+1}$ .



*Note:* Figure plots the partial- and general-equilibrium causal effects of an increase in  $\bar{E}_t \pi_{t+1}$ , following the formulae in Propositions 2 and 3, and a calibration derived from the estimates in Baumann et al. (2026) (detail in Appendix C.2), with and without cross-learning from  $\pi_{t+1}$  to  $mc_{t+1}$ . Gray shaded bands show 90% pointwise confidence intervals without cross-learning to  $mc_{t+1}$ , hatched regions show them with cross-learning to  $mc_{t+1}$  (detail in Appendix C.3).

experiment, including which treatments to run, and which posterior expectations to elicit.

For the causal effect of inflation expectations in the Calvo model, Baumann et al. (2026) come close to the ideal setup, hence its use here: their treatments are about future inflation (the expectation of interest) and current inflation (the source of general-equilibrium learning). They also elicit posteriors on both inflation and the components of marginal cost (the two variables over which expectations are formed).

There are just two other details suggested by temporary-information equilibrium in this context. First, if the experiment also elicited posterior nowcasts of current inflation, I could replace the assumption of perfect  $\pi_t$  observation with a measured  $\bar{\tau}_{\pi_t}$ , improving the calculation of general-equilibrium learning effects. Second, if the experiment included an open-ended question or a hypothetical to independently measure average cross-learning, I could decompose the overall effect into representative-agent and narrative-heterogeneity components (see above).

## 5 Survey evidence on narrative heterogeneity

In Section 4, I measured the total causal effects of firm inflation expectations on realized inflation using the temporary-information equilibrium of the Calvo model. This total effect combined the representative-agent and narrative-heterogeneity channels identified in Sections 2 and 3. In this section I measure the narrative heterogeneity channel directly, in the context of household beliefs around inflation. I document two empirical results that reveal the existence of a time-varying narrative heterogeneity channel in this context. These results inform the model in Section 6, from which I observe the aggregate consequences of this instance of the narrative heterogeneity channel.

## 5.1 Data

Expectations combine both information and subjective models (Sections 2 and 3), so commonly-used survey data on realized expectations cannot be used to identify the narrative heterogeneity channel. However, the Bank of England Inflation Attitudes Survey (IAS) contains several unique questions that overcome this limitation.

The IAS is a quarterly survey of a repeated cross-section of UK households, run since 2001 (annual until 2003). After weighting, the sample is representative of the UK adult population. I use the individual-level response data from 2001-2019, as the implementation of the survey changed substantially in 2020 (Bank of England, 2020).

The first survey question I use asks households about their subjective model of the relationship between inflation and the ‘strength of the economy’.

**Question 1** *If prices started to rise faster than they are now, do you think Britain’s economy would end up stronger, or weaker, or would it make little difference?*

This differs from standard questions on expected future economic outcomes because it does not invoke the use of information about the state of the world. The answers are informative about cross-learning only ( $\chi_{ijk}$  in the models above).<sup>19</sup> In the analysis below, I will refer to a respondent answering that inflation would make the economy stronger/little difference/weaker as having a positive/neutral/negative subjective model of inflation respectively.

There are two possible interpretations of this question. Households may view it as asking about the causal effects of inflation on the economy (as in the model of Spiegler, 2021). Alternatively, they could see it as asking about the most likely source of a rise in inflation, if they believe supply- and demand-driven inflation is associated with different real outcomes (as in e.g. Kamdar and Ray, 2024). For the purposes of this section, this distinction does not matter.  $\chi_{ijk}$  is defined above as the degree to which households update their expectations of variable  $j$  when their expectation of variable  $k$  changes, irrespective of why that link exists. In this case, it is the updating of expectations about the strength of the economy when expected inflation rises. The sign of this updating is captured by the question, whether it occurs because of a perceived direct causal link from inflation to the real economy, or a belief about the type of shocks hitting the economy.

The next set of questions concern the information households use to form their inflation expectations, without asking what those expectations are. This allows us to learn about household information precision (encoded in  $\tau_{ik}$ ) without contamination from cross-learning ( $\chi_{ijk}$ ).

**Question 2a** *What were the most important factors in getting to your expectation for how prices in the shops would change over the next 12 months?*

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<sup>19</sup>This is similar to how Andre et al. (2022) interpret the answers to the hypothetical vignettes in their surveys.

*Please select up to 4 [from a list reproduced in Appendix D.1].*

Question 2a was only asked in 2016Q1, but very similar questions were asked at other times. In each, the respondent is asked about the information sources they used to arrive at their expected inflation, or that led them to change that expectation over the previous year. Some of the options that respondents can choose refer to information specifically on inflation, while others refer to information on other variables, which can only be used to forecast inflation if the household engages in some cross-learning. For each such question I construct a dummy variable equal to 1 if the respondent reports using direct information about inflation, and equal to 0 if they do not. Full details of these questions, and the options representing direct information, are in Appendix D.1. Combining these dummy variables gives an indicator for if the respondent used direct information on inflation in forming their expectations, that is whether  $\tau_{i\pi_t} > 0$ . This indicator is observed for 8 separate quarters between 2009Q1-2019Q1. I confirm below that the key results of this section do not vary substantially with the changes in question wording over these periods.

In Appendix D.2 I confirm that these measures of information and subjective models correlate with planned household consumption, in ways that are consistent with the measures picking up the desired elements of household beliefs. A further possible test of the information indicator would ask if households who obtain direct information about inflation make more accurate forecasts. However, if beliefs about the level of inflation affect subjective models, that may in turn change the incentives to acquire further information, complicating the predicted correlation between information and forecast accuracy.<sup>20</sup> Indeed, recent evidence from Link et al. (2025) suggests that greater household information acquisition is often associated with larger forecast errors about inflation.

## 5.2 Information and subjective models in the cross-section

With this data, I first explore the cross-sectional relationship between the distributions of information and subjective models, which is the source of any narrative heterogeneity channel. Table 1 shows the estimated average marginal effects from a probit regression of the information indicator defined in Section 5.1 on the respondent's subjective model of inflation, represented by their answer to Question 1. This gives us our first empirical result.

**Empirical Result 1** *Households who believe inflation makes no difference to the economy acquire less information about inflation on average than households who believe inflation does affect the economy (in either direction).*

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<sup>20</sup>In a previous version of this paper (Macaulay, 2022) I explicitly model this process and show that the data line up qualitatively with the model.

**Table 1:** Information correlates with subjective models

|                           | (1)                    | (2)                    |
|---------------------------|------------------------|------------------------|
| end up stronger           | -0.0102<br>(0.0191)    | -0.00827<br>(0.0192)   |
| make little<br>difference | -0.0356***<br>(0.0128) | -0.0315**<br>(0.0129)  |
| dont know                 | -0.0627***<br>(0.0172) | -0.0605***<br>(0.0172) |
| Controls                  | None                   | All                    |
| Time FE                   | Yes                    | Yes                    |
| Observations              | 8270                   | 8270                   |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ 

*Note:* The table reports average marginal effects from estimating a probit regression of the information indicator on the responses to Question 1. The information indicator equals 1 if the household reports using a direct source of information about inflation when forming their expectations, as defined in Appendix D.1. The omitted category is the belief that inflation makes the economy weaker. All regressions are weighted using the survey weights provided in the IAS. All regressions include period fixed-effects. The controls in column 2 are gender, age, class, employment status, income, education, region, and home-ownership status. Age, class, income and education are all reported in bands, and included as categorical variables.

The probability of using direct inflation information is 3-3.5 percentage points lower for those with a neutral model of the effects of inflation than for those who believe inflation weakens the economy. Over the whole population, 23% of respondents use direct inflation information, so this difference is non-trivial. More important than the magnitude, however, is that this shows a systematic cross-sectional relationship between information and subjective models, indicating a role for the narrative heterogeneity channel. Quantifying the effect requires a model, as developed in Section 6 below.

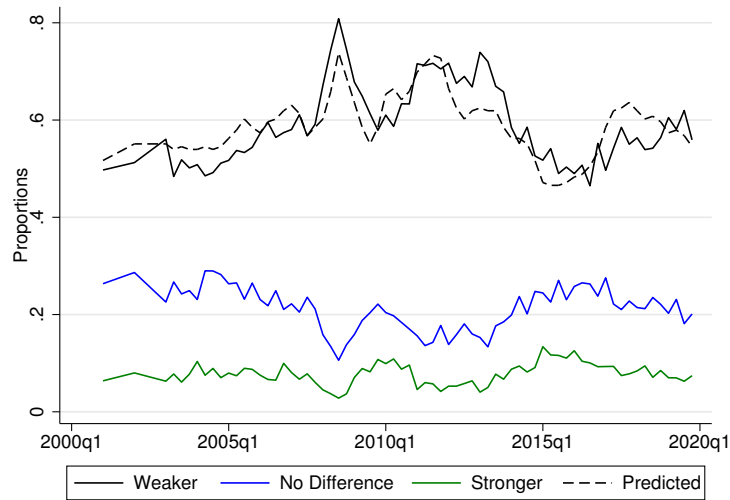
The information indicator is composed of answers to several slightly different questions, with small wording and definition changes across survey waves. In Appendix D.3 I repeat Table 1 for subsets of the questions. As some respondents do not answer the unique survey questions used here, I also account for the concern that there may be selection bias in whose answers are observed, using a selection model as in Heckman (1979). In both cases the results are robust.

Result 1 is not consistent with models with exogenous information, as there would be no reason for information to be systematically correlated with household subjective models. It is, however, consistent with models of endogenous information acquisition, as explored in Section 2.4. Intuitively, the value of inflation information is lower for households who believe inflation makes little difference to other variables that matter for their decisions.

### 5.3 Subjective models over time

Figure 3 shows the proportions answering Question 1 with each subjective model of inflation over time ('don't know' omitted for figure clarity).

**Figure 3:** Proportions giving each answer to Question 1: “If prices started to rise faster than they are now, do you think Britain’s economy would end up stronger, or weaker, or would it make little difference?”



*Note:* Proportions shown are calculated using the survey weights provided in the IAS. Proportion answering ‘Don’t know’ is omitted for figure clarity. The dashed line shows the predicted values from regressing the proportion reporting that inflation makes the economy weaker on annual CPI inflation:  $\Pr(\hat{\text{weaker}})_t = 0.057 \times \text{CPI inflation}_t + 0.466$ . The coefficient on inflation is significant at the 1% level.

In all quarters, most households answer that inflation would make the economy weaker, consistent with the findings in [Shiller \(1997\)](#), [Andre et al. \(2022\)](#), and [Kamdar and Ray \(2024\)](#). The second-largest group report that inflation makes no difference. Combined with Empirical Result 1, this suggests that the covariance of inflation information and cross-learning from inflation to the strength of the economy is *negative*. If households consume more when they believe the economy is strong, the narrative heterogeneity channel will therefore reduce the consumption response to inflationary shocks in partial equilibrium.

The relatively long time series of the IAS also allows us to see that the distribution of answers varies substantially over time, which gives us our second empirical result.

**Empirical Result 2** *A greater proportion of households believe inflation weakens the economy when realized inflation is high.*

The correlation between annual CPI inflation and the proportion of respondents with negative models of inflation is extremely high, at 0.8.<sup>21</sup> The dashed line in Figure 3 plots the predicted values

<sup>21</sup>Although I do not use later data due to the structural break induced by the methodology change in 2020, this pattern is robust through the 2021-22 inflation surge. Using data from 2020Q1-2024Q3, the equivalent correlation is 0.65.

from regressing this proportion on CPI inflation, showing that this correlation is strong across the whole sample. Tests in Appendix D.4 show that the correlation is robust to the addition of various macroeconomic controls, which themselves explain far less of the variation in the distribution of responses than realized inflation. The correlations are also robust to using inflation measures split by various household characteristics.

A key advantage of the temporary-information equilibrium approach is that we do not need to take a stand on the source of this time-series variation in order to assess how it affects macroeconomic dynamics. In Section 6 below, I take the evolving distribution of subjective models from this data as given, and use that as an input into the quantification of the causal effects of inflation expectations. In Appendix D.5, however, I show that perceptions and expectations of current and future inflation are systematically higher for those with negative subjective models of inflation, consistent with a mechanism in which information about high inflation causes households to update their subjective models towards more negative views. This parsimoniously accounts for the cross-sectional and time-series evidence: within a period, those who receive signals that inflation is high shift to more negative subjective models, and when realized inflation rises more households receive such signals.

## 6 The causal effects of household inflation expectations

Having documented direct evidence of a time-varying narrative heterogeneity channel in household beliefs around inflation, I now interpret that evidence using a New Keynesian model.

### 6.1 Households

A representative household chooses consumption  $C_t$ , labor supply  $N_t$ , and real one-period bond holdings  $B_t$  to maximize

$$\bar{E}_t U_t = \bar{E}_t \sum_{s=0}^{\infty} \beta^s \left( \frac{C_{t+s}^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}} - \varphi \frac{N_{t+s}^{1+\psi}}{1+\psi} + \xi \frac{B_{t+s}^{1-\eta}}{1-\eta} \right) \quad (35)$$

subject to:

$$C_t + B_t = \frac{R_{t-1}}{\Pi_t} B_{t-1} + W_t N_t + D_t - T_t \quad (36)$$

where  $R_t$  is the gross nominal interest rate,  $\Pi_t$  is the per-period rate of inflation,  $W_t$  is the real wage,  $D_t$  is profits transferred from firms, and  $T_t$  are lump-sum taxes.

There are two differences to the household problem in standard New Keynesian models. First, I include bonds in the utility function, as in Hagedorn (2018) and Michaillat and Saez (2019). This allows the model to generate plausible marginal propensities to consume out of current and

expected income (Kaplan and Violante, 2018; Auclert et al., 2024), which is important here as the key channel operates through the effect of inflation on income expectations and thus demand.

The second departure from standard models is the expectations operator  $\bar{E}_t$ . Rather than having the household form expectations directly, the model features a continuum of ‘agents’ within the household. Each agent  $i$  forms their own idiosyncratic expectations of all variables, using potentially heterogeneous signals and cross-learning, and the household aggregates these before deciding  $C_t, N_t$ . This simplification allows the model to isolate the role of heterogeneous expectations through household actions, without generating further effects from heterogeneous expectations causing heterogeneities in wealth.<sup>22</sup>

In the notation of Section 3, the agent actions  $x_{it}$  are to choose a value for the expectation of each variable in the system. The household consumption choice, after log-linearization, then depends on the aggregate of these  $x_{it}$  through the following equation.

**Proposition 9** *Normalize steady state output and bond holdings to  $Y = B = 1$ . The log-linear consumption function is*

$$c_t = \mu b_{t-1} + \Lambda \sum_{s=0}^{\infty} \Lambda^s (\zeta \bar{E}_t y_{t+s} - \sigma (\bar{E}_t r_{t+s} - \bar{E}_t \pi_{t+s+1})) \quad (37)$$

where lower-case letters denote log-deviations of corresponding capitalized variables from their steady states,  $y_t$  is output,  $R$  is the steady state nominal interest rate, and  $\mu, \zeta, \Lambda$  are coefficients defined in Appendix E.1.

**Proof.** Appendix E.1. ■

This fits into the form of equation (25). Equation (37) is similar to the standard form in models that do not impose rational expectations (e.g. Eusepi and Preston, 2018), though notably the responses to future income and real interest rates differ from the standard form due to the bonds in the utility function. Labor supply is then determined by a standard intratemporal labor supply condition, presented (along with all other model equations) in Appendix E.2.

## 6.2 Firms, policy, and market clearing

As the focus of this model is the behavior of households, I keep the rest of the model simple. Full details and derivations are in Appendix E.2.

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<sup>22</sup>This is analogous to how McKay and Wolf (2023) and Debortoli and Galí (2025) in the HANK literature use labor-aggregating unions to abstract from effects of wealth on labor supply, to focus on core demand effects. See Broer et al. (2026) and Macaulay and Moberly (2026) for evidence that the correlation between wealth and inflation expectations is small, especially outside the hand-to-mouth region.



Monopolistically competitive intermediate goods producers set prices subject to quadratic adjustment costs, and produce using labor as the only input to a linear production function. They supply a perfectly competitive final goods producer, who combines intermediate goods varieties using a CES aggregator. Log-linearizing the solution to the intermediate goods firm pricing problem about the zero-inflation steady state yields the Phillips curve. As in [Bilbiie \(2025\)](#),<sup>23</sup> I make the simplification that the Phillips curve is static, to focus on the mechanisms of interest operating through demand:

$$\pi_t = \kappa y_t \quad (38)$$

Monetary policy is set according to a standard Taylor rule:

$$r_t = \phi_\pi \pi_t + \phi_y y_t + \nu_{rt} \quad (39)$$

where  $\nu_{rt}$  is an AR(1) monetary policy shock with Gaussian innovations.

The fiscal authority issues a constant supply of real bonds  $B = 1$ , and levies lump-sum taxes to pay the interest on those bonds. Goods market clearing implies

$$y_t = c_t \quad (40)$$

### 6.3 Agent expectations

In principle, the temporary-information equilibrium approach can be applied for a broad range of assumptions on agent information and cross-learning. However, in this case I choose a particularly simple structure, that allows me to isolate the role of the correlations documented in [Section 5](#) without introducing a large number of free parameters in expectation formation, for which we have no reliable disciplinary information. Documenting other aspects of household information and cross-learning that are abstracted from here would be a fruitful avenue for future research.

**Information.** Agents observe realized  $y_t$  and  $r_t$ . However, they only observe a noisy signal about  $\pi_t$ , as in e.g. [Coibion and Gorodnichenko \(2015\)](#). That signal takes the form used in [\(15\)](#) and [\(28\)](#), which in this context is

$$s_{it}(\pi_t) = \pi_t + \varepsilon_{\pi t} + \tilde{\varepsilon}_{i\pi t} \quad (41)$$

where  $\varepsilon_{\pi t}$  and  $\tilde{\varepsilon}_{i\pi t}$  are both mean zero Gaussian variables. As is common in this literature, I abstract from any direct signals about the future when computing the causal effect of current inflation perceptions  $\bar{E}_t \pi_t$ .<sup>24</sup> However, for the exercises involving the causal effect of inflation *expectations*

<sup>23</sup>As shown in [Bilbiie \(2025\)](#), this can be microfounded by assuming firms pay a Rotemberg adjustment cost relative to last-period's aggregate price level, rather than their own past prices.

<sup>24</sup>See [Goldstein and Gorodnichenko \(2022\)](#) for an exception.

$\bar{E}_t\pi_{t+1}$  we will need there to be a signal about  $\pi_{t+1}$ . In those cases I assume a one-time signal  $s_{it}(\pi_{t+1})$ , with the same functional form as  $s_{it}(\pi_t)$ .

**Subjective models.** Households know the true monetary policy rule (39). They also know the functional form of the relationship between output and inflation (38), but may hold incorrect and heterogeneous beliefs about the coefficient. Thus their expectations of future output and inflation are related by

$$E_{it}y_{t+s} = \chi_{iy\pi} E_{it}\pi_{t+s} \quad (42)$$

where  $\chi_{iy\pi}$  is the cross-learning from inflation to output, which I discipline below with the data in Section 5. Finally, agents believe that inflation follows a simple AR(1) process, so that:

$$E_{it}\pi_{t+s} = \rho E_{it}\pi_{t+s-1} \quad (43)$$

where the perceived persistence  $\rho$  is common across agents. This is a simplification of the true inflation dynamics in equilibrium, made here to minimize the number of parameters in the subjective models which would otherwise need to be calibrated. It is also consistent with evidence that households typically use simple perceived laws of motion to forecast inflation (e.g. Adam, 2007).

## 6.4 Temporary-information equilibrium

This model fits into the framework of Section 3. As explained above, agent-specific expectations are the choice variables  $x_{it}$  here. The state variables  $c_t, y_t, \pi_t, r_t$  depend on aggregate expectation choices through equations (37), (38), (39) and (40), which map into equation (25), with shocks  $\nu_{rt}$ .<sup>25</sup> A formal statement of the temporary-information equilibrium of the model, along with algebraic expressions for the causal effects of  $\bar{E}_t\pi_t$  and  $\bar{E}_t\pi_{t+1}$ , is presented in Appendix E.3.

## 6.5 Calibration

I set most of the parameters to standard values in the New Keynesian literature for a quarterly calibration. The curvature parameters in the utility function are set to  $\sigma = \psi = 1$ , and (following the suggestion of Kaplan and Violante, 2018)  $\eta = 2.5$ . Price stickiness is set to target an average price duration of 5 quarters. The Taylor rule parameters are set to  $(\phi_\pi, \phi_y) = (1.5, 0.125)$ .

Next, I choose the discount factor  $\beta$  and the weight on bonds in utility  $\xi$  to jointly hit two targets: a quarterly steady state interest rate of 1%,<sup>26</sup> and an impact marginal propensity to consume of 0.35, which is the average estimate across the empirical studies surveyed by Sokolova (2023).

<sup>25</sup>Since bond supply is constant, in equilibrium  $b_t = 0 \forall t$ , so we do not need to consider  $b_t$  as a state variable.

<sup>26</sup>Note that with bonds in the utility function  $\beta$  alone does not pin down the steady state  $R$ .

To calibrate  $\rho$ , I estimate an AR(1) regression using quarterly UK CPI inflation from Q2 2001-Q4 2019, as in the survey data sample. Since at the time of writing the UK does not publish seasonally-adjusted CPI data (Dixon and Michail, 2025), I include quarter-of-the-year fixed effects in the regression. This yields a persistence parameter of  $\rho = 0.44$ . More sophisticated seasonal adjustments give very similar results (see Appendix E.4).

I calibrate the joint distribution of  $\chi_{iy\pi}, \tau_{i\pi_t}$  to the data in Section 5. Since that data partitions information and subjective models into discrete groups, I impose that agents are divided into three such groups:

$$(\chi_{iy\pi}, \tau_{i\pi_t}) = \begin{cases} (\chi_+, \tau_H) & \text{with proportion } p_{+,t} \\ (0, \tau_L) & \text{with proportion } 1 - p_{+,t} - p_{-,t} \\ (\chi_-, \tau_H) & \text{with proportion } p_{-,t} \end{cases} \quad (44)$$

with  $\chi_+ > 0 > \chi_-$ , and  $\tau_H > \tau_L$ . This distribution reflects Empirical Result 1, that those with positive and negative models of inflation’s effects on “the strength of the economy” (interpreted here as output) have similar information precision, while those who believe inflation has no effect on output use less precise information. In Section 6.6, I set the proportions in each group to match the average proportions giving each subjective model answer over the sample. In Section 6.7, I calibrate them period-by-period to the proportions observed in each wave of the survey. In each case, I pool households responding to Question 1 with “make no difference” and “don’t know” into the  $(\chi_{iy\pi}, \tau_{i\pi_t}) = (0, \tau_L)$  group.

The survey data is not informative about the magnitude of households’ perceived relationship between inflation and output. I therefore proceed by assuming that households with  $\chi_+$  know the true slope of the Phillips curve, implying  $\chi_{iy\pi} = \kappa^{-1}$ .<sup>27</sup> I then choose  $\chi_-$  such that the squared passthrough from  $E_{it}\pi_t$  to  $c_t$  is equal across households with positive and negative models. This is motivated by models of rational inattention: with standard quadratic objective functions, optimal information precision is proportional to the square of the elasticity of the action to the state (see the example in Appendix B). Thus if the households with  $\chi_+$  and  $\chi_-$  make the same information choices, they should have the same squared passthrough.

Those same rational inattention models typically imply that optimal information precision is 0 for agents who do not believe the state is sufficiently valuable to learn about. I assume that  $\chi_{iy\pi} = 0$  falls in this range, so  $\tau_L = 0$ . I then calibrate  $\tau_H$  to target an estimate of the average Kalman gain across households obtained from the survey data, in a procedure described in Appendix E.4. Finally, when computing the causal effects of  $\bar{E}_t\pi_{t+1}$  I assume that households are equally well-informed about current and one-period-ahead inflation, so  $\tau_{i\pi_{t+1}} = \tau_{i\pi_t}$ . This is consistent with the finding

<sup>27</sup> Since this is a strong structural assumption from which the extent of cross-learning heterogeneity follows, I explore the robustness of the quantitative results to alternative calibrations in Appendix E.5. The narrative heterogeneity channel remains substantial for a wide range of  $\chi_+$  values.

that Empirical Result 1 holds, with quantitatively similar magnitudes, for questions about inflation at various horizons (see Appendix D.3).

In both exercises below I also compute the causal effects of interest in a quasi-full-information rational-expectations (FIRE) benchmark, in which  $\tau_{i\pi_t}$  and  $\tau_{i\pi_{t+1}} \rightarrow 1$  and all households know the true slope of the Phillips curve, implying  $\chi_{iy\pi} = \kappa^{-1}$ . Note this is not a true FIRE calculation, as I maintain the AR(1) perceived law of motion for inflation, with  $\rho$  as calibrated above.

## 6.6 The causal effects of inflation expectations in steady state

The first exercise I conduct with the calibrated temporary-information equilibrium model is to find the causal effects of an inflation perception or expectation shock when the economy is in steady state. I calibrate the steady states of  $p_+$  and  $p_-$  to the average proportions giving positive and negative answers to Question 1 in the survey data. Using these probabilities, and applying Propositions 6 and 7, I find the causal effects of inflation expectations on output. The results are displayed in Table 2.

**Table 2:** The causal effect of aggregate inflation perceptions  $\bar{E}_t\pi_t$  and expectations  $\bar{E}_t\pi_{t+1}$  on output  $y_t$ .

|                                                           | Total  | RA channel | NH channel | FIRE  |
|-----------------------------------------------------------|--------|------------|------------|-------|
| <i>Panel A: Effect of <math>\bar{E}_t\pi_t</math></i>     |        |            |            |       |
| Partial equilibrium                                       | -0.167 | -0.092     | -0.076     | 1.150 |
| General equilibrium                                       | -0.128 | -0.070     | -0.058     | 0.882 |
| <i>Panel B: Effect of <math>\bar{E}_t\pi_{t+1}</math></i> |        |            |            |       |
| Partial equilibrium                                       | -0.379 | -0.208     | -0.172     | 2.607 |
| General equilibrium                                       | -0.291 | -0.159     | -0.132     | 2.000 |

*Note:* The table reports the causal effect of aggregate inflation perceptions and expectations on output, in partial and general equilibrium. The total effect is decomposed into the representative-agent (RA) channel and the narrative heterogeneity (NH) channel, as described in Appendix E.3. The FIRE column reports a benchmark in which  $\tau_{i\pi_t}, \tau_{i\pi_{t+1}} \rightarrow 1$  and  $\chi_{iy\pi} = \kappa^{-1} \forall i$ . The calibration is described in Section 6.5 and Appendix E.4.

Greater inflation perceptions and expectations both reduce aggregate output, contrary to the intuition popularly put forward for models with rational expectations (see e.g. Eggertsson and Woodford, 2003; Adam and Billi, 2007), and the findings in the FIRE benchmark. This is because the majority of households believe inflation weakens the economy in the survey, in line with e.g. Hajdini et al. (2023). Since all variables here are in log-deviations, the magnitudes reflect percentages: a 1% increase in  $\bar{E}_t\pi_{t+1}$  causes a 0.3% fall in output in general equilibrium. This is noticeably smaller in magnitude than the predictions under FIRE, because greater inflation expectations now feature two competing effects: lower expected future income through the subjective model is partially offset by the standard positive demand effect of lower expected real interest rates.

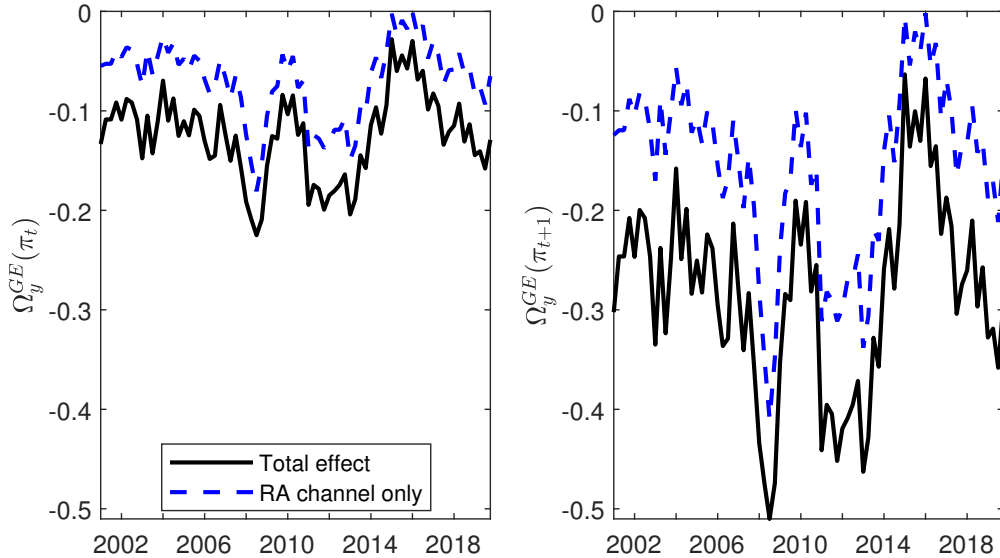
As observed in the IAS data, there is a negative correlation between information  $\tau_i$  and  $\chi_i$ , so the narrative heterogeneity channel is also negative for both shocks I consider. Quantitatively, this amplifies the steady state GE effects substantially, accounting for 45% of the total effect of

$\bar{E}_t\pi_{t+1}$ .<sup>28</sup> For both perceptions and expectations, general equilibrium feedback is also critical to the magnitudes: since an increase in each belief is contractionary, it causes the monetary policymaker to cut nominal interest rates, which weakens the overall effect.

## 6.7 The causal effects of inflation expectations over time

I now feed into the model the time series of  $p_{+,t}$ ,  $p_{-,t}$  observed in the IAS. As shown in Figure 3,  $p_{-,t}$  rises and  $p_{+,t}$  falls when realized inflation is high. Figure 4 shows the resulting causal effects of  $\bar{E}_t\pi_t$  and  $\bar{E}_t\pi_{t+1}$  on  $y_t$  in general equilibrium. The solid black line is the total effect. The dashed blue line is the representative-agent channel only.

**Figure 4:** The causal effect of inflation perceptions  $\bar{E}_t\pi_t$  and expectations  $\bar{E}_t\pi_{t+1}$  on output  $y_t$  over time.



*Note:* Figure plots the general-equilibrium causal effect of aggregate inflation perceptions (left panel) and expectations (right panel) on output over time. In each panel, the solid black line is the total causal effect and the dashed blue line gives the representative-agent channel only. The difference between the two lines reflects the narrative heterogeneity channel. The proportions  $p_{+,t}$  and  $p_{-,t}$  are calibrated period-by-period to the proportions observed in each wave of the IAS. All other parameters are calibrated as described in Section 6.5 and Appendix E.4.

The causal effects of inflation perceptions and expectations vary substantially over time. When annualized inflation peaked at 4.8% in 2008 Q3, the causal effect of  $\bar{E}_t\pi_t$  on  $y_t$  was -0.22. By 2015 Q1, when annualized inflation was 0.1%, the effect was -0.03. For  $\bar{E}_t\pi_{t+1}$ , these numbers are -0.51 and -0.06. Both causal effects decreased by almost an order of magnitude over this horizon.

However, for both variables the volatility of the overall causal effect is somewhat greater than that of the representative-agent channel. This is because when inflation increases, many of the households who previously engaged in positive cross-learning from inflation to output ( $\chi_{iy\pi} = \chi_+ > 0$ ) shift to the neutral subjective model ( $\chi_{iy\pi} = 0$ ). This strengthens the negative correlation between  $\tau_{i\pi_t}$  and

<sup>28</sup>Under the baseline calibration this ratio is identical for perceptions and expectations. Appendix E.5 shows that relaxing  $\tau_L = 0$  breaks this equivalence, but the differences remain quantitatively negligible.

$\chi_{iy\pi}$ , and thus strengthens the narrative heterogeneity channel. When realized inflation falls, the distribution of subjective models becomes more dispersed, and the ‘positive model’ group reduces the magnitude of this systematic covariance. The narrative heterogeneity channel increases the variance of  $\Omega_y^{GE}(\pi_{t+1})$  by 21%.

The narrative heterogeneity channel is therefore an important driver of the causal effect of inflation expectations. In fact, the magnitudes reported here are likely to be an underestimate of its effects, because in imposing the discrete-group structure in equation (44) I have removed much of the potential heterogeneity in  $\tau_{i\pi_t}$  and  $\chi_{iy\pi}$  across households, and thus limited the scope for narrative heterogeneity effects. In particular, the cross-sectional evidence in Appendix D.5 indicates that subjective models may respond to inflation perceptions and expectations, which are themselves highly dispersed across households. Incorporating this feedback would introduce additional covariation between information and cross-learning beyond what is captured here.

## 7 Conclusion

This paper studies the causal effects of heterogeneously-formed expectations in general equilibrium. To do this I develop a new approach, which I label the *temporary-information equilibrium*. This builds on the core insight from the recent literature on information-provision survey experiments in economics: namely, that one can study the causal effect of expectations by creating exogenous variation in a part of agent information sets. Taking this idea from the empirical literature and applying it to theoretical macroeconomic modeling retains the precise causal interpretation of temporary equilibrium analysis, while allowing for cross-learning, general-equilibrium feedback into expectations, and heterogeneity in the reception and response to information.

I show how to apply this approach to a general class of linear macroeconomic models, and to the specific cases of a Calvo pricing model, and a New Keynesian model. In doing so, I find that the causal effects of expectations, and by implication the role of expectations in the transmission of fundamental shocks, depend in general on a novel *narrative heterogeneity channel*. This operates whenever information and subjective models covary systematically across agents. Heterogeneous subjective models imply heterogeneous responses to information, so systematic patterns in the distribution of information across agents with different subjective models distort the aggregate response to shocks. Moreover, many common assumptions about expectation formation popular in current literature generate transmission channels of this kind whenever there is non-zero heterogeneity in both information and subjective models.

In the Calvo model, the presence of this channel can have large quantitative and even qualitative effects on the conclusions from temporary-equilibrium analysis that necessarily assumes no expectations heterogeneity. For the New Keynesian model, I use unique features of a UK

household survey to show that household subjective models and information about inflation covary systematically with each other, implying that the narrative heterogeneity channel will indeed be present in the causal effects of household inflation expectations. When matching the model to the empirical patterns, the narrative heterogeneity channel substantially amplifies the negative effects of inflation expectations on output, and the time-variation in that effect.

Finally, an effective use of temporary equilibrium has been to combine “theory with measured expectations” (Piazzesi and Schneider, 2016; Ludwig et al., 2024). The drawback of this approach is that the measured expectations are informative about the level of expectations, but not how they change in response to shocks. Since temporary-information equilibrium is conceptually related to information-provision experiments, it can be used to combine theory with measured expectation changes. I provide one example of this, combining the temporary-information equilibrium of the Calvo model with the RCT results in Baumann et al. (2026). Open-ended survey data on narratives and subjective models (reviewed in Haaland et al., 2025) offers a similar opportunity, as the responses provide a direct source of data on the cross-learning parameters that are central to temporary-information equilibrium and the narrative heterogeneity channel. More broadly, this method opens up the possibility of combining state-of-the-art expectations data with macroeconomic theory to answer a wide range of macroeconomic questions.

## References

- Adam, K. (2007). Experimental Evidence on the Persistence of Output and Inflation. *The Economic Journal*, 117(520):603–636.
- Adam, K. and Billi, R. M. (2007). Discretionary monetary policy and the zero lower bound on nominal interest rates. *Journal of Monetary Economics*, 54(3):728–752.
- Alaoui, L. and Penta, A. (2016). Endogenous Depth of Reasoning. *The Review of Economic Studies*, 83(4):1297–1333.
- Alaoui, L. and Penta, A. (2022). Cost-Benefit Analysis in Reasoning. *Journal of Political Economy*, 130(4):881–925.
- Andrade, P., Crump, R. K., Eusepi, S., and Moench, E. (2016). Fundamental disagreement. *Journal of Monetary Economics*, 83:106–128.
- Andrade, P., Gaballo, G., Mengus, E., and Mojon, B. (2019). Forward guidance and heterogeneous beliefs. *American Economic Journal: Macroeconomics*, 11(3):1–29.
- Andre, P., Haaland, I., Roth, C., Wiederholt, M., and Wohlfart, J. (2025). Narratives about the Macroeconomy. *Review of Economic Studies*, forthcoming.



- Andre, P., Pizzinelli, C., Roth, C., and Wohlfart, J. (2022). Subjective Models of the Macroeconomy: Evidence from Experts and a Representative Sample. *The Review of Economic Studies*.
- Angeletos, G.-M. and Lian, C. (2018). Forward Guidance without Common Knowledge. *American Economic Review*, 108(9):2477–2512.
- Armantier, O., Sbordone, A., Topa, G., van der Klaauw, W., and Williams, J. C. (2022). A new approach to assess inflation expectations anchoring using strategic surveys. *Journal of Monetary Economics*, 129:S82–S101.
- Ascari, G., Fasani, S., Grazzini, J., and Rossi, L. (2023). Endogenous uncertainty and the macroeconomic impact of shocks to inflation expectations. *Journal of Monetary Economics*, 140:S48–S63.
- Ascari, G. and Rossi, L. (2012). Trend Inflation and Firms Price-Setting: Rotemberg Versus Calvo\*. *The Economic Journal*, 122(563):1115–1141.
- Auclert, A. (2019). Monetary Policy and the Redistribution Channel. *American Economic Review*, 109(6):2333–67.
- Auclert, A., Rognlie, M., and Straub, L. (2024). The Intertemporal Keynesian Cross. *Journal of Political Economy*, 132(12):4068–4121.
- Bachmann, R., Topa, G., and Van Der Klaauw, W. (2023). *Handbook of Economic Expectations*. Elsevier.
- Baley, I. and Veldkamp, L. (2022). Bayesian Learning. In Bachmann, R., Topa, G., and van der Klaauw, W., editors, *Handbook of Economic Expectations*.
- Balla-Elliott, D. (2025). Identifying Causal Effects in Information Provision Experiments.
- Bank of England (2020). Inflation Attitudes Survey Methodology and notes - online survey.
- Baumann, U., Ferrando, A., Georgarakos, D., Gorodnichenko, Y., and Reinelt, T. (2026). Firms’ Inflation Expectations in a Monetary Union. *Journal of the European Economic Association*.
- Beutel, J. and Weber, M. (2025). Beliefs and Portfolios: Causal Evidence. *Review of Financial Studies*.
- Bhandari, A., Borovička, J., and Ho, P. (2024). Survey Data and Subjective Beliefs in Business Cycle Models. *The Review of Economic Studies*.
- Bianchi, F., Ilut, C., and Saijo, H. (2024). Diagnostic Business Cycles. *Review of Economic Studies*, 91(1).
- Bilbiie, F. O. (2019). The New Keynesian cross. *Journal of Monetary Economics*, 114:90–108.

- Bilbiie, F. O. (2025). Monetary Policy and Heterogeneity: An Analytical Framework. *Review of Economic Studies*, 92(4).
- Blanchard, O. J. and Kahn, C. M. (1980). The Solution of Linear Difference Models under Rational Expectations. *Econometrica*, 48(5):1305–1311.
- Bonaglia, E., d'Arienzo, D., Fallico, N., Gennaioli, N., and Iovino, L. (2025). The Volatility of Inflation Expectations and Interest Rates.
- Bordalo, P., Conlon, J. J., Gennaioli, N., Kwon, S. Y., and Shleifer, A. (2023). Memory and Probability. *The Quarterly Journal of Economics*, 138(1):265–311.
- Bordalo, P., Gennaioli, N., Ma, Y., and Shleifer, A. (2020). Overreaction in Macroeconomic Expectations. *American Economic Review*, 110(9):2748–2782.
- Branch, W. A. and Evans, G. W. (2006). Intrinsic heterogeneity in expectation formation. *Journal of Economic Theory*, 127(1):264–295.
- Broer, T., Kohlhas, A., Mitman, K., and Schlafmann, K. (2026). Expectation and Wealth Heterogeneity in the Macroeconomy. *Journal of Political Economy*, forthcoming.
- Bullard, J. and Suda, J. (2016). The stability of macroeconomic systems with Bayesian learners. *Journal of Economic Dynamics and Control*, 62:1–16.
- Candia, B., Coibion, O., and Gorodnichenko, Y. (2022). The Macroeconomic Expectations of Firms. *Handbook of Economic Expectations*, 30042.
- Carroll, C. D. (2003). Macroeconomic Expectations of Households and Professional Forecasters. *The Quarterly Journal of Economics*, 118(1):269–298.
- Carvalho, V. M. and Tahbaz-Salehi, A. (2019). Production Networks: A Primer. *Annual Review of Economics*, 11:635–663.
- Chahrour, R. and Gaballo, G. (2021). Learning from House Prices: Amplification and Business Fluctuations. *The Review of Economic Studies*, 88(4):1720–1759.
- Chahrour, R. and Jurado, K. (2025). Expectation Response Functions in Dynamic Linear Economies.
- Chahrour, R., Nimark, K., and Pitschner, S. (2021). Sectoral Media Focus and Aggregate Fluctuations. *American Economic Review*, 111(12):3872–3922.
- Cogley, T., Primiceri, G. E., and Sargent, T. J. (2010). Inflation-Gap Persistence in the US. *American Economic Journal: Macroeconomics*, 2(1):43–69.
- Coibion, O. and Gorodnichenko, Y. (2015). Information rigidity and the expectations formation process: A simple framework and new facts. *American Economic Review*, 105(8):2644–2678.

- Coibion, O., Gorodnichenko, Y., and Weber, M. (2022). Monetary Policy Communications and their Effects on Household Inflation Expectations. *Journal of Political Economy*, 130(6).
- D’Acunto, F., Malmendier, U., and Weber, M. (2023). What do the data tell us about inflation expectations? In Bachmann, R., Topa, G., and van der Klaauw, W., editors, *Handbook of Economic Expectations*, pages 133–161.
- d’Arienzo, D. (2020). Maturity Increasing Overreaction and Bond Market Puzzles.
- Dawber, J., Würz, N., Smith, P. A., Flower, T., Thomas, H., Schmid, T., and Tzavidis, N. (2022). Experimental UK Regional Consumer Price Inflation with Model-Based Expenditure Weights. *Journal of Official Statistics*, 38(1):213–237.
- Debortoli, D. and Galí, J. (2025). Heterogeneity and Aggregate Fluctuations: Insights from TANK Models. *NBER Macroeconomics Annual*, 39(1):307–357.
- Dixon, H. and Michail, M. G. (2025). Seasonal Adjustment of CPI and CPIH. *NIESR Policy Paper*, 45.
- Donaldson, P. (2025). Cross-sectional Identification with Heterogeneous Exposure to General Equilibrium Effects.
- Dräger, L., Lamla, M. J., and Pfajfar, D. (2022). The Hidden Heterogeneity of Inflation Expectations and its Implications. *CESifo Working Paper*, (9637).
- Eggertsson, G. B. and Woodford, M. (2003). The Zero Bound on Interest Rates and Optimal Monetary Policy. *Brookings Papers on Economic Activity*, 1.
- Eliaz, K. and Spiegler, R. (2026). News Media as Suppliers of Narratives (and Information). *Theoretical Economics*, 21(3).
- Eusepi, S. and Preston, B. (2018). The science of monetary policy: An imperfect knowledge perspective. *Journal of Economic Literature*, 56(1):3–59.
- Evans, G. W. and McGough, B. (2020). Adaptive Learning in Macroeconomics. *Oxford Research Encyclopedia of Economics and Finance*.
- Farboodi, M. and Veldkamp, L. (2025). A Model of the Data Economy. *Review of Economic Studies*, forthcoming.
- Farhi, E. and Werning, I. (2019). Monetary policy, bounded rationality, and incomplete markets. *American Economic Review*, 109(11):3887–3928.
- Fuster, A. and Zafar, B. (2023). Survey experiments on economic expectations. In Bachmann, R., Topa, G., and van der Klaauw, W., editors, *Handbook of Economic Expectations*, pages 107–130.
- Gabaix, X. (2020). A behavioral new Keynesian model. *American Economic Review*, 110(8).

- Gaballo, G. (2018). Price Dispersion, Private Uncertainty, and Endogenous Nominal Rigidities. *The Review of Economic Studies*, 85(2):1070–1110.
- Galí, J. (2008). *Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework*. Princeton University Press, Princeton, USA and Oxford, UK.
- Gennaioli, N., Leva, M., Schoenle, R., and Shleifer, A. (2024). How Inflation Expectations De-Anchor: The Role of Selective Memory Cues. *NBER Working Papers*, 32633.
- Gibbons, R. and Prusak, L. (2020). Knowledge, Stories, and Culture in Organizations. *AEA Papers and Proceedings*, 110:187–92.
- Goldstein, N. and Gorodnichenko, Y. (2022). Expectations Formation and Forward Information. *NBER Working Paper Series*, 29711.
- Grandmont, J. M. (1977). Temporary General Equilibrium Theory. *Econometrica*, 45(3):535.
- Grossman, S. J. and Stiglitz, J. E. (1980). On the Impossibility of Informationally Efficient Markets. *The American Economic Review*, 70(3):393–408.
- Guerreiro, J. (2023). Belief Disagreement and Business Cycles.
- Guren, A., McKay, A., Nakamura, E., and Steinsson, J. (2020). What Do We Learn from Cross-Regional Empirical Estimates in Macroeconomics? *NBER Macroeconomics Annual*, 35.
- Haaland, I., Roth, C., Stantcheva, S., and Wohlfart, J. (2025). Understanding Economic Behavior Using Open-Ended Survey Data. *Journal of Economic Literature*, 63(4):1244–1280.
- Haaland, I., Roth, C., and Wohlfart, J. (2023). Designing Information Provision Experiments. *Journal of Economic Literature*, 61(1):3–40.
- Hagedorn, M. (2018). Prices and Inflation when Government Bonds are Net Wealth.
- Hajdini, I., Knotek, E. S., Leer, J., Pedemonte, M., Rich, R. W., and Schoenle, R. S. (2023). Low Passthrough from Inflation Expectations to Income Growth Expectations: Why People Dislike Inflation. *Federal Reserve Bank of Cleveland Working Paper*, (22-21R).
- Heckman, J. J. (1979). Sample Selection Bias as a Specification Error. *Econometrica*, 47(1).
- Hoffmann, M., Moench, E., Pavlova, L., and Schulte Frankenfeld, G. (2022). Would households understand average inflation targeting? *Journal of Monetary Economics*, 129:S52–S66.
- Iovino, L. and Sergeyev, D. (2023). Central Bank Balance Sheet Policies without Rational Expectations. *Review of Economic Studies*, 90(6):3119–3152.
- Kamdar, R. and Ray, W. (2024). Attention-Driven Sentiment and the Business Cycle.

- Kaplan, G. and Schulhofer-Wohl, S. (2017). Inflation at the household level. *Journal of Monetary Economics*, 91:19–38.
- Kaplan, G. and Violante, G. L. (2018). Microeconomic Heterogeneity and Macroeconomic Shocks. *NBER Working Papers*, 24734.
- Kiss, R. and Strasser, G. (2024). Inflation Heterogeneity Across Households. *ECB Working Paper Series*, 2898.
- Leombroni, M., Piazzesi, M., Rogers, C., and Schneider, M. (2020). Inflation and the Price of Real Assets.
- Link, S., Peichl, A., Pfäuti, O., Roth, C., and Wohlfart, J. (2025). Attention to the Macroeconomy. *ECONtribute Discussion Papers*, 256.
- Link, S., Peichl, A., Roth, C., and Wohlfart, J. (2023). Information Frictions Among Firms and Households. *Journal of Monetary Economics*, 135:99–115.
- Lucas, R. E. (1972). Expectations and the neutrality of money. *Journal of Economic Theory*, 4(2):103–124.
- Ludwig, A., Mankart, J., Quintana, J., and Wiederholt, M. (2024). Heterogeneity in Expectations and House Price Dynamics.
- Macaulay, A. (2021). The Attention Trap: Rational Inattention, Inequality, and Fiscal Policy. *European Economic Review*, 135:103716.
- Macaulay, A. (2022). Heterogeneous Information, Subjective Model Beliefs, and the Time-Varying Transmission of Shocks. *CESifo Working Paper*, 9733.
- Macaulay, A. and Moberly, J. (2026). Heterogeneity in Imperfect Inflation Expectations: Theory and Evidence from a Novel Survey. *Journal of Money, Credit and Banking*, 58(4).
- Macaulay, A. and Song, W. (2022). Narrative-Driven Fluctuations in Sentiment: Evidence Linking Traditional and Social Media. *University of Oxford Economics Discussion Paper Series*.
- Maćkowiak, B., Matějka, F., and Wiederholt, M. (2023). Rational Inattention: A Review. *Journal of Economic Literature*, 61(1):226–73.
- Maćkowiak, B. and Wiederholt, M. (2009). Optimal Sticky Prices under Rational Inattention. *American Economic Review*, 99(3):769–803.
- Maćkowiak, B. and Wiederholt, M. (2015). Business Cycle Dynamics under Rational Inattention. *Review of Economic Studies*, 82(4):1502–1532.
- Maćkowiak, B. and Wiederholt, M. (2024). Rational Inattention during an RCT. *American Economic Review: Insights*.

- Mankiw, N. G. and Reis, R. (2002). Sticky Information versus Sticky Prices: A Proposal to Replace the New Keynesian Phillips Curve. *The Quarterly Journal of Economics*, 117(4):1295–1328.
- Mankiw, N. G., Reis, R., and Wolfers, J. (2004). Disagreement about Inflation Expectations. *NBER Macroeconomics Annual*, 18:209–270.
- Matthes, C., Nagasaka, N., and Schwartzman, F. (2025). Estimating the Missing Intercept.
- McKay, A. and Wolf, C. K. (2023). Optimal Policy Rules in HANK.
- Michaillat, P. and Saez, E. (2019). Resolving New Keynesian Anomalies with Wealth in the Utility Function. *The Review of Economics and Statistics*.
- Michelacci, C. and Paciello, L. (2024). Ambiguity Aversion and Heterogeneity in Households’ Beliefs. *American Economic Journal: Macroeconomics*, 16(2):95–126.
- Milani, F. (2023). Chapter 18 - Expectational data in DSGE models. In Bachmann, R., Topa, G., and van der Klaauw, W., editors, *Handbook of Economic Expectations*, pages 541–567.
- Nakamura, E. and Steinsson, J. (2018). Identification in Macroeconomics. *Journal of Economic Perspectives*, 32(3):59–86.
- Nimark, K. P. (2014). Man-Bites-Dog Business Cycles. *American Economic Review*, 104(8).
- Nimark, K. P. and Pitschner, S. (2019). News media and delegated information choice. *Journal of Economic Theory*, 181:160–196.
- Nord, L. (2022). Who Cares about Inflation? Endogenous Expectation Formation of Heterogeneous Households.
- Nunes, R. and Park, D. (2020). Inflation Expectations, Interest Rates, and Consumption Behavior.
- Pfajfar, D. and Santoro, E. (2010). Heterogeneity, learning and information stickiness in inflation expectations. *Journal of Economic Behavior & Organization*, 75(3):426–444.
- Piazzesi, M. and Schneider, M. (2016). Chapter 19 - Housing and Macroeconomics. In Taylor, J. B. and Uhlig, H., editors, *Handbook of Macroeconomics*, volume 2, pages 1547–1640. Elsevier.
- Reis, R. (2006). Inattentive Producers. *The Review of Economic Studies*, 73(3):793–821.
- Rotemberg, J. J. (1982). Sticky Prices in the United States. *Journal of Political Economy*, 90(6).
- Roth, C., Wiederholt, M., and Wohlfart, J. (2021). The Effects of Forward Guidance: Theory with Measured Expectations. *ECONtribute Discussion Papers*, (126).
- Shiller, R. J. (1997). Why Do People Dislike Inflation? In Romer, C. D. and Romer, D. H., editors, *Reducing Inflation: Motivation and Strategy*, pages 13–70. University of Chicago Press.
- Sims, C. A. (2003). Implications of rational inattention. *Journal of Monetary Economics*, 50(3).

- Sokolova, A. (2023). Marginal propensity to consume and unemployment: A meta-analysis. *Review of Economic Dynamics*, 51:813–846.
- Spiegler, R. (2020). Behavioral implications of causal misperceptions. *Annual Review of Economics*, 12.
- Spiegler, R. (2021). A Simple Model of Monetary Policy under Phillips-Curve Causal Disagreements.
- Weber, M., Candia, B., Afrouzi, H., Ropele, T., Lluberas, R., Frache, S., Meyer, B., Kumar, S., Gorodnichenko, Y., Georgarakos, D., Coibion, O., Kenny, G., and Ponce, J. (2025). Tell Me Something I Don't Already Know: Learning in Low- and High-Inflation Settings. *Econometrica*, 93(1).
- Weber, M., D'Acunto, F., Gorodnichenko, Y., and Coibion, O. (2022). The Subjective Inflation Expectations of Households and Firms: Measurement, Determinants, and Implications. *Journal of Economic Perspectives*, 36(3):157–184.
- Werning, I. (2022). Expectations and the Rate of Inflation. *NBER Working Papers*, 30260.
- Wolf, C. K. (2023). The Missing Intercept: A Demand Equivalence Approach. *American Economic Review*, 113(8):2232–2269.
- Woodford, M. (2003). Imperfect common knowledge and the effects of monetary policy. In *Knowledge, Information, and Expectations in Modern Macroeconomics: In Honor of Edmund S. Phelps*.
- Woodford, M. (2013). Macroeconomic Analysis Without the Rational Expectations Hypothesis. *Annual Review of Economics*, 5:303–346.



## A Proofs for Sections 2 and 3

**Relaxing upper-triangularity of  $M_i$ .** In Section 2, we restricted attention to upper-triangular  $M_i$  matrices. This implied that  $\chi_i$  had a unit diagonal. We now relax that assumption. Doing this implies that  $\tau_i$  can no longer be interpreted as the extent of expectations updating after a direct signal.<sup>29</sup> However, it is helpful to re-scale the  $\chi_i$  and  $\tau_i$  such that the original interpretation is restored. As in the first stage of an information-treatment RCT, we would like  $\tau_i$  to reflect the updating from signals to expectations of the variables being signaled.<sup>30</sup>

To do that, define a new square matrix  $D_i$ , where the diagonal elements are the diagonal elements of  $\chi_i$ , but all off-diagonal elements are equal to 0. Then define  $\hat{\chi}_i = \chi_i D_i^{-1}$  and  $\hat{\tau}_i = D_i \tau_i$ , so

$$E_{it}(z) = \hat{\chi}_i \hat{\tau}_i s_{it} \quad (45)$$

This is of the form in equation (9), but  $\hat{\chi}_i$  is now guaranteed to have a unit diagonal, such that  $\hat{\tau}_i$  captures the updating from direct signals. I drop the hat-notation for the rest of the paper to simplify notation, but with this rescaling it is therefore possible to allow for non-upper-triangular  $M_i$  in all of the paper's methods and results.

**Proposition 1.** Differentiating equation (11) with respect to the information shock yields

$$\frac{d\pi_t}{d\varepsilon_{\pi_t}} = (1 - \lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i\pi_t+s\pi_t} \tau_{i\pi_t} + (1 - \beta\lambda) \chi_{imc_t+s\pi_t} \tau_{i\pi_t}) di \quad (46)$$

Aggregating equation (9) across firms and then differentiating yields

$$\frac{d\bar{E}_t(\pi_t)}{d\varepsilon_{\pi_t}} = \int_i \tau_{i\pi_t} di \equiv \bar{\tau}_{\pi_t} \quad (47)$$

where we have used the fact that the diagonal elements of  $\chi_i$  are equal to 1 by construction.

Since the model is linear, scaling the shock by  $\bar{\tau}_{\pi_t}^{-1}$  implies a unit change in  $\bar{E}_t(\pi_t)$ , which then generates equation (12).

<sup>29</sup>For example, in the Calvo model a signal about  $\pi_{t+1}$  has a direct effect on  $E_{it}\pi_{t+1}$ , but also may affect  $E_{it}mc_{t+1}$ . If  $M_i$  is not upper-triangular, this change in  $E_{it}mc_{t+1}$  may cause a further change in  $E_{it}\pi_{t+1}$ .  $\tau_{i\pi_{t+1}}$  only captures the first of these effects, which is no longer the full update after the signal.

<sup>30</sup>When  $M_i$  is not upper-triangular, in principle  $(I - (I - \tau_i)M_i)$  may not be invertible, and thus  $\chi_i$  may not be well-defined. A sufficient condition for invertibility is that the spectral radius of  $(I - \tau_i)M_i$  is  $< 1$ . Economically, this condition implies that feedback loops in agent  $i$ 's subjective model, in which  $E_{it}(z_j)$  affects  $E_{it}(z_k)$  affects  $E_{it}(z_j)$ , cannot be so powerful as to be explosive. I restrict attention throughout to cases where this condition holds.

**Proposition 2.** The proof follows the same steps as that for Proposition 1. Differentiating equation (11) with respect to the information shock yields

$$\frac{d\pi_t}{d\varepsilon_{\pi_{t+1}}} = (1 - \lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i\pi_{t+s}\pi_{t+1}} \tau_{i\pi_{t+1}} + (1 - \beta\lambda) \chi_{imc_{t+s}\pi_{t+1}} \tau_{i\pi_{t+1}}) di \quad (48)$$

Aggregating equation (9) across firms and then differentiating yields

$$\frac{d\bar{E}_t(\pi_{t+1})}{d\varepsilon_{\pi_{t+1}}} = \int_i \tau_{i\pi_{t+1}} di \equiv \bar{\tau}_{\pi_{t+1}} \quad (49)$$

Since the model is linear, scaling the shock by  $\bar{\tau}_{\pi_{t+1}}^{-1}$  implies a unit change in  $\bar{E}_t(\pi_{t+1})$ , which then generates equation (13).

**GE effects of  $\varepsilon_{\pi_t}$ .** With the assumptions on signals in equation (15), equation (11) can be written

$$\pi_t = (1 - \lambda) \left( \sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i\pi_{t+s}\pi_t} + (1 - \beta\lambda) \chi_{imc_{t+s}\pi_t}) \tau_{i\pi_t} di \right) (\pi_t + \varepsilon_{\pi_t}) + a \quad (50)$$

where  $a$  collects the steady-state markup  $\mu$  and all terms involving signals of  $m_{c_{t+s}}$ , and  $\pi_{t+s}$  for  $s \geq 1$ . Differentiating with respect to  $\varepsilon_{\pi_t}$  yields

$$\frac{d\pi_t}{d\varepsilon_{\pi_t}} = (1 - \lambda) \left( \sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i\pi_{t+s}\pi_t} + (1 - \beta\lambda) \chi_{imc_{t+s}\pi_t}) \tau_{i\pi_t} di \right) \left( \frac{d\pi_t}{d\varepsilon_{\pi_t}} + 1 \right) \quad (51)$$

which rearranges to

$$\frac{d\pi_t}{d\varepsilon_{\pi_t}} = \frac{\bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)}{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)} \quad (52)$$

where

$$\Omega^{PE}(\pi_t) \equiv (1 - \lambda) \left( \sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i\pi_{t+s}\pi_t} + (1 - \beta\lambda) \chi_{imc_{t+s}\pi_t}) \frac{\tau_{i\pi_t}}{\bar{\tau}_{\pi_t}} di \right) \quad (53)$$

denotes the (scaled) partial-equilibrium effect of  $\pi_t$  expectations, as in equation (12).

Aggregating equation (9) across firms and then differentiating yields

$$\frac{d\bar{E}_t(\pi_t)}{d\varepsilon_{\pi_t}} = \bar{\tau}_{\pi_t} \left( \frac{d\pi_t}{d\varepsilon_{\pi_t}} + 1 \right) \quad (54)$$

$$= \bar{\tau}_{\pi_t} \left( \frac{\bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)}{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)} + 1 \right) \quad (55)$$

$$= \frac{\bar{\tau}_{\pi_t}}{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)} \quad (56)$$

where the second equality uses equation (52). To ensure the impact on  $\bar{E}_t(\pi_t) = 1$ , we therefore scale the shock by  $\frac{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)}{\bar{\tau}_{\pi_t}}$ . This implies

$$\Omega^{GE}(\pi_t) = \frac{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)}{\bar{\tau}_{\pi_t}} \frac{d\pi_t}{d\varepsilon_{\pi_t}} = \Omega^{PE}(\pi_t) \quad (57)$$

After scaling, general equilibrium effects of expected current inflation are irrelevant.

**Proposition 3.** The proof follows the same steps as for the GE effects of expected  $\pi_t$  above. Equation (11) can be written

$$\begin{aligned} \pi_t = & (1 - \lambda) \left( \sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i\pi_{t+s}\pi_t} + (1 - \beta\lambda)\chi_{imc_{t+s}\pi_t}) \tau_{i\pi_t} di \right) (\pi_t + \varepsilon_{\pi_t}) \\ & + (1 - \lambda) \left( \sum_{s=0}^{\infty} (\beta\lambda)^s \int_i (\chi_{i\pi_{t+s}\pi_{t+1}} + (1 - \beta\lambda)\chi_{imc_{t+s}\pi_{t+1}}) \tau_{i\pi_{t+1}} di \right) (\pi_{t+1} + \varepsilon_{\pi_{t+1}}) + a' \end{aligned} \quad (58)$$

where  $a'$  collects the steady-state markup  $\mu$  and all terms involving signals of  $mc_{t+s}$ , and  $\pi_{t+s}$  for  $s \geq 2$ . Differentiating with respect to  $\varepsilon_{\pi_{t+1}}$  yields

$$\frac{d\pi_t}{d\varepsilon_{\pi_{t+1}}} = \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t) \frac{d\pi_t}{d\varepsilon_{\pi_{t+1}}} + \bar{\tau}_{\pi_{t+1}} \Omega^{PE}(\pi_{t+1}) \quad (59)$$

where the realized value of  $\pi_{t+1}$  is unaffected by the shock. By the time period  $t + 1$  arrives, the firms realize that the change in  $s_{it}(\pi_{t+1})$  they observed was noise, so  $E_{it+1}$  expectations return to steady state. Note that this is an equilibrium outcome, not an additional assumption: the model is purely forward-looking, so once expectations return to steady state, per-period inflation  $\pi_{t+1}$  also returns to steady state. Rearranging, we obtain

$$\frac{d\pi_t}{d\varepsilon_{\pi_{t+1}}} = \frac{\bar{\tau}_{\pi_{t+1}} \Omega^{PE}(\pi_{t+1})}{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)} \quad (60)$$

Aggregating equation (9) across firms and then differentiating yields

$$\frac{d\bar{E}_t(\pi_{t+1})}{d\varepsilon_{\pi_{t+1}}} = \bar{\tau}_{\pi_{t+1}} + \frac{d\pi_t}{d\varepsilon_{\pi_{t+1}}} \int_i \chi_{i\pi_{t+1}\pi_t} \tau_{i\pi_t} di \quad (61)$$

$$= \bar{\tau}_{\pi_{t+1}} + \frac{\bar{\tau}_{\pi_{t+1}} \Omega^{PE}(\pi_{t+1})}{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)} \int_i \chi_{i\pi_{t+1}\pi_t} \tau_{i\pi_t} di \quad (62)$$

This is more complicated than the case of a shock to  $\varepsilon_{\pi_t}$  because the general equilibrium feedback into aggregate  $\bar{E}_t(\pi_{t+1})$  occurs through cross-learning from changes in realized  $\pi_t$ , and thus in  $s_{it}(\pi_t)$ . Define the constant  $K$  as

$$K \equiv \left( \bar{\tau}_{\pi_{t+1}} + \frac{\bar{\tau}_{\pi_{t+1}} \Omega^{PE}(\pi_{t+1})}{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)} \int_i \chi_{i\pi_{t+1}\pi_t} \tau_{i\pi_t} di \right)^{-1} \quad (63)$$

$$= \frac{1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)}{\bar{\tau}_{\pi_{t+1}} (1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)) + \bar{\tau}_{\pi_{t+1}} \Omega^{PE}(\pi_{t+1}) (\bar{\chi}_{\pi_{t+1}\pi_t} \bar{\tau}_{\pi_t} + Cov_I[\chi_{i\pi_{t+1}\pi_t}, \tau_{i\pi_t}])} \quad (64)$$

The scaled aggregate effect of the shock is therefore

$$K \cdot \frac{d\pi_t}{d\varepsilon_{\pi_{t+1}}} = \frac{\bar{\tau}_{\pi_{t+1}} \Omega^{PE}(\pi_{t+1})}{\bar{\tau}_{\pi_{t+1}} (1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)) + \bar{\tau}_{\pi_{t+1}} \Omega^{PE}(\pi_{t+1}) (\bar{\chi}_{\pi_{t+1}\pi_t} \bar{\tau}_{\pi_t} + Cov_I[\chi_{i\pi_{t+1}\pi_t}, \tau_{i\pi_t}])} \quad (65)$$

which after dividing through by  $\bar{\tau}_{\pi_{t+1}}$  becomes equation (16).

**Proposition 4.** This follows from equation (20). Using this and the definition of the counterfactual, the fraction of transmission due to expectations is

$$\frac{\left. \frac{d\pi_t}{dmc_t} - \frac{d\pi_t}{dmc_t} \right|_{\sim E}}{\frac{d\pi_t}{dmc_t}} = 1 - \frac{\left. \frac{d\pi_t}{dmc_t} \right|_{\sim E}}{\frac{d\pi_t}{dmc_t}} = 1 - (1 - \bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t)) \quad (66)$$

which implies equation (21).

**Proposition 5.** The parameters in equation (22) have been chosen so that the representative-agent effects of inflation expectations (line 1 in equation (14)) are equal to those in equation (6). Since  $\chi_{imc_t+s\pi_t}$  is set to 0, it only remains to find, for  $s \geq 1$ ,

$$Cov_I[\chi_{i\pi_{t+s}\pi_t}, \tau_{i\pi_t}] = E_I[\chi_{i\pi_{t+s}\pi_t} \tau_{i\pi_t}] - \bar{\chi}_{\pi_{t+s}\pi_t} \bar{\tau}_{\pi_t} \quad (67)$$

$$= E_I[\chi_{i\pi_{t+s}\pi_t} | \tau_{i\pi_t} = 1] \Pr(\tau_{i\pi_t} = 1) - \frac{1}{2} \quad (68)$$

$$= \frac{2q}{2} - \frac{1}{2} = \frac{1}{2}(2q - 1) \quad (69)$$

where  $E_I[\cdot]$  denotes a cross-sectional expectation across firms, and  $q \equiv \Pr(\chi_{i\pi_t+s\pi_t} = 2 | \tau_{i\pi_t} = 1)$ . With this, the narrative heterogeneity effects (line 2 in equation (14)) are

$$\frac{(1-\lambda)}{\frac{1}{2}} \sum_{s=1}^{\infty} (\beta\lambda)^s \text{Cov}_I[\chi_{i\pi_t+s\pi_t}, \tau_{i\pi_t}] = (1-\lambda)(2q-1) \sum_{s=1}^{\infty} (\beta\lambda)^s \quad (70)$$

$$= \frac{\beta\lambda(1-\lambda)(2q-1)}{1-\beta\lambda} \quad (71)$$

Adding this together with the representative-agent effects from equation (6) yields

$$\frac{1}{\bar{\tau}_{\pi_t}} \frac{d\pi_t}{d\varepsilon_{\pi_t}} = \frac{1-\lambda}{1-\beta\lambda} + \frac{\beta\lambda(1-\lambda)(2q-1)}{1-\beta\lambda} = \frac{1-\lambda}{1-\beta\lambda} \cdot (1 + \beta\lambda(2q-1)) \quad (72)$$

and taking  $\beta \rightarrow 1$  then generates equation (23).

**Proposition 6.** The proof follows the same steps as for Proposition 1. The  $k$ th row of equation (27) can be written as

$$x_{ikt} = \sum_{n=0}^{N_z} A_{ikn} x_{int-1} + \sum_{n=0}^{N_z} \sum_{m=0}^{N_z} B_{ikm} \chi_{imn} \tau_{in} s_{int} \quad (73)$$

where  $A_{ikn}$  denotes the  $k, n$ th element of  $A_i$ , and  $B_{ikm}$  denotes the  $k, m$ th element of  $B_i$ .  $\chi_{imn}$  and  $\tau_{in}$  are the  $m, n$ th element of  $\chi_i$  and the  $n$ th diagonal element of  $\tau_i$  respectively.

Aggregating across agents  $i$  and differentiating with respect to  $\varepsilon_{jt}$  we obtain

$$\frac{dx_{kt}}{d\varepsilon_{jt}} = \int_i \left( \sum_{m=0}^{N_z} B_{ikm} \chi_{imj} \tau_{ij} \right) di \quad (74)$$

$$= \sum_{m=0}^{N_z} (\bar{B}_{km} \bar{\chi}_{mj} \bar{\tau}_j + \text{Cov}_I[B_{ikm}, \chi_{imj} \tau_{ij}] + \bar{B}_{km} \text{Cov}_I[\chi_{imj}, \tau_{ij}]) \quad (75)$$

where the second equality uses the definition of a covariance.

Similarly, aggregating equation (9) across agents and differentiating yields

$$\frac{d\bar{E}_t(z_{jt})}{d\varepsilon_{jt}} = \int_i \tau_{ij} di \equiv \bar{\tau}_j \quad (76)$$

where we have used the fact that diagonal elements of  $\chi_i$  are equal to 1 by construction.

Since the model is linear, scaling the shock by  $\bar{\tau}_j^{-1}$  implies a unit change in  $\bar{E}_t(z_{jt})$ , which then implies equation (29).

**Nested cases in Proposition 6.** Setting  $\tau_{ij} = 1$  for all  $i, j$  in equation (29) recovers the effect in standard temporary equilibrium, in which  $E_{it}(z_{jt})$  shifts equally for all agents, but no other expectations change. This abstracts from all effects of heterogeneity through the response heterogeneity and narrative heterogeneity channels. All that matters is how, on average, expectations of  $z_{jt}$  enter the policy function for  $x_{kt}$ .

$$\Omega_k^{PE}(z_{jt}|\tau_{ij} = 1\forall i, j) = \bar{B}_{kj} \quad (77)$$

Setting  $\tau_{ij} = \bar{\tau}_j < 1$  and  $\chi_{iz_{jt+s}z_{jt}} = 1 + s$  for all  $i, j$  recovers the effect in extended temporary equilibrium, as in Werning (2022). Again, this abstracts from all effects of heterogeneity, though now a broader range of policy function coefficients determine the aggregate response, as signals on  $z_{jt}$  also impact expectations of  $z_{jt+1}$ ,  $z_{jt+2}$ , etc.

$$\Omega_k^{PE}(z_{jt}|\tau_{ij} = \bar{\tau}_j < 1, \chi_{iz_{jt+s}z_{jt}} = 1 + s, \forall i, j) = \sum_{s=0}^{\infty} \bar{B}_{kjs}(1 + s) \quad (78)$$

where  $\bar{B}_{kjs}$  denotes the average coefficient on  $z_{jt+s}$  in the row of the policy function (27) corresponding to  $x_{ikt}$ .

**Proposition 7.** Aggregating equation (27) across agents, we can express the aggregate policy function as

$$\mathbf{x}_t = \int_i A_i \mathbf{x}_{it-1} di + \sum_{s=0}^{\infty} [\Omega_{xs}^{PE} \bar{\tau}_{xs}(\mathbf{x}_{t+s} + \boldsymbol{\varepsilon}_{xt+s}) + \Omega_{ws}^{PE} \bar{\tau}_{ws}(\mathbf{w}_{t+s} + \boldsymbol{\varepsilon}_{wt+s})] \quad (79)$$

where  $\Omega_{xs}^{PE}$  is an  $N_x \times N_x$  matrix collecting the partial-equilibrium causal effects of expectations about each element of the future aggregate choice vector  $\mathbf{x}_{t+s}$  on each aggregate choice (i.e. each element of  $\mathbf{x}_t$ ), as defined in Proposition 6. Similarly,  $\Omega_{ws}^{PE}$  is an  $N_x \times N_w$  matrix collecting the partial-equilibrium causal effects of expectations about each element of the future state vector  $\mathbf{w}_{t+s}$  on each aggregate choice.  $\bar{\tau}_{xs}$  and  $\bar{\tau}_{ws}$  are square diagonal matrices with diagonals equal to the average  $\tau_{ij}$  parameters across agents for signals on  $\mathbf{x}_{t+s}$  and  $\mathbf{w}_{t+s}$  respectively. Finally,  $\boldsymbol{\varepsilon}_{xt+s}$  and  $\boldsymbol{\varepsilon}_{wt+s}$  are vectors collecting the public component of noise in signals on  $\mathbf{x}_{t+s}$  and  $\mathbf{w}_{t+s}$ . Note that the idiosyncratic noise terms in equation (28) cancel out when we aggregate across agents.

Rearranging equation (25) we obtain

$$\mathbf{w}_t = -D^{-1}C\mathbf{x}_t - D^{-1}F\boldsymbol{\xi}_t \quad (80)$$

Substituting (80) into (79) and rearranging yields

$$\mathbf{x}_t = (I - \Lambda_0)^{-1} \left( \int_i A_i \mathbf{x}_{it-1} di + \sum_{s=1}^{\infty} [\Lambda_s \mathbf{x}_{t+s}] + \sum_{s=0}^{\infty} [\Omega_{xs}^{PE} \bar{\tau}_{xs} \varepsilon_{xt+s} + \Omega_{ws}^{PE} \bar{\tau}_{ws} (\varepsilon_{wt+s} - D^{-1} F \xi_{t+s})] \right) \quad (81)$$

where

$$\Lambda_s \equiv \Omega_{xs}^{PE} \bar{\tau}_{xs} - \Omega_{ws}^{PE} \bar{\tau}_{ws} D^{-1} C \quad (82)$$

Differentiating with respect to  $\varepsilon_{xt+h}$  yields

$$\frac{d\mathbf{x}_t}{d\varepsilon_{xt+h}} = (I - \Lambda_0)^{-1} \left( \Omega_{xh}^{PE} \bar{\tau}_{xh} + \sum_{s=1}^{\infty} \left[ \Lambda_s \frac{d\mathbf{x}_{t+s}}{d\varepsilon_{xt+h}} \right] \right) \quad (83)$$

Next, we turn to  $\frac{d\mathbf{x}_{t+s}}{d\varepsilon_{xt+h}}$ . For this, note that the terms relating  $\mathbf{x}_t$  to  $\mathbf{x}_{it-1}$  in equation (79) can be written as

$$\int_i A_i \mathbf{x}_{it-1} di = \tilde{A} \mathbf{x}_{t-1} \quad (84)$$

where

$$\tilde{A} \equiv \int_i A_i di + Cov_I[A_i, \mathbf{x}_{it-1}] \mathbf{x}_{t-1}^+ \quad (85)$$

and  $\mathbf{x}_{t-1}^+$  is the Moore-Penrose pseudoinverse of  $\mathbf{x}_{t-1}$ , such that  $\mathbf{x}_{t-1}^+ \mathbf{x}_{t-1} = 1$ .<sup>31</sup>

To proceed, we will make the simplifying assumption that  $Cov_I[A_i, \mathbf{x}_{it-1}] \mathbf{x}_{t-1}^+$  is constant over time. That is, the correlation between an agent's weight on lagged actions in their policy rule and their deviation from the aggregate actions may be non-zero, but it is unaffected by shocks. With this simplification,  $\tilde{A}$  is constant.

Let the matrix  $R$  be defined by the recursion

$$R = (I - \Lambda_0)^{-1} \left( \tilde{A} + \sum_{k=1}^{\infty} \Lambda_k R^{k+1} \right) \quad (86)$$

Given this, we can show the following.

---

<sup>31</sup>Note that  $\tilde{A}$  is only well-defined if  $\mathbf{x}_{t-1} \neq 0$ . For a shock arriving in steady state,  $\mathbf{x}_{t-1} = 0$  and the first term in (81) drops out, so the value of  $\tilde{A}$  is immaterial in that case.

**Lemma 1** *If the series  $\sum_{k=1}^{\infty} \Lambda_k R^{k+1}$  converges, and if the spectral radius of  $R$  is  $\rho(R) < 1$ , then*

$$\frac{d\mathbf{x}_{t+s}}{d\boldsymbol{\varepsilon}_{xt+h}} = R^s \frac{d\mathbf{x}_t}{d\boldsymbol{\varepsilon}_{xt+h}} \quad (87)$$

**Proof.** We prove equation (87) with a guess-and-verify approach. Rolling equation (81) forwards and differentiating, we have

$$\frac{d\mathbf{x}_{t+s}}{d\boldsymbol{\varepsilon}_{xt+h}} = (I - \Lambda_0)^{-1} \left( \tilde{A} \frac{d\mathbf{x}_{t+s-1}}{d\boldsymbol{\varepsilon}_{xt+h}} + \sum_{k=1}^{\infty} \left[ \Lambda_k \frac{d\mathbf{x}_{t+s+k}}{d\boldsymbol{\varepsilon}_{xt+h}} \right] \right) \quad (88)$$

Substituting in equation (87) this becomes

$$\frac{d\mathbf{x}_{t+s}}{d\boldsymbol{\varepsilon}_{xt+h}} = (I - \Lambda_0)^{-1} \left( \tilde{A} R^{s-1} \frac{d\mathbf{x}_t}{d\boldsymbol{\varepsilon}_{xt+h}} + \sum_{k=1}^{\infty} \left[ \Lambda_k R^{s+k} \frac{d\mathbf{x}_t}{d\boldsymbol{\varepsilon}_{xt+h}} \right] \right) \quad (89)$$

$$= (I - \Lambda_0)^{-1} \left( \tilde{A} + \sum_{k=1}^{\infty} [\Lambda_k R^{k+1}] \right) R^{s-1} \frac{d\mathbf{x}_t}{d\boldsymbol{\varepsilon}_{xt+h}} \quad (90)$$

which when combined with (86) verifies the guess.

The requirement that  $\sum_{k=1}^{\infty} \Lambda_k R^{k+1}$  converges in some consistent matrix norm ensures that  $R$  is a well-defined finite matrix.  $\rho(R) < 1$  ensures uniqueness of the bounded solution in (87), ruling out explosive alternatives. We assume these conditions hold, in analogy with the determinacy conditions in rational-expectations models (existence and uniqueness of a bounded solution). ■

Substituting equation (87) into equation (83) and rearranging yields

$$\frac{d\mathbf{x}_t}{d\boldsymbol{\varepsilon}_{xt+h}} = \left( I - \Lambda_0 - \sum_{s=1}^{\infty} [\Lambda_s R^s] \right)^{-1} \Omega_{xh}^{PE} \bar{\tau}_{xh} \quad (91)$$

Applying the same steps to a shock to information about state variables yields

$$\frac{d\mathbf{x}_t}{d\boldsymbol{\varepsilon}_{wt+h}} = \left( I - \Lambda_0 - \sum_{s=1}^{\infty} [\Lambda_s R^s] \right)^{-1} \Omega_{wh}^{PE} \bar{\tau}_{wh} \quad (92)$$

Defining the  $N_x \times N_z$  matrix  $\Omega^{PE} = (\Omega_{x0}^{PE}, \Omega_{w0}^{PE}, \Omega_{x1}^{PE}, \Omega_{w1}^{PE}, \dots)$ , we can combine these into one expression:

$$\frac{d\mathbf{x}_t}{d\boldsymbol{\varepsilon}_t} = \Lambda^{GE} \Omega^{PE} \bar{\tau} \quad (93)$$



where

$$\Lambda^{GE} \equiv \left( I - \Lambda_0 - \sum_{s=1}^{\infty} [\Lambda_s R^s] \right)^{-1} \quad (94)$$

To extract the causal effect of the  $j$ th element of  $\varepsilon_t$  on the  $k$ th aggregate action ( $x_{kt}$ ), we simply take the  $k, j$ th element of the matrix  $\frac{d\mathbf{x}_t}{d\varepsilon_t}$ . Since  $\bar{\tau}$  is diagonal, this implies

$$\frac{dx_{kt}}{d\varepsilon_{jt}} = [\Lambda^{GE} \Omega^{PE}]_{k,j} \bar{\tau}_j. \quad (95)$$

All that remains is to scale this so the impact on the relevant aggregate expectation is equal to 1. The  $j$ th row of equation (9) is

$$E_{it}(z_{jt}) = \sum_{n=0}^{N_z} \chi_{ijn} \tau_{in} (z_{nt} + \tilde{\varepsilon}_{int} + \varepsilon_{nt}) \quad (96)$$

Aggregating across agents and differentiating with respect to  $\varepsilon_{jt}$  we find

$$\frac{d\bar{E}_t(z_{jt})}{d\varepsilon_{jt}} = \bar{\tau}_j (1 + \Gamma_j) \quad (97)$$

where

$$\Gamma_j \equiv \frac{1}{\bar{\tau}_j} \sum_{n=0}^{N_z} [\bar{\chi}_{jn} \bar{\tau}_n + Cov_I[\chi_{ijn}, \tau_{in}]] \frac{dz_{nt}}{d\varepsilon_{jt}} \quad (98)$$

$$= \sum_{n=0}^{N_z} [\bar{\chi}_{jn} \bar{\tau}_n + Cov_I[\chi_{ijn}, \tau_{in}]] \cdot [P \Lambda^{GE} \Omega^{PE}]_{n,j} \quad (99)$$

where

$$P \equiv [I, -D^{-1}C, R, -D^{-1}CR, R^2, -D^{-1}CR^2, \dots]'. \quad (100)$$

The second equality giving rise to equation (99) comes from using equations (93) and (80) to obtain  $\frac{dw_t}{d\varepsilon_{jt}}$  in terms of  $\frac{d\mathbf{x}_t}{d\varepsilon_{jt}}$ , and using this to substitute out for  $\frac{dz_{nt}}{d\varepsilon_{jt}}$ . Since  $\rho(R) < 1$  by the assumption in Lemma 1, the powers  $R^s$  converge to zero and the matrix  $P$  is well-defined.

To ensure a unit change in  $\bar{E}_t(z_{jt})$ , the un-scaled effect  $\frac{dx_{kt}}{d\varepsilon_{jt}}$  must be multiplied by  $K = (\bar{\tau}_j(1 + \Gamma_j))^{-1}$ . Combined with equation (95) this implies equation (30).

**Corollary 1** *If there are no signals about any variables realized after the current period, the general equilibrium feedback from an exogenous change in expectations of a contemporaneous*

variable is determined by the matrix

$$\Lambda^{GE} = (I - \Omega_{x0}^{PE} \bar{\tau}_{x0} + \Omega_{w0}^{PE} \bar{\tau}_{w0} D^{-1} C)^{-1} \quad (101)$$

where  $\Omega_{x0}^{PE}$  and  $\Omega_{w0}^{PE}$  collect the partial-equilibrium effects of each expectation  $\bar{E}_t(z_{jt})$  for  $z_{jt} \in \{\mathbf{x}_t, \mathbf{w}_t\}$ , and  $\bar{\tau}_{x0}, \bar{\tau}_{w0}$  are matrices collecting the corresponding elements of average  $\tau_i$ .

**Proof.** If there is no forward information, then  $\bar{\tau}_{xs}$  and  $\bar{\tau}_{ws}$  are zero matrices for all  $s \geq 1$ . Equation (82) therefore implies  $\Lambda_s$  is a zero matrix for all  $s \geq 1$ . Equation (94) then becomes

$$\Lambda^{GE} = (I - \Lambda_0)^{-1} = (I - \Omega_{x0}^{PE} \bar{\tau}_{x0} + \Omega_{w0}^{PE} \bar{\tau}_{w0} D^{-1} C)^{-1} \quad (102)$$

where the second equality uses equation (82). ■

From this, we can see that an exogenous change in  $\bar{E}_t(z_{jt})$  causes two related general equilibrium feedback effects. First, as in the Calvo model in Section 2, the shock causes a change in aggregate actions  $\mathbf{x}_t$ , which feeds back into the information agents receive.

Second, this change in  $\mathbf{x}_t$  also affects the state variables  $\mathbf{w}_t$  through the equilibrium condition (25), which *also* affects agent information. This was not present in Section 2 because we abstracted there from all responses of the state (marginal costs) to firm actions.

Finally, note that outside of the special case in Corollary 1, there is a further category of effects, as changes in  $\bar{E}_t(z_{jt})$  affect future actions and states through the lagged-action term  $A_i \mathbf{x}_{it-1}$  in the policy function (27). If agents get signals about future aggregate actions or states, those signals will also therefore react to the change in  $\bar{E}_t(z_{jt})$ .

**The role of expectations in the transmission of fundamental shocks.** The causal effect of an exogenous shock  $\xi_t$  on aggregate choices is given by Proposition 10, which follows Section 2.3.

**Proposition 10** *The causal effect of a fundamental shock  $\xi_t$  on aggregate choices is*

$$\frac{d\mathbf{x}_t}{d\xi_t} = -\Lambda^{GE} \Omega_{w0}^{PE} \bar{\tau}_{w0} D^{-1} F \quad (103)$$

where the matrix  $\Omega_{w0}^{PE}$  collects the partial-equilibrium effects of each element of expectations about current state variables ( $\bar{E}_t(\mathbf{w}_t)$ ), and  $\bar{\tau}_{w0}$  is a diagonal matrix collecting the corresponding elements of average  $\tau_i$ .

**Proof.** Differentiating equation (81) with respect to the fundamental shock vector  $\xi_t$  yields

$$\frac{d\mathbf{x}_t}{d\xi_t} = (I - \Lambda_0)^{-1} \left( -\Omega_{w0}^{PE} \bar{\tau}_{w0} D^{-1} F + \sum_{s=1}^{\infty} \left[ \Lambda_s \frac{d\mathbf{x}_{t+s}}{d\xi_t} \right] \right) \quad (104)$$

Following analogous reasoning to that in Lemma 1 we can substitute out for  $\frac{d\mathbf{x}_{t+s}}{d\boldsymbol{\xi}_t} = R^s \frac{d\mathbf{x}_t}{d\boldsymbol{\xi}_t}$ , where  $R$  is defined in (86). Rearranging the resulting expression implies equation (103). ■

Intuitively, the general equilibrium condition (25) implies that  $-D^{-1}F$  is the response of state variables  $\mathbf{w}_t$  to a  $\boldsymbol{\xi}_t$  shock, in the absence of any  $\mathbf{x}_t$  response.  $\Omega_{w0}^{PE} \bar{\boldsymbol{\tau}}_{w0}$  then gives the partial-equilibrium effect of that  $\mathbf{w}_t$  change on  $\mathbf{x}_t$ , through changes in expectations of  $\mathbf{w}_t$ . Finally,  $\Lambda^{GE}$  accounts for general equilibrium feedback through both  $\mathbf{w}_t$  and  $\mathbf{x}_t$ .

Next, construct a counterfactual transmission in which there is no response of  $E_{it}(z_{jt})$  for some set of variables defined by  $j \in J$ . To do this, replace  $\tau_{ij} = \chi_{ijk} = 0$  for all  $i, k$ , and for all  $j \in J$ , then re-compute  $\Lambda^{GE}, \Omega_{w0}^{PE}, \boldsymbol{\tau}_{w0}$ . The share of shock transmission due to expectations of  $z_{jt}$  for  $j \in J$  is then given by  $\left( \frac{d\mathbf{x}_t}{d\boldsymbol{\xi}_t} - \frac{d\mathbf{x}_t}{d\boldsymbol{\xi}_t} \Big|_{\sim E_{j \in J}} \right) \cdot \left( \frac{d\mathbf{x}_t}{d\boldsymbol{\xi}_t} \right)^{-1}$ .

## B Narrative heterogeneity in common models of expectation formation

I focus here on the model in Section 2 under the simplifying assumptions that  $\tau_{ij} = 0$  for all variables except  $\pi_t$ , and  $\varepsilon_{\pi_t} = 0$ ,  $\tilde{\varepsilon}_{i\pi_t} \sim N(0, \sigma_i^2)$ ,  $\pi_t \sim N(0, \sigma_\pi^2)$ . The reset price of firm  $i$  is then given by

$$p_{it}^* - P_{t-1} = \left( \sum_{s=0}^{\infty} (\beta\lambda)^s [\chi_{i\pi_{t+s}\pi_t} + (1 - \beta\lambda)\chi_{imc_{t+s}\pi_t}] \right) \tau_{i\pi_t} (\pi_t + \tilde{\varepsilon}_{i\pi_t}) \quad (105)$$

It will be helpful here to define  $\tilde{\chi}_i$  as the sum of cross-learning parameters as they enter the reset price first order condition:

$$\tilde{\chi}_i \equiv \sum_{s=0}^{\infty} (\beta\lambda)^s [\chi_{i\pi_{t+s}\pi_t} + (1 - \beta\lambda)\chi_{imc_{t+s}\pi_t}] \quad (106)$$

I also assume that firms incorporate their signals into expectations of  $\pi_t$  optimally. Standard Bayesian updating therefore implies

$$\tau_{i\pi_t} = \frac{\sigma_\pi^2}{\sigma_\pi^2 + \sigma_i^2} \quad (107)$$

**Rational inattention.** Following Sims (2003), models of rational inattention assume that agents choose the precision of their signals, subject to a cost (see Maćkowiak et al., 2023, for a review).

To study this in our Calvo model, we make two assumptions. First, a resetting firm  $i$ 's expected payoff is proportional to  $-E_{it-1}(p_{it}^* - p_{it}^{FI})^2$ , where  $p_{it}^{FI}$  is the reset price that they would choose if

they precisely observed  $\pi_t$ .<sup>32</sup> Second, before the signal  $s_{it}(\pi_t)$  is observed, each firm can choose the variance of  $\tilde{\varepsilon}_{i\pi_t}$ , subject to a cost that is proportional to the expected entropy reduction between their prior and the resulting posterior beliefs about  $\pi_t$ .<sup>33</sup>

Since there is a one-to-one mapping between  $\sigma_i^2$  and  $\tau_{i\pi_t}$  (equation (107)), we can formulate the information-choice problem as

$$\arg \max_{\tau_{i\pi_t} \in [0,1]} -E_{it-1}(p_{it}^* - p_{it}^{FI})^2 - \mu \log \left( \frac{1}{1 - \tau_{i\pi_t}} \right) \quad (108)$$

where  $\mu > 0$  is the marginal cost of information.

Substituting out for  $p_{it}^*$  and  $p_{it}^{FI}$  using (105) and evaluating the expectations, the problem becomes

$$\arg \max_{\tau_{i\pi_t} \in [0,1]} -\tilde{\chi}_i^2 ((\tau_{i\pi_t} - 1)^2 \sigma_\pi^2 + \tau_{i\pi_t}^2 \sigma_i^2) - \mu \log \left( \frac{1}{1 - \tau_{i\pi_t}} \right) \quad (109)$$

which after substituting out for  $\sigma_i^2$  using (107) and differentiating gives the first order condition:

$$\tau_{i\pi_t} = \max \left\{ 0, 1 - \frac{\mu}{\sigma_\pi^2 \tilde{\chi}_i^2} \right\} \quad (110)$$

This clearly generates correlations between  $\tau_{i\pi_t}$  and  $\chi_{i\pi_t+s\pi_t}$ ,  $\chi_{imc_t+s\pi_t}$ , as long as some firms choose to process strictly positive amounts of information ( $\tau_{i\pi_t} > 0$ ). Intuitively, firms who believe there is a stronger pass-through from current inflation to future inflation and to marginal costs (larger  $\tilde{\chi}_i^2 = [\sum_{s=0}^{\infty} (\beta\lambda)^s [\chi_{i\pi_t+s\pi_t} + (1 - \beta\lambda)\chi_{imc_t+s\pi_t}]]^2$ ) find information more valuable, so process more of it – implying a  $\tau_{i\pi_t}$  closer to 1.

**Learning.** Models of rational inattention endogenize  $\tau_i$ . In contrast, models of learning make subjective models endogenous: agents estimate the coefficients of aggregate laws of motion from their observations (Evans and McGough, 2020). This also generates a narrative heterogeneity effect, because agents learning from systematically different information estimate different laws of motion.

To incorporate this in a simple way into the Calvo framework used here, assume that all firms have perceived laws of motion that specify there is no link between  $\pi_t$  and future  $\pi_{t+s}$  or  $mc_{t+s}$  for  $s \geq 1$ , that  $\pi_t \sim N(\mu_\pi, \sigma_\pi^2)$ , and that current  $mc_t$  is perfectly correlated with  $\pi_t$ :

$$\pi_t = \gamma mc_t \quad (111)$$

These assumptions imply that  $\tilde{\chi}_i$  is equal to the firm's belief about the parameter  $\gamma$ .

<sup>32</sup>An objective of this form can be obtained by taking a log-quadratic approximation to the profit function (Maćkowiak and Wiederholt, 2009).

<sup>33</sup>Since the firm's objective function is quadratic, and priors are Gaussian, the signal structure in equation (15) without public noise is in fact optimal (Maćkowiak and Wiederholt, 2009).

Furthermore, suppose that at the end of period  $t$  (i.e. after resetting firms have chosen  $p_{it}$ ) firms observe the realized  $mc_t$ . They can therefore use the combination of  $mc_t$  and  $s_{it}(\pi_t)$  to learn about the true slope  $\gamma$ .

There are a range of possible learning technologies present in the literature. I proceed here with a tractable application of Bayesian learning (reviewed in [Baley and Veldkamp, 2022](#)). Using their observation of  $mc_t$  and their inflation signal from period  $t$ , firm  $i$  can construct an unbiased signal about  $\gamma$ :

$$\frac{s_{it}(\pi_t)}{mc_t} = \gamma + \frac{\epsilon_{i\pi_t}}{mc_t} \quad (112)$$

To keep the learning problem simple for this example, suppose that firms start period  $t$  with a Gaussian prior belief about  $\gamma$ , with mean  $\mu_{\gamma it}$  and variance  $\sigma_{\gamma it}^2$ . Learning about  $\gamma$  is then straightforward: Bayesian updating implies that posterior beliefs about  $\gamma$  at the end of period  $t$  are:

$$E(\gamma | s_{it}(\pi_t), mc_t) = \mu_{\gamma it} + K_{it} \left( \gamma + \frac{\epsilon_{i\pi_t}}{mc_t} - \mu_{\gamma it} \right) \quad (113)$$

where the gain  $K_{it}$  is given by:

$$K_{it} = \frac{mc_t^2 \sigma_{\gamma it}^2}{mc_t^2 \sigma_{\gamma it}^2 + \sigma_{\pi}^2 \left( \frac{1 - \tau_{i\pi_t}}{\tau_{i\pi_t}} \right)} \quad (114)$$

This gain is strictly increasing in  $\tau_{i\pi_t}$ . Firms with more precise information update their subjective models more strongly in response to it. At the extremes, a firm with  $\tau_{i\pi_t} = 1$  is able to exactly back out the realized  $\gamma$ , so  $K_{it} = 1$  and their posterior belief matches the true correlation between  $\pi_t$  and  $mc_t$ . A firm with  $\tau_{i\pi_t} \rightarrow 0$  has  $K_{it} \rightarrow 0$ : their signal is uninformative, so they never update their subjective model. If period- $t$  posteriors persist to influence the prior beliefs used in pricing decisions in period  $t + 1$ , this difference in learning speeds will create a narrative heterogeneity channel.

We can see the effects of this more concretely if we impose further structure. Assume that in an initial period 0, all firms share the same priors, with mean  $\mu_0$  and variance  $\sigma_{\gamma 0}^2$ . Firms are split equally between  $\tau_{i\pi_t} = \{0, 0.5, 1\}$ . Finally, assume all firms believe  $\gamma$  is constant, so prior beliefs in period 1 are centered on their period-0 posteriors.

In period 0, resetting firms choose prices before observing  $mc_t$ , so use their prior beliefs on  $\gamma_0$ .<sup>34</sup> Since there is no heterogeneity in these beliefs, the narrative heterogeneity term is equal to 0.

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<sup>34</sup>In all periods, I assume that firms make pricing decisions as if they were certain of their estimate of  $\gamma$ , following the standard ‘anticipated utility’ assumption in the learning literature ([Bullard and Suda, 2016](#)).

After  $mc_0$  is realized, firms update their subjective models using equation (113), which yields:

$$E_i(\gamma|s_{i0}(\pi_t), mc_0) = \begin{cases} \mu_0 & \text{if } \tau_{i\pi_t} = 0 \\ \mu_0 + \frac{mc_t^2 \sigma_{\gamma_0}^2}{mc_t^2 \sigma_{\gamma_0}^2 + \sigma_{\pi}^2} \left( \gamma + \frac{\epsilon_{i\pi_t}}{mc_0} - \mu_0 \right) & \text{if } \tau_{i\pi_t} = 0.5 \\ \gamma & \text{if } \tau_{i\pi_t} = 1 \end{cases} \quad (115)$$

In period 1, resetting firms choose their price using  $\tilde{\chi}_{i1} = E_i(\gamma|s_{i0}(\pi_t), mc_0)$  as their subjective model. The narrative heterogeneity channel is then:

$$Cov_I[\tilde{\chi}_{i1}, \tau_{i\pi_t}] = \frac{1}{6}(\gamma - \mu_0) \quad (116)$$

If prior beliefs in period 0 are below the true  $\gamma$ , then in period 1 the narrative heterogeneity channel is positive. Firms with precise information update their beliefs about  $\gamma$  upwards, while those with imprecise information do not, so more precise information is associated with higher  $\tilde{\chi}_{i1}$ . If period-0 priors are above the true  $\gamma$ , the narrative heterogeneity channel has the opposite sign. The only way for the narrative heterogeneity channel to disappear is if period-0 priors are centered on  $\gamma$ .

**Sticky information.** Following [Mankiw and Reis \(2002\)](#), firms observe information only sporadically. With probability  $q_i \in (0, 1)$ , firm  $i$  observes  $\pi_t$  precisely, which through equation (107) implies  $\tau_{i\pi_t} = 1$ . With probability  $1 - q_i$ , firm  $i$  observes no information at all ( $\tau_i = 0$ ). As in [Reis \(2006\)](#) and others, firms can pay a cost to increase  $q_i$ .<sup>35</sup>

Applying the same steps as used to obtain equation (109), expected utility net of the increasing, convex cost of increasing  $q_i$  ( $C(q_i)$ ) is:

$$-\tilde{\chi}_i^2 \sigma_{\pi}^2 (1 - q_i) - C(q_i) \quad (117)$$

The firm chooses  $q_i$  to maximize this objective. The optimal  $q_i$  satisfies the first order condition:

$$\tilde{\chi}_i^2 \sigma_{\pi}^2 = C'(q_i) \quad (118)$$

which implies  $q_i$  is strictly increasing in  $\tilde{\chi}_i^2 = [\sum_{s=0}^{\infty} (\beta\lambda)^s [\chi_{i\pi_t+s\pi_t} + (1 - \beta\lambda)\chi_{imc_t+s\pi_t}]]^2$ . The intuition is the same as the rational inattention model: information is more valuable to firms who believe that current inflation has larger effects on future inflation and marginal costs.

<sup>35</sup>Specifically, [Reis \(2006\)](#) assumes firms can pay a cost to observe information, which is equivalent to paying to increase  $q_i$  to 1.

By the law of iterated expectations, each narrative heterogeneity covariance can be expressed as:

$$Cov_I(\chi_{ij\pi_t}, \tau_{i\pi_t}) = E(E(\chi_{ij\pi_t} \tau_{i\pi_t} | q_i)) - \bar{\chi}_{j\pi_t} E(E(\tau_{i\pi_t} | q_i)) \quad (119)$$

$$= E(\chi_{ij\pi_t} q_i) - \bar{\chi}_{j\pi_t} E(q_i) \quad (120)$$

$$= Cov_I(\chi_{ij\pi_t}, q_i) \quad (121)$$

That is, the narrative heterogeneity channel appears whenever the probability of observing information is correlated with a firm's subjective model. Equation (118) implies that this is indeed the case. In the case where  $\chi_{ij\pi_t} > 0$  for all  $i$ , then  $Cov_I(\chi_{ij\pi_t}, q_i) > 0$ .

**Information delegation to media outlets.** As in [Nimark \(2014\)](#), [Chahrour et al. \(2021\)](#), and others, assume that firms receive their signals from media outlets, who perform an editorial role. Media outlet  $m$  observes  $\pi_t$  precisely, and chooses whether or not to report it to their readers depending on its realization.

As is common in the literature, I focus on editorial decision rules in which large shocks are deemed more newsworthy. Media outlet  $m$  reports on  $\pi_t$  if  $|\pi_t| > \theta_m$ , where  $\theta_m > 0$  is an outlet-specific threshold. If outlet  $m$  reports on  $\pi_t$ , any firm that reads outlet  $m$  observes  $\pi_t$  precisely ( $\tau_{i\pi_t} = 1$ ). Using the same quadratic objective as in the rational inattention case, the firm has a payoff of 0. If  $m$  does not report  $\pi_t$ , readers do not observe any information about  $\pi_t$  ( $\tau_{i\pi_t} = 0$ ). Instead of  $\pi_t$ , the firms read about a non-economic news story, from which they gain utility  $x$ . Each firm only reads the news from one outlet. The expected utility from reading outlet  $m$  is:

$$E_{imt-1} U_{it} = \Pr(|\pi_t| > \theta_m) \cdot 0 + \Pr(|\pi_t| < \theta_m) \cdot (x - \tilde{\chi}_i^2 Var(\pi_t | |\pi_t| < \theta_m)) \quad (122)$$

Importantly, the variance of  $\pi_t$  conditional on  $\pi_t$  not being reported is not equal to the unconditional variance  $\sigma_\pi^2$ . This is the key insight of [Nimark \(2014\)](#): the fact that  $\pi_t$  is not reported allows firms to rule out extreme realizations of  $\pi_t$ .

As  $\theta_m$  rises,  $\Pr(|\pi_t| < \theta_m)$  rises, but  $Var(\pi_t | |\pi_t| < \theta_m)$  also rises as fewer extreme realizations can be ruled out. This means that as  $\pi_t$  becomes less likely to be reported, the expected utility costs of it going unreported also rise. These costs are proportional to  $\tilde{\chi}_i^2$ , so firms with larger cross-learning parameters prefer  $\pi_t$  to be reported more frequently. This is the key mechanism that generates a narrative heterogeneity effect, as firms with greater  $\chi_{ij\pi_t}$  parameters demand news with a lower threshold  $\theta_m$ .

The exact value of the narrative heterogeneity channel depends on the nature of the media market. For simplicity, here I follow [Nimark and Pitschner \(2019\)](#) and assume that there is a continuum of media outlets, one for each firm, and that each outlet maximizes the expected utility

of their associated firm. The optimal  $\theta_i$  for firm  $i$  is given by the first order condition

$$\frac{x}{\tilde{\chi}_i^2} - Var(\pi_t | |\pi_t| < \theta_i) = Pr(|\pi_t| < \theta_i) \cdot \left( \frac{\partial Var(\pi_t | |\pi_t| < \theta_i)}{\partial \theta_i} \right) \left( \frac{\partial Pr(|\pi_t| < \theta_i)}{\partial \theta_i} \right)^{-1} \quad (123)$$

which for any symmetric distribution of  $\pi_t$  straightforwardly implies that  $\theta_i$  is decreasing in  $\tilde{\chi}_i^2$ .

To see this, note that for any distribution of  $\pi_t$  that is symmetric about zero,  $E(\pi_t | |\pi_t| < \theta) = 0$ , and so

$$Var(\pi_t | |\pi_t| < \theta_i) = \frac{\int_{-\theta_i}^{\theta_i} \pi^2 f(\pi) d\pi}{Pr(|\pi_t| < \theta_i)} \quad (124)$$

where  $f$  is the density of  $\pi_t$ . By symmetry,  $\frac{\partial Pr(|\pi_t| < \theta_i)}{\partial \theta_i} = 2f(\theta_i)$ . Differentiating the conditional variance with respect to  $\theta_i$  gives

$$\frac{\partial Var(\pi_t | |\pi_t| < \theta_i)}{\partial \theta_i} = \frac{2f(\theta_i)}{Pr(|\pi_t| < \theta_i)} \left[ \theta_i^2 - Var(\pi_t | |\pi_t| < \theta_i) \right] \quad (125)$$

Putting these together, we obtain

$$Pr(|\pi_t| < \theta_i) \cdot \left( \frac{\partial Var(\pi_t | |\pi_t| < \theta_i)}{\partial \theta_i} \right) \left( \frac{\partial Pr(|\pi_t| < \theta_i)}{\partial \theta_i} \right)^{-1} = \theta_i^2 - Var(\pi_t | |\pi_t| < \theta_i) \quad (126)$$

Substituting into equation (123), the conditional variance terms cancel, to give

$$\theta_i = \sqrt{\frac{x}{\tilde{\chi}_i^2}} \quad (127)$$

which is decreasing in  $\tilde{\chi}_i^2$ .

For any given realization of  $\pi_t$ , we therefore have  $\tau_{i\pi_t} = 1$  if  $\tilde{\chi}_i^2$  is above a certain threshold, and  $\tau_{i\pi_t} = 0$  if not. This generates a narrative heterogeneity effect because information is therefore concentrated among those with large cross-learning parameters, as in the rational inattention and sticky information models. Unlike those models, the extent of this covariance is now state-dependent: at  $\pi_t$  close to 0, almost no firms receive news about it, so there is little heterogeneity in information and the covariance is small. At very large (positive or negative) realizations of  $\pi_t$ , almost all firms receive the news, and again there is little heterogeneity. The narrative heterogeneity effect is therefore strongest for intermediate realizations of  $\pi_t$ , when news coverage is more divided.

$$Cov_I[\tilde{\chi}_i, \tau_{i\pi_t}] = [E_I(\tilde{\chi}_i | \tilde{\chi}_i^2 > \chi^*(\pi_t)) - \tilde{\chi}] \cdot Pr(\tilde{\chi}_i^2 > \chi^*(\pi_t)) \quad (128)$$

where  $\chi^*(\pi_t)$  is the threshold such that firm  $i$  receives news on  $\pi_t$  if  $\tilde{\chi}_i^2 > \chi^*(\pi_t)$ . The results above



imply that  $\chi^*(\pi_t)$  is decreasing in  $|\pi_t|$ . For small  $|\pi_t|$ , the first component is close to 0. For large  $|\pi_t|$ , the second component is close to 0.

This analysis abstracts from two further channels through which media provision of information may affect the joint distribution of information and subjective models. First, I considered only news reporting functions which were symmetric about  $\pi_t = 0$ . This implied that  $E_{it}\pi_t = 0$  whenever the firm's media outlet did not report  $\pi_t$ . If there were a bias (e.g. if negative shocks are reported more often than positive shocks, as in [Chahrour et al. \(2021\)](#)), then a firm observing their outlet not reporting  $\pi_t$  will adjust their posterior expectation of  $\pi_t$ , as well as their posterior variance. Two firms may form different expectations even if neither of their outlets reports on  $\pi_t$ , if their outlets have different reporting functions. This creates further information heterogeneity between firms based on their media consumption. Second, recent work has suggested that media may also transmit subjective models as well as information ([Macaulay and Song, 2022](#); [Eliaz and Spiegel, 2026](#)). While not considered here, it provides a further reason why the media sector may induce correlations between information and subjective models.

**Selective recall.** Following [Bordalo et al. \(2023\)](#) and [Gennaioli et al. \(2024\)](#), firms form beliefs about the probability of future inflation by recalling previous inflation experiences. Assume that the bank of experiences that can be recalled is a finite history of inflation realizations  $\{\pi_{t-n}\}_{n=1}^N$ . The probability of recalling a given past inflation episode is determined by its similarity to the cues that the firm is exposed to. In [Gennaioli et al. \(2024\)](#), the 'numerical context' cue is the current realized inflation rate. In the model developed here, firms do not observe that current inflation within the period, but rather observe the noisy signal  $s_{it}(\pi_t)$ . I take this noisy signal as the cue, which gives the recall probability function

$$\Pr(\text{recall } \pi_{t-n} | s_{it}(\pi_t)) = \frac{S(s_{it}(\pi_t), \pi_{t-n})}{\sum_{m=1}^N S(s_{it}(\pi_t), \pi_{t-m})} \quad (129)$$

where  $S$  is a similarity function, often chosen to be exponential:

$$S(s_{it}(\pi_t), \pi_{t-n}) = \exp(-(s_{it}(\pi_t) - \pi_{t-n})^2) \quad (130)$$

Firms recall a finite number of experiences, to give them a recalled sample  $\mathcal{R}_i$ . In [Gennaioli et al. \(2024\)](#), agents form expectations of future inflation by averaging over their recalled sample. However, to fit with the framework outlined above (especially equation (7)), I instead model firms using this selectively-recalled sample to form beliefs about inflation persistence  $\chi_{i\pi_{t+1}\pi_t}$ . In this case, the perceived inflation persistence of firm  $i$  is the estimated first-order autocorrelation of

inflation over the sample of inflation episodes in  $\mathcal{R}_i$ .

$$\chi_{i\pi_{t+1}\pi_t} = \frac{\sum_{t-n \in \mathcal{R}_i} \pi_{t-n+1} \pi_{t-n}}{\sum_{t-n \in \mathcal{R}_i} \pi_{t-n}^2} \quad (131)$$

where I am assuming that for every recalled episode, the firm recalls the level of inflation, and the inflation one period afterwards, so they recall how persistent inflation was in that recalled event.

This naturally raises the possibility of a narrative heterogeneity channel. Firms with precise signals on  $\pi_t$  (high  $\tau_{i\pi_t}$ ) recall past inflation episodes with inflation close to  $\pi_t$ . Firms with less precise signals (lower  $\tau_{i\pi_t}$ ) have a greater variance of observed signals, and thus greater dispersion in their recalled samples. If the persistence of inflation varies over time or with realized inflation (as found in e.g. [Cogley et al., 2010](#)), then these firms sampling from different parts of the historical inflation distribution will form systematically different  $\chi_{i\pi_{t+1}\pi_t}$ , and thus will extrapolate from signals in systematically different ways.

A simple example makes this clear. Suppose that firms only recall a single past inflation experience, and the similarity function is such that they recall the most similar past experience to their numerical cue  $s_{it}(\pi_t)$  with certainty. Furthermore, assume that the support of the history of available inflation rates is  $\pi_{t-n} \in [\pi^-, \pi^+]$ , realized  $\pi_t$  is also within this support, and the sample autocorrelation of inflation is increasing in  $\pi_{t-n}$ .

If the inflation history is dense, then the recalled inflation rate of household  $i$  is

$$\pi_{t-n_i^*} = \begin{cases} \pi^- & \text{if } s_{it}(\pi_t) \leq \pi^- \\ s_{it}(\pi_t) & \text{if } s_{it}(\pi_t) \in [\pi^-, \pi^+] \\ \pi^+ & \text{if } s_{it}(\pi_t) \geq \pi^+ \end{cases} \quad (132)$$

The average recalled inflation for firms with information precision  $\tau_{i\pi_t}$  is then

$$E(\pi_{t-n_i^*} | \tau_{i\pi_t}) = \pi_t + \sigma_\pi \sqrt{\frac{1 - \tau_{i\pi_t}}{\tau_{i\pi_t}}} \cdot [a\Phi(a) + b(1 - \Phi(b)) + \phi(a) - \phi(b)] \quad (133)$$

where  $a \equiv \frac{\pi^- - \pi_t}{\sigma_i}$ ,  $b \equiv \frac{\pi^+ - \pi_t}{\sigma_i}$ , and  $\Phi(\cdot)$ ,  $\phi(\cdot)$  are the CDF and PDF of  $N(0, 1)$ .

When  $\pi_t = \frac{\pi^- + \pi^+}{2}$ , the probabilities of hitting either boundary of the available inflation history are equal, and  $E(\pi_{t-n_i^*} | \tau_{i\pi_t}) = \pi_t$  independent of  $\tau_{i\pi_t}$ . However, when  $\pi_t > \frac{\pi^- + \pi^+}{2}$ , firms are more likely to be constrained by the highest inflation in their memory database, and so  $E(\pi_{t-n_i^*} | \tau_{i\pi_t}) < \pi_t$ . The greater the variance of signal noise (i.e. the lower  $\tau_{i\pi_t}$ ), the more firms are constrained by the limits of this database, and so the greater this under-estimation will be. Since we assumed the sample autocorrelation of inflation increased with the level of inflation, this means that  $Cov_I[\tau_{i\pi_t}, \chi_{i\pi_{t+1}\pi_t}] >$

0. In contrast, when  $\pi_t < \frac{\pi^- + \pi^+}{2}$ , the logic is reversed, and  $Cov_I[\tau_{i\pi_t}, \chi_{i\pi_{t+1}\pi_t}] < 0$ . The narrative heterogeneity channel is therefore state-dependent, as realized inflation affects firm signals, and thus the distribution of recalled experiences.

**Data generated through production.** Farboodi and Veldkamp (2025) have recently introduced models in which firms generate information (data) through production. To model this in our framework, note that log-linearizing the firm-level demand equation used to derive the Calvo first order condition (105) implies (relative) production is a decreasing function of relative price:

$$y_{it} - Y_t = -\gamma(p_{it} - P_t) \quad (134)$$

where  $y_{it}$  and  $Y_t$  denote firm  $i$  and aggregate output respectively, and  $\gamma$  is the elasticity of substitution between firms' products.

If a firm has a low relative price in period  $t$ , they therefore produce more, and thus receive a more precise signal about inflation in the following period. As in the model with media, this introduces time-variation in signal-to-noise ratios. Specifically, I follow Farboodi and Veldkamp (2025) and assume that signal precision is linearly increasing in production (i.e. is linearly decreasing in the relative price), which implies firm  $i$ 's signal-to-noise ratio in period  $t$  is:

$$\tau_{i\pi_t, t} = \frac{\sigma_\pi^2}{\sigma_\pi^2 + \frac{\sigma_\epsilon^2}{z(p_{it-1} - P_{t-1})}}, \quad z(p_{it-1} - P_{t-1}) = \max\{\delta, z^* - (p_{it-1} - P_{t-1})\} \quad (135)$$

where  $\sigma_\epsilon^2$  is constant across firms,  $z^* > 0$  is a constant, and  $\delta > 0$  is arbitrarily close to 0. I focus here on the tractable case where firms are myopic, so resetting firms do not account for the value of future signals when choosing prices in period  $t$ , and the pricing decision is as in equation (105). The precision of a firm's signal in period  $t^* + 1$  is therefore:<sup>36</sup>

$$\sigma_\epsilon^{-2} z(p_{it^*} - P_{t^*}) = \sigma_\epsilon^{-2} \cdot \max(\delta, z^* - (p_{it^*} - P_{t^*-1}) + (P_{t^*} - P_{t^*-1})) \quad (136)$$

$$= \sigma_\epsilon^{-2} \cdot \max(\delta, z^* + (P_{t^*} - P_{t^*-1}) - \tilde{\chi}_i \tau_{i\pi_t, t^*} s_{it^*}(\pi_{t^*})) \quad (137)$$

If  $s_{it^*}(\pi_{t^*}) > 0$ , this is decreasing in  $\tilde{\chi}_i$ . If  $s_{it^*}(\pi_{t^*}) < 0$ , it is increasing. The signal-to-noise ratio  $\tau_{i\pi_t, t^*+1}$  is strictly increasing in this precision.

Consider the case where  $\pi_{t^*} > 0$ , and all  $\tilde{\chi}_i > 0$ . The majority of resetting firms will receive  $s_{it^*}(\pi_{t^*}) > 0$ , and will increase prices. Firms with larger  $\tilde{\chi}_i$  increase prices by more than those with small  $\tilde{\chi}_i$ . As a result, large  $\tilde{\chi}_i$  firms have less precise information in period  $t^* + 1$  than those with

<sup>36</sup>In the second equality I substitute out for the realized price  $p_{it^*}$  with the first order condition for the optimal reset price (105). This is only valid for those firms who reset their prices in period  $t^*$ , but since those are the firms who determine inflation in that period they are the relevant firms of interest here.

small  $\tilde{\chi}_i$ : there is a negative  $Cov_I[\tilde{\chi}_i, \tau_{i,t^*+1}]$ . The reverse would be true if  $\pi_{t^*} < 0$ . Overall, the narrative heterogeneity channel arises because subjective models affect pricing decisions, which in turn affects information production. As in the model with media, the narrative heterogeneity channel produced here is therefore state-dependent.

If firms were not myopic, there would also be a second channel here: firms would set prices below the within-period optimum to increase the precision of signals in the next period. As discussed extensively above, the value of precise information is larger for firms with large  $\tilde{\chi}_i^2$ , so this is also likely to induce a correlation between  $\tilde{\chi}_i$  and information precision.

**Heterogeneous price indices.** Recent literature (e.g. [Kaplan and Schulhofer-Wohl, 2017](#)) has found that households in particular face highly heterogeneous inflation rates, and so some heterogeneity in survey expectations may come because households are forecasting their own idiosyncratic inflation rates. We now adapt that idea to our simple firm model, and show that it can also generate a narrative heterogeneity channel, even if all agents have rational expectations.

Suppose that each firm  $i$  belongs to a sector  $k$ , and that each sector faces its own inflation process, which is a weighted average of past sector-specific inflation and an aggregate i.i.d. shock  $u_t \sim N(0, \sigma_u^2)$ :

$$\pi_{kt} = \rho_k \pi_{kt-1} + (1 - \rho_k) u_t \quad (138)$$

where  $\rho_k \in (0, 1)$  is the persistence of that sector's inflation process.

Firms observe their own sector's inflation at the end of each period, and receive signals of the form:

$$s_{i\pi_{kt}} = \pi_{kt} + \epsilon_{i\pi_{kt}}, \quad \epsilon_{i\pi_{kt}} \sim N(0, \sigma_\epsilon^2) \quad (139)$$

The posterior expectation of  $\pi_{kt}$  after observing this signal is:

$$E_{it}\pi_{kt} = (1 - \tau_{k\pi_t})\rho_k\pi_{kt-1} + \tau_{k\pi_t}(\pi_{kt} + \epsilon_{i\pi_{kt}}) \quad (140)$$

where

$$\tau_{k\pi_t} = \frac{(1 - \rho_k)^2 \sigma_u^2}{(1 - \rho_k)^2 \sigma_u^2 + \sigma_\epsilon^2} \quad (141)$$

From this, we see that  $\tau_k$  is strictly decreasing in  $\rho_k$  for all  $\rho_k < 1$ . This is because greater persistence in inflation means that in any given period, prior beliefs are more informative, leading to less reaction to current signals. This heterogeneity in  $\tau_{k\pi_t}$  arises even though the precision of inflation signals is assumed to be equal across sectors.

Moreover, inflation persistence also affects how firms cross-learn from current to future inflation.

It is straightforward to see that

$$E_{it}\pi_{kt+s} = \rho_k^s E_{it}\pi_{kt} \quad (142)$$

and thus that  $\chi_{i\pi_{t+s}\pi_t} = \rho_k^s$ , which is increasing in  $\rho_k$ . The covariance between  $\chi_{i\pi_{t+s}\pi_t}$  and  $\tau_{i\pi_t}$  is therefore negative. Firms in sectors with less persistent inflation react more to inflation information, but extrapolate less strongly from current inflation expectations to the future. Responsiveness to information is therefore concentrated among those who cross-learn the least from it, which is a narrative heterogeneity effect that weakens the aggregate causal effect of inflation expectations.

**Level-k thinking.** For this, we switch the setup slightly to facilitate tractability. Specifically, now assume that the only signals available to firms are on current marginal costs  $m_{ct}$ , not inflation. There is cross-learning from  $m_{ct}$  to  $\pi_t$ , but no cross-learning to longer horizons. That is,  $\chi_{i\pi_{t+s}m_{ct}} = \chi_{im_{ct+s}m_{ct}} = 0$  for all  $s \geq 1$ . As in the other cases above, assume there is no public noise in signals. In this case, the optimal price of a resetting firm is

$$p_{it}^* - P_{t-1} = [1 - \beta\lambda + \chi_{i\pi_t m_{ct}}]\tau_{im_{ct}}(m_{ct} + \varepsilon_{im_{ct}}) \quad (143)$$

This can alternatively be written as a static beauty contest

$$p_{it}^* - P_{t-1} = E_{it}\pi_t + (1 - \beta\lambda)E_{it}m_{ct} = (1 - \beta\lambda)E_{it}m_{ct} + (1 - \lambda)E_{it} \int (p_{jt}^* - P_{t-1})dj \quad (144)$$

Following [Farhi and Werning \(2019\)](#) and [Iovino and Sergeyev \(2023\)](#), a firm that reasons to level 0 observes their signal about  $m_{ct}$ , and updates their expectations of the shock accordingly, but fails to update their beliefs about the prices of others. That is, they ignore general equilibrium effects of the shock. The price choice of such a level-0 firm is therefore:

$$p_{it}^{*0} - P_{t-1} = (1 - \beta\lambda)E_{it}m_{ct} = (1 - \beta\lambda)\tau_{im_{ct}}s_{it}(m_{ct}) \quad (145)$$

A firm that reasons to level 1 realizes that others may react to the shock as well, but does not reason all the way to general equilibrium. Rather, they assume that all other firms are level 0. In this case, the optimal reset price is:

$$p_{it}^{*1} - P_{t-1} = (1 - \beta\lambda)E_{it}m_{ct} + (1 - \lambda)E_{it} \int (p_{jt}^{*0} - P_{t-1})dj \quad (146)$$

$$= (1 - \beta\lambda)[1 + (1 - \lambda)\bar{\tau}_{m_{ct}}]\tau_{im_{ct}}s_{it}(m_{ct}) \quad (147)$$

For the purpose of this example, I stop at  $k = 1$ . Letting  $\chi^k$  be the  $\chi_{i\pi_t m_{ct}}$  of a firm engaged in

$k$  levels of reasoning, we have:

$$\chi^0 = 0, \quad \chi^1 = (1 - \lambda)(1 - \beta\lambda)\bar{\tau}_{mc_t} \quad (148)$$

Level-1 reasoners have a strictly greater  $\chi_{i\pi_t mc_t}$  than level-0 reasoners. If the depth of reasoning is correlated with the precision of information  $\tau_i$ , then there will be a narrative heterogeneity effect. I now show that such a correlation arises naturally if firms are able to choose their depth of reasoning, as in [Alaoui and Penta \(2016\)](#). In the model they develop, agents decide whether to complete an extra level of reasoning by weighing up the perceived costs and benefits. Specifically, in their model, two games are said to be cognitively equivalent if they entail the same sequence of actions as reasoning develops ( $k$  increases). To fit the firm problem into this framework, denote the choice of  $\chi_{i\pi_t mc_t}$  as the firm's action, with pricing following mechanically using equation (143).  $\chi^0$  and  $\chi^1$  are independent of all firm-level variables, so framed in this way the problem facing firm  $i$  is cognitively equivalent to the one facing firm  $j$ . As a result, the costs of reasoning are the same for all firms, and the depth of reasoning only differs according to the value of doing so.

[Alaoui and Penta \(2016\)](#) propose a number of functions for the value of reasoning. I demonstrate the narrative heterogeneity channel for the general 'expected gain' formulation from their Example 2, which is a generalization of the form they use for their examples, and is given an axiomatic foundation in [Alaoui and Penta \(2022\)](#). Adapting it to the firm problem, the value of reasoning to level 1 is given by:

$$v_i^1 = \int p(\chi') [E_{it-1}U_{i,t}(BR(\chi'), \chi') - E_{it-1}U_{i,t}(\chi^0, \chi')] d\chi' \quad (149)$$

where  $p(\chi')$  are weights such that  $p(\chi') \in [0, 1]$  and  $\int p(\chi')d\chi' = 1$ ,  $U_{i,t}(\chi_i, \chi')$  is the payoff to choosing  $\chi_i$  when all other firms choose  $\chi'$ , and  $BR(\chi')$  is the  $\chi_i$  that maximizes  $E_{it-1}U_{i,t}(\chi_i, \chi')$  for a given  $\chi'$ . That is, the value of reasoning is the expected utility gain from switching from  $\chi^0$  to the best-response  $\chi_i$ , weighted by the firm's perceived probability that reasoning will reveal  $\chi_j = \chi'$ . Reasoning is undertaken before the observation of  $s_{i,t}$ , so the expectations are indexed  $t - 1$ .

If  $\chi_j = \chi'$  for all firms  $j \neq i$ , the optimal price is (from equation (144)):

$$p_{it}^* - P_{t-1} = [1 - \beta\lambda + (1 - \lambda)(1 - \beta\lambda + \chi')\bar{\tau}_{mc_t}] \tau_{imc_t} s_{it}(mc_t) \quad (150)$$

The best-response  $\chi_i$  is therefore:

$$BR(\chi') = (1 - \lambda)(1 - \beta\lambda + \chi')\bar{\tau}_{mc_t} \quad (151)$$

Following the steps used to compute indirect utility in the rational inattention case above, the

value of reasoning to level 1 is:

$$v_i^1 = (1 - \lambda)^2 \sigma_{mc}^2 \bar{\tau}_{mc_t}^2 \tau_{imc_t} \int p(\chi') (1 - \beta\lambda + \chi')^2 d\chi' \quad (152)$$

The value of reasoning is therefore linearly increasing in  $\tau_{imc_t}$ , whatever perceptions  $p(\chi')$  the firm holds. If we further assume that all firms use the same  $p(\chi')$  when evaluating the benefits of reasoning (as in [Alaoui and Penta, 2016](#)), then there exists a threshold  $\tau^*$  above which firms decide to reason to level 1, and below which they do not. This implies that high  $\tau_{imc_t}$  is associated with high  $\chi_{i\pi_t mc_t}$ , so the narrative heterogeneity channel is positive. Formally:

$$Cov_I[\chi_{i\pi_t mc_t}, \tau_{imc_t}] = q\chi^0\tau^- + (1 - q)\chi^1\tau^+ - \bar{\tau}(q\chi^0 + (1 - q)\chi^1) \quad (153)$$

where  $q$  is the proportion with  $\tau_{imc_t} < \tau^*$ , and  $\tau^-$ ,  $\tau^+$  are the average  $\tau_{imc_t}$  conditional on  $\tau_{imc_t} < \tau^*$  and  $\tau_{imc_t} > \tau^*$  respectively. Simplifying, this becomes:

$$Cov_I[\chi_{i\pi_t mc_t}, \tau_{imc_t}] = -q(1 - q)(\tau^+ - \tau^-)(\chi^0 - \chi^1) > 0 \quad (154)$$

Intuitively, firms with more precise information on shocks have more to gain from understanding how to use that information, and so they will choose a higher  $k$  than firms with less information. Information is therefore concentrated among firms with large  $\chi_{i\pi_t mc_t}$ .

## C Mapping RCT estimates to the Calvo model: details

### C.1 Proof of Proposition 8

Denote the treated variable as  $z_n$ , and order it last in the vector  $z$ . Since  $M_i$  is upper-triangular with zero diagonal, the last row of  $M_i$  is identically zero, meaning there is no cross-learning to  $z_n$  from other variables. The prior for  $z_n$  therefore depends only on the agent's pre-experiment signal:

$$E_{it}^{prior}(z_n) = \tau_{in} s_{it}(z_n) \quad (155)$$

From equation (33), the posterior for a treated agent is:

$$E_{it}^{post}(z_n) = \tau_{in}^{RCT} s^{RCT} + (1 - \tau_{in}^{RCT}) E_{it}^{prior}(z_n) \quad (156)$$

so the revision in expectations of  $z_n$  is:

$$\Delta_{in} = E_{it}^{post}(z_n) - E_{it}^{prior}(z_n) = \tau_{in}^{RCT} (s^{RCT} - E_{it}^{prior}(z_n)) \quad (157)$$

For any other variable  $z_j$  with  $j < n$ , the expectation is determined by the system:

$$E_{it}(z_j) = \tau_{ij}s_{it}(z_j) + (1 - \tau_{ij}) \sum_{k \neq j} M_{ijk} E_{it}(z_k) \quad (158)$$

which follows from combining equations (7) and (8). When moving from pre-experiment priors to posteriors, the only change is that  $E_{it}^{prior}(z_n)$  in the prior is replaced by  $E_{it}^{post}(z_n)$ . This implies:

$$\Delta_{ij} = \chi_{ijn} \Delta_{in} = \chi_{ijn} \tau_{in}^{RCT} (s^{RCT} - E_{it}^{prior}(z_n)) \quad (159)$$

where  $\chi_{ijn}$  is the  $(j, n)$  element of the cross-learning matrix  $\chi_i = (I - (I - \tau_i)M_i)^{-1}$  from the prior system (equation (10)).

To verify (159), partition  $\chi_i^{-1} = I - (I - \tau_i)M_i$  into blocks corresponding to the first  $n - 1$  variables and  $z_n$ :

$$\chi_i^{-1} = \begin{pmatrix} I_{n-1} - (I_{n-1} - \tau_{i,<n})M_{i,<n,<n} & -(I_{n-1} - \tau_{i,<n})M_{i,<n,n} \\ \mathbf{0}' & 1 \end{pmatrix} \quad (160)$$

Where  $\tau_{i,<n}$  represents the upper-left  $(n - 1) \times (n - 1)$  block of the diagonal matrix  $\tau_i$ , and  $M_{i,<n,<n}$  is the corresponding block of  $M_i$ . The last row is  $(\mathbf{0}', 1)$  because the last row of  $M_i$  contains only zeros. Since  $\chi_i^{-1}$  is block-triangular, inverting it yields:

$$\chi_i = \begin{pmatrix} (I_{n-1} - (I_{n-1} - \tau_{i,<n})M_{i,<n,<n})^{-1} & (I_{n-1} - (I_{n-1} - \tau_{i,<n})M_{i,<n,<n})^{-1}(I_{n-1} - \tau_{i,<n})M_{i,<n,n} \\ \mathbf{0}' & 1 \end{pmatrix} \quad (161)$$

Now, subtracting the prior from the posterior version of (158) for all  $j < n$  simultaneously:

$$\Delta_{i,<n} = (I_{n-1} - \tau_{i,<n})M_{i,<n,<n}\Delta_{i,<n} + (I_{n-1} - \tau_{i,<n})M_{i,<n,n}\Delta_{in} \quad (162)$$

Solving:

$$\Delta_{i,<n} = (I_{n-1} - (I_{n-1} - \tau_{i,<n})M_{i,<n,<n})^{-1}(I_{n-1} - \tau_{i,<n})M_{i,<n,n}\Delta_{in} \quad (163)$$

Comparing with (161), the coefficient on  $\Delta_{in}$  is precisely the last column of  $\chi_i$  (excluding the bottom entry of 1), confirming  $\Delta_{ij} = \chi_{ijn}\Delta_{in}$  for all  $j < n$ .

**Normalization.** The ‘everyday’ signal on  $z_n$  received before the experiment has the form in equation (15):  $s_{it}(z_n) = z_n + \varepsilon_{nt} + \tilde{\varepsilon}_{int}$ , where  $\varepsilon_{nt}$  is public noise (common across agents) and  $\tilde{\varepsilon}_{int}$  is idiosyncratic noise, mean zero, with variance  $\sigma_i^2$  that may vary across agents. The experimental



signal takes the form  $s^{RCT} = z_n + \varepsilon^{RCT}$ , where  $\varepsilon^{RCT}$  is independent of  $\varepsilon_{nt}$ . Since the linearization point of the model has not been specified, we normalize  $z_n + \varepsilon_{nt} = 0$ . This is equivalent to linearizing around the average prior belief about the treated variable  $z_n$  as long as at least one agent has  $\tau_{in} > 0$ .<sup>37</sup> Since the model is linear, this normalization does not affect any equilibrium objects. Under this normalization:

$$E_{it}^{prior}(z_n) = \tau_{in}\tilde{\varepsilon}_{int}, \quad s^{RCT} = \varepsilon^{RCT} - \varepsilon_{nt} \neq 0 \quad (164)$$

and the revisions become:

$$\Delta_{in} = \tau_{in}^{RCT}(\varepsilon^{RCT} - \varepsilon_{nt} - \tau_{in}\tilde{\varepsilon}_{int}) \quad (165)$$

$$\Delta_{ij} = \chi_{ijn}\tau_{in}^{RCT}(\varepsilon^{RCT} - \varepsilon_{nt} - \tau_{in}\tilde{\varepsilon}_{int}) \quad (166)$$

Note that  $\tilde{\varepsilon}_{int}$  is mean zero and independent of all agent characteristics  $(\chi_{ijn}, \tau_{in}, \tau_{in}^{RCT}, \sigma_i^2)$  by assumption.

**Part (i): Wald estimand.** Using only  $D_{ij}$  as the excluded instrument, with  $E_{it}^{prior}(z_n)$  as an included exogenous control, the IV coefficient is the Wald estimand. By random assignment,  $D_{ij}$  is independent of all agent characteristics and priors, so:

$$\gamma_{1,j}^{Wald} = \frac{E_I[E_{it}^{post}(z_j) | D_{ij} = 1] - E_I[E_{it}^{post}(z_j) | D_{ij} = 0]}{E_I[E_{it}^{post}(z_n) | D_{ij} = 1] - E_I[E_{it}^{post}(z_n) | D_{ij} = 0]} = \frac{E_I[\Delta_{ij}]}{E_I[\Delta_{in}]} \quad (167)$$

where the  $E_{it}^{prior}(z_j)$  and  $E_{it}^{prior}(z_n)$  terms cancel because they are identically distributed in treatment and control by random assignment.

Evaluating the numerator and denominator using (166) and (165) respectively, the terms in  $\varepsilon^{RCT} - \varepsilon_{nt} - \tau_{in}\tilde{\varepsilon}_{int}$  cancel, giving:

$$\gamma_{1,j}^{Wald} = \frac{\int_i \chi_{ijn}\tau_{in}^{RCT} di}{\bar{\tau}_n^{RCT}} = \int_i \chi_{ijn} \frac{\tau_{in}^{RCT}}{\bar{\tau}_n^{RCT}} di \quad (168)$$

Setting  $j = Y$  and  $n = \pi_{t+h}$  gives equation (34).

**Remark on normalization.** The derivation of (168) relies on the normalization  $z_n + \varepsilon_{nt} = 0$ . As discussed above, this is equivalent to taking the linearization of the model about the average prior belief on  $z_n$ . In other words,  $\gamma_{1,j}^{Wald}$  is helpful in identifying the causal effect of shocks to

<sup>37</sup>Before the normalization, the average prior is  $\bar{E}_t^{prior}(z_n) = \bar{\tau}_n(z_n + \varepsilon_{nt})$ . Since  $\tau_{in} \in [0, 1]$ , if one agent has  $\tau_{in} > 0$ , then  $\bar{\tau}_n > 0$ , and  $\bar{E}_t^{prior}(z_n) = 0$  (i.e. average priors are at the linearization point) requires  $z_n + \varepsilon_{nt} = 0$ .

expectations, for shocks arriving when prior beliefs are in steady state. Without this normalization, equation (168) becomes:

$$\gamma_{1,j}^{Wald} = \frac{\left(\int_i \chi_{ijn} \tau_{in}^{RCT} di\right) (s^{RCT} - \bar{E}_t^{prior}(z_n)) - \bar{\tau}_n^{-1} \bar{E}_t^{prior}(z_n) Cov_I [\chi_{ijn} \tau_{in}^{RCT}, \tau_{in}]}{\bar{\tau}_n^{RCT} (s^{RCT} - \bar{E}_t^{prior}(z_n)) - \bar{\tau}_n^{-1} \bar{E}_t^{prior}(z_n) Cov_I [\tau_{in}^{RCT}, \tau_{in}]} \quad (169)$$

In the particular exercises conducted in Section 4.2, we only require this normalization for expectations (and signals) of  $\pi_{t+1}$ , even for the general equilibrium exercises where responses to  $\pi_t$  enter the calculations. This is because we assume  $\tau_{i\pi_t} = 1$  for all agents, and thus the covariance terms in (169) equal 0 without further normalization. However, if this assumption is relaxed, then interpreting RCT estimates in general equilibrium would require assuming that the shock hits when priors on all treated variables (in this case both  $\pi_{t+1}$  and  $\pi_t$ ) are at steady state.

An alternative approach could be to relax this normalization using other information available from the RCT. We can, for instance, observe  $\bar{\tau}_n^{RCT}$  from equation (31), and  $s^{RCT} - \bar{E}_t^{prior}(z_n)$  from the priors stated by survey participants before any information treatment (and the treatment design). Theoretical work such as Maćkowiak and Wiederholt (2024) could provide discipline for  $Cov_I[\tau_{in}^{RCT}, \tau_{in}]$ . Using these, and perhaps some changes to RCT design, it may be possible to infer  $\frac{\int_i \chi_{ijn} \tau_{in}^{RCT} di}{\bar{\tau}_n^{RCT}}$  without the normalization imposed here. This could be a fruitful avenue for future research, especially for interpreting RCTs aimed specifically at comparing between different baseline environments, such as e.g. Weber et al. (2025).

**Part (ii): 2SLS with both instruments.** Using both  $D_{ij}$  and  $D_{ij} \cdot E_{it}^{prior}(z_n)$  as excluded instruments, with  $E_{it}^{prior}(z_n)$  included as an exogenous control, the 2SLS coefficient is computed (by Frisch–Waugh–Lovell) after partialling out the included exogenous variables ( $1, E_{it}^{prior}(z_n)$ ) from all quantities.

To simplify notation, define  $p_i \equiv E_{it}^{prior}(z_n) = \tau_{in} \tilde{\varepsilon}_{int}$  and  $\delta_i \equiv \Delta_{in} = \tau_{in}^{RCT} (s^{RCT} - \tau_{in} \tilde{\varepsilon}_{int})$ . The first stage projects  $\tilde{X}_i$  (the residualized endogenous regressor) onto the residualized excluded instruments. Since  $D_{ij}$  is independent of  $p_i$ , the residualized fitted value takes the form:

$$\hat{\tilde{X}}_i = (D_{ij} - \bar{D}) \hat{\delta}(p_i) \quad (170)$$

where  $\hat{\delta}(p_i) = c_0 + c_1 p_i$  is the best linear predictor (BLP) of  $\delta_i$  given  $p_i$ . After partialling out  $(1, p_i)$  from the dependent variable, the only component correlated with  $D_{ij}$  is  $D_{ij} \chi_{ijn} \delta_i$  (by random assignment). The 2SLS estimand is therefore:

$$\gamma_{1,j}^{2SLS} = \frac{E_I[\hat{\delta}(p_i) \chi_{ijn} \delta_i]}{E_I[\hat{\delta}(p_i) \delta_i]} \quad (171)$$

*Computing the BLP.* The slope of the BLP is  $c_1 = \text{Cov}(\delta_i, p_i) / \text{Var}(p_i)$ . Using  $E_I[\tilde{\varepsilon}_{int}] = 0$  and the independence of  $\tilde{\varepsilon}_{int}$  from agent characteristics:

$$E_I[\delta_i p_i] = E_I[\tau_{in}^{RCT} (s^{RCT} - \tau_{in} \tilde{\varepsilon}_{int}) \tau_{in} \tilde{\varepsilon}_{int}] = -E_I[\tau_{in}^{RCT} \tau_{in}^2 \sigma_i^2] \quad (172)$$

$$E_I[p_i^2] = E_I[\tau_{in}^2 \sigma_i^2] \quad (173)$$

Since  $E_I[\delta_i] = s^{RCT} \bar{\tau}_n^{RCT}$  and  $E_I[p_i] = 0$  under the  $z_n + \varepsilon_{nt} = 0$  normalization, we have:

$$c_1 = \frac{-E_I[\tau_{in}^{RCT} \tau_{in}^2 \sigma_i^2]}{E_I[\tau_{in}^2 \sigma_i^2]}, \quad c_0 = s^{RCT} \bar{\tau}_n^{RCT} \quad (174)$$

*Evaluating the numerator and denominator.* The remaining required moments are:

$$E_I[\chi_{ijn} \delta_i] = s^{RCT} E_I[\chi_{ijn} \tau_{in}^{RCT}] \quad (175)$$

$$E_I[p_i \chi_{ijn} \delta_i] = -E_I[\chi_{ijn} \tau_{in}^{RCT} \tau_{in}^2 \sigma_i^2] \quad (176)$$

Substituting into (171):

$$\begin{aligned} E_I[\hat{\delta}(p_i) \chi_{ijn} \delta_i] &= c_0 E_I[\chi_{ijn} \delta_i] + c_1 E_I[p_i \chi_{ijn} \delta_i] \\ &= (s^{RCT})^2 \bar{\tau}_n^{RCT} E_I[\chi_{ijn} \tau_{in}^{RCT}] + \frac{E_I[\tau_{in}^{RCT} \tau_{in}^2 \sigma_i^2]}{E_I[\tau_{in}^2 \sigma_i^2]} E_I[\chi_{ijn} \tau_{in}^{RCT} \tau_{in}^2 \sigma_i^2] \end{aligned} \quad (177)$$

$$\begin{aligned} E_I[\hat{\delta}(p_i) \delta_i] &= c_0 E_I[\delta_i] + c_1 E_I[p_i \delta_i] \\ &= (s^{RCT})^2 (\bar{\tau}_n^{RCT})^2 + \frac{(E_I[\tau_{in}^{RCT} \tau_{in}^2 \sigma_i^2])^2}{E_I[\tau_{in}^2 \sigma_i^2]} \end{aligned} \quad (178)$$

Dividing (177) by (178) and rearranging:

$$\gamma_{1,j}^{2SLS} = \omega \cdot \frac{E_I[\chi_{ijn} \tau_{in}^{RCT}]}{\bar{\tau}_n^{RCT}} + (1 - \omega) \cdot \frac{E_I[\chi_{ijn} \tau_{in}^{RCT} \tau_{in}^2 \sigma_i^2]}{E_I[\tau_{in}^{RCT} \tau_{in}^2 \sigma_i^2]} \quad (179)$$

where

$$\omega = \frac{(s^{RCT})^2 (\bar{\tau}_n^{RCT})^2}{(s^{RCT})^2 (\bar{\tau}_n^{RCT})^2 + \frac{(E_I[\tau_{in}^{RCT} \tau_{in}^2 \sigma_i^2])^2}{E_I[\tau_{in}^2 \sigma_i^2]}} \in (0, 1) \quad (180)$$

The first component of (179) is the Wald estimand from equation (168). The second component is a weighted average of  $\chi_{ijn}$  with weights proportional to  $\tau_{in}^{RCT} \tau_{in}^2 \sigma_i^2$ , which arises from the interaction instrument's exploitation of the noise-driven covariance between priors and revisions.

**When are the two IV estimates close to one another?** Clearly, as  $(s^{RCT})^2 \rightarrow \infty$ ,  $\omega \rightarrow 1$  and  $\gamma_{1,j}^{2SLS} \rightarrow \gamma_{1,j}^{Wald}$ . To go further, we can write the weight  $\omega$  as

$$\omega = \frac{1}{1 + F} \quad (181)$$

where

$$F = \frac{(E_I[\tau_{in}^{RCT} \tau_{in}^2 \sigma_i^2])^2}{(s^{RCT})^2 (\bar{\tau}_n^{RCT})^2 E_I[\tau_{in}^2 \sigma_i^2]} \quad (182)$$

Since  $\tau_i^{RCT} \in [0, 1]$ , we can bound  $F$  from above with

$$F \leq \frac{(E_I[\tau_{in}^2 \sigma_i^2])^2}{(s^{RCT})^2 (\bar{\tau}_n^{RCT})^2 E_I[\tau_{in}^2 \sigma_i^2]} = \frac{E_I[\tau_{in}^2 \sigma_i^2]}{(s^{RCT})^2 (\bar{\tau}_n^{RCT})^2} \quad (183)$$

Moreover, since  $\tau_{in} \in [0, 1]$ , we can simplify the bound further, to

$$F \leq \frac{E_I[\sigma_i^2]}{(s^{RCT})^2 (\bar{\tau}_n^{RCT})^2} \quad (184)$$

Therefore as  $(s^{RCT})^2 = (\varepsilon^{RCT} - \varepsilon_{nt})^2$  becomes large relative to  $E_I[\sigma_i^2]$ ,  $F$  approaches 0, and  $\omega$  approaches 1, implying  $\gamma_{1,j}^{2SLS} \rightarrow \gamma_{1,j}^{Wald}$ . Formally:

$$\frac{(\varepsilon^{RCT} - \varepsilon_{\pi_{t+h}})^2}{E_I[\sigma_i^2]} \rightarrow \infty \implies \gamma_{1,Y}^{2SLS} \rightarrow \gamma_{1,Y}^{Wald}. \quad (185)$$

*Example calibration.* Since the application in Section 4.2 uses the approximation  $\gamma_{1,j}^{2SLS} \approx \gamma_{1,j}^{Wald}$ , it is helpful to get a sense of how close this approximation is in this context. From the estimates of equation (31), we have that  $\bar{\tau}_n^{RCT} = 0.53$ . We then have:

$$\begin{aligned} \varepsilon^{RCT} - \varepsilon_{nt} &= z_n + \varepsilon^{RCT} - z_n - \varepsilon_{nt} = s^{raw} - \frac{1}{\bar{\tau}_n^{RCT}} \bar{E}_t^{prior}(z_n) \\ &= 2.8 - \frac{1}{0.53} \times 5.83 = -8.2 \end{aligned} \quad (186)$$

where  $s^{raw}$  is the un-normalized signal given to the treatment group in the survey, calibrated to 2.8 using the survey design (i.e. the survey participants were shown a forecast that  $\pi_{t+1}$  would be 2.8%).  $\bar{E}_t^{prior}(z_n) = 5.83$  comes from the pre-treatment summary statistics.

For the bound in (183), we now just need to calibrate  $E_I[\tau_{in}^2 \sigma_i^2]$ . For this, we can use the dispersion of point forecasts: the variance of  $E_{it}^{prior}(\pi_{t+1})$  is reported in [Baumann et al. \(2026\)](#) as

2.95<sup>2</sup>. Using equation (155), this can be expressed as

$$\begin{aligned}
Var_I[E_{it}^{prior}(\pi_{t+1})] &= Var_I[\tau_{i\pi_{t+1}}(\pi_{t+1} + \varepsilon_{\pi_{t+1}} + \tilde{\varepsilon}_{i\pi_{t+1}t})] \\
&= Var_I\left[\frac{\bar{E}_t^{prior}(\pi_{t+1})}{\bar{\tau}_{\pi_{t+1}}} \tau_{i\pi_{t+1}} + \tau_{i\pi_{t+1}} \tilde{\varepsilon}_{i\pi_{t+1}t}\right] \\
&= \left(\frac{\bar{E}_t^{prior}(\pi_{t+1})}{\bar{\tau}_{\pi_{t+1}}}\right)^2 Var_I[\tau_{i\pi_{t+1}}] + Var_I[\tau_{i\pi_{t+1}} \tilde{\varepsilon}_{i\pi_{t+1}t}] \quad (187)
\end{aligned}$$

where the final equality uses  $Cov_I[\tau_{i\pi_{t+1}}, \tau_{i\pi_{t+1}} \tilde{\varepsilon}_{i\pi_{t+1}t}] = 0$  because  $\tilde{\varepsilon}_{i\pi_{t+1}t}$  is mean 0 for all  $\tau_{i\pi_{t+1}}$ .

Expanding the second term, we have

$$Var_I[\tau_{i\pi_{t+1}} \tilde{\varepsilon}_{i\pi_{t+1}t}] = E_I[\tau_{i\pi_{t+1}}^2 \tilde{\varepsilon}_{i\pi_{t+1}t}^2] - (E_I[\tau_{i\pi_{t+1}} \tilde{\varepsilon}_{i\pi_{t+1}t}])^2 = E_I[\tau_{i\pi_{t+1}}^2 \tilde{\varepsilon}_{i\pi_{t+1}t}^2] \quad (188)$$

Employing the law of iterated expectations, this becomes

$$Var_I[\tau_{i\pi_{t+1}} \tilde{\varepsilon}_{i\pi_{t+1}t}] = E_I[E_I[\tau_{i\pi_{t+1}}^2 \tilde{\varepsilon}_{i\pi_{t+1}t}^2 | \tau_{i\pi_{t+1}}]] = E_I[\tau_{i\pi_{t+1}}^2 \sigma_i^2] \quad (189)$$

While this is the object of interest we wish to calibrate, equation (187) makes it clear that we cannot do that from the variance of prior point estimates, as the cross-sectional variance of  $\tau_{i\pi_{t+1}}$  contaminates the mapping. We can, however, make further progress if we impose for this indicative calibration that agents follow Bayes' rule in forming their prior beliefs, and all uncertainties are Gaussian. In that case, we have

$$\tau_{i\pi_{t+1}} = \frac{\sigma_{\pi_{t+1}}^2}{\sigma_{\pi_{t+1}}^2 + \sigma_{\varepsilon\pi_{t+1}}^2 + \sigma_i^2} \quad (190)$$

where  $\sigma_{\pi_{t+1}}^2$  is the variance of  $\pi_{t+1}$  before any signals, and  $\sigma_{\varepsilon\pi_{t+1}}^2$  is the variance of the public noise  $\varepsilon_{\pi_{t+1}t}$ . Rearranging this we obtain

$$\sigma_i^2 = \sigma_{\pi_{t+1}}^2 \left( \frac{1 - \tau_{i\pi_{t+1}}}{\tau_{i\pi_{t+1}}} \right) - \sigma_{\varepsilon\pi_{t+1}}^2 \quad (191)$$

$$\implies \tau_{i\pi_{t+1}}^2 \sigma_i^2 = \sigma_{\pi_{t+1}}^2 \tau_{i\pi_{t+1}} (1 - \tau_{i\pi_{t+1}}) - \tau_{i\pi_{t+1}}^2 \sigma_{\varepsilon\pi_{t+1}}^2 \quad (192)$$

which is maximized, giving the most conservative bound for  $F$  in equation (183), when there is no public noise ( $\sigma_{\varepsilon\pi_{t+1}}^2 = 0$ ). In this case, we have

$$Var_I[\tau_{i\pi_{t+1}} \tilde{\varepsilon}_{i\pi_{t+1}t}] = E_I[\tau_{i\pi_{t+1}}^2 \sigma_i^2] = \sigma_{\pi_{t+1}}^2 (\bar{\tau}_{\pi_{t+1}} - \bar{\tau}_{\pi_{t+1}}^2 - Var_I[\tau_{i\pi_{t+1}}]) \quad (193)$$

where the final equality uses the definition of a variance to substitute out for  $E_I[\tau_{i\pi_{t+1}}^2]$ . Substituting this into equation (187), we obtain:

$$Var_I[E_{it}^{prior}(\pi_{t+1})] = \left( \frac{\bar{E}_t^{prior}(\pi_{t+1})}{\bar{\tau}_{\pi_{t+1}}} \right)^2 Var_I[\tau_{i\pi_{t+1}}] + \sigma_{\pi_{t+1}}^2 (\bar{\tau}_{\pi_{t+1}} - \bar{\tau}_{\pi_{t+1}}^2 - Var_I[\tau_{i\pi_{t+1}}]) \quad (194)$$

$$\Rightarrow Var_I[\tau_{i\pi_{t+1}}] = \frac{Var_I[E_{it}^{prior}(\pi_{t+1})] - \sigma_{\pi_{t+1}}^2 (\bar{\tau}_{\pi_{t+1}} - \bar{\tau}_{\pi_{t+1}}^2)}{\left( \frac{\bar{E}_t^{prior}(\pi_{t+1})}{\bar{\tau}_{\pi_{t+1}}} \right)^2 - \sigma_{\pi_{t+1}}^2} \quad (195)$$

$$= \frac{2.95^2 - 3.1 \times (0.53 - 0.53^2)}{\left( \frac{5.83}{0.53} \right)^2 - 3.1} = 0.067 \quad (196)$$

where for the last equality I assume that average information use before the survey is similar to that within the survey, so  $\bar{\tau}_{\pi_{t+1}} = \bar{\tau}_{\pi_{t+1}}^{RCT} = 0.53$ , and I use  $\sigma_{\pi_{t+1}}^2 = 3.1$ , which is the variance of 12-month Euro-area HICP inflation rates over Jan 1997 - May 2023 (the month before the survey was run). Substituting this value into equation (193), we obtain:

$$E_I[\tau_{i\pi_{t+1}}^2 \sigma_i^2] = 3.1 \times (0.53 - 0.53^2 - 0.067) = 0.564 \quad (197)$$

which in equation (183) implies  $F \leq 0.03$ , or equivalently that  $\omega \geq 0.97$ . This suggests that the approximation of  $\gamma_{1,j}^{2SLs} \approx \gamma_{1,j}^{Wald}$  is likely to be reasonable in this application.

## C.2 RCT-implied calibration

**Figure 1: abstracting from marginal cost cross-learning.** Estimating equation (31) with priors and posteriors of  $\pi_{t+1}$ , and the treatment on  $\pi_{t+1}$  only, gives a coefficient of  $\hat{b}_{\pi_{t+1}} = -0.53$ , significantly different from 0 at the 1% level. In our model, that translates to  $\int_i \chi_{i\pi_{t+1}\pi_{t+1}} \tau_{i\pi_{t+1}} di = \tau_{\pi_{t+1}}^- = 0.53$ , where the first equality uses the fact that  $\chi_{ijj} = 1$ .

Estimating equation (32) with  $E_{it}\pi_{t+3}$  as the dependent variable, and instrumenting  $E_{it}^{post}\pi_{t+1}$  with the  $\pi_{t+1}$  treatment only, gives an estimated  $\gamma_{1,\pi_{t+3}}$  of 0.742. Doing the same with  $E_{it}\pi_{t+5}$  as the dependent variable gives an estimated  $\gamma_{1,\pi_{t+5}}$  of 0.755.<sup>38</sup> Both of these coefficients are significantly different from 0 at the 1% level.<sup>39</sup>

To adapt these estimates to the particular model set out in Section 2, we need to interpolate between estimates at different horizons to get the total effect of a given signal on inflation at horizons

<sup>38</sup>Although these are not reported in Baumann et al. (2026), my thanks go to Timo Reinelt for confirming that these estimates correspond closely with those obtained in the authors' own implementation.

<sup>39</sup>Using the  $\pi_t$  treatment only these coefficients are 0.19 and 0.14 respectively, and neither is significantly different from 0. For this reason we set cross-learning from  $\pi_t$  to future  $\pi_{t+h}$  to 0.

$t + 2$  and  $t + 4$ , where there are no estimates available. I use a simple linear interpolation, so e.g.  $\gamma_{1,\pi_{t+2}} = 0.5(\gamma_{1,\pi_{t+1}} + \gamma_{1,\pi_{t+3}})$ . The resulting coefficients implied by the [Baumann et al. \(2026\)](#) experiment are presented in Table 3.

**Table 3:** Response-weighted cross-learning parameters implied by [Baumann et al. \(2026\)](#), mapped to the Calvo model.

| Model object                                                                                                            | $\pi_t$ signal | $\pi_{t+1}$ signal |
|-------------------------------------------------------------------------------------------------------------------------|----------------|--------------------|
| $\bar{\tau}_{\text{signal period}}^{-1} \int_i (\chi_{\pi_t, \text{signal period}} \tau_{\text{signal period}}) di$     | 1              | 0                  |
| $\bar{\tau}_{\text{signal period}}^{-1} \int_i (\chi_{\pi_{t+1}, \text{signal period}} \tau_{\text{signal period}}) di$ | 0              | 1                  |
| $\bar{\tau}_{\text{signal period}}^{-1} \int_i (\chi_{\pi_{t+2}, \text{signal period}} \tau_{\text{signal period}}) di$ | 0              | 0.871              |
| $\bar{\tau}_{\text{signal period}}^{-1} \int_i (\chi_{\pi_{t+3}, \text{signal period}} \tau_{\text{signal period}}) di$ | 0              | 0.742              |
| $\bar{\tau}_{\text{signal period}}^{-1} \int_i (\chi_{\pi_{t+4}, \text{signal period}} \tau_{\text{signal period}}) di$ | 0              | 0.748              |
| $\bar{\tau}_{\text{signal period}}^{-1} \int_i (\chi_{\pi_{t+5}, \text{signal period}} \tau_{\text{signal period}}) di$ | 0              | 0.755              |

The ‘ $\pi_t$  signal’ column contains a 1 in the first row (the diagonal element  $\chi_{i\pi_t\pi_t} = 1$ ) and 0s elsewhere, reflecting the assumption of no cross-learning from  $\pi_t$  to per-period inflation at future horizons. The 0 in the first row of the ‘ $\pi_{t+1}$  signal’ column reflects the assumption discussed in the main text that there is no cross-learning from  $\pi_{t+1}$  signals back to  $\pi_t$ , in line with the literature that focuses on forward rather than backward passthrough of inflation expectations (e.g. [Armantier et al., 2022](#)).

In principle, we could extrapolate to parameters further into the future, but I will cut off firm planning horizons at 5 years here, for this and the two benchmarks.

**Including cross-learning to marginal costs.** [Baumann et al. \(2026\)](#) Table 4 contains results from estimating equation (32) for  $Y_i \in \{1\text{-yr-ahead expected wage, non-wage costs}\}$ . To go from those variables to the marginal cost that enters the model, we need to specify the firm’s production function, before then applying the temporary-information equilibrium approach. For simplicity, I specify the following production function:

$$y_{it} = \min \left( \frac{n_{it}}{\alpha_n}, \frac{x_{it}}{\alpha_x} \right) \quad (198)$$

where labor  $n_{it}$  and other inputs  $x_{it}$  are perfect complements, so the optimal factor ratio is

$$\frac{n_{it}}{x_{it}} = \frac{\alpha_n}{\alpha_x} \quad (199)$$

In this case, real marginal costs are given by

$$mc_t = \alpha_n \frac{w_t}{P_t} + \alpha_x \frac{p_{xt}}{P_t} \quad (200)$$

where  $w_t$  is the nominal wage and  $p_{xt}$  is the price of  $x_{it}$ . Log-linearizing we obtain

$$mc_t = \tilde{\alpha}_n w_t + \tilde{\alpha}_x p_{xt} - (\tilde{\alpha}_n + \tilde{\alpha}_x) P_t \quad (201)$$

where

$$\tilde{\alpha}_n \equiv \frac{\alpha_n w^{ss}}{\alpha_n w^{ss} + \alpha_x p_x^{ss}}, \quad \tilde{\alpha}_x \equiv \frac{\alpha_x p_x^{ss}}{\alpha_n w^{ss} + \alpha_x p_x^{ss}} \quad (202)$$

are the steady-state shares of labor and other costs in marginal costs. Since this model features a constant marginal cost, these are also the steady state shares of labor costs and other costs in total costs. The average update to expected marginal costs after a  $\pi_{t+1}$  signal is therefore

$$\frac{dE_{it} mc_{t+1}}{d\varepsilon_{\pi_{t+1}}} = \tilde{\alpha}_n \frac{dE_{it} w_{t+1}}{d\varepsilon_{\pi_{t+1}}} + \tilde{\alpha}_x \frac{dE_{it} p_{xt+1}}{d\varepsilon_{\pi_{t+1}}} - \frac{dE_{it} (\pi_t + \pi_{t+1})}{d\varepsilon_{\pi_{t+1}}} \quad (203)$$

$$= \underbrace{(\tilde{\alpha}_n \chi_{iw_{t+1}\pi_{t+1}} + \tilde{\alpha}_x \chi_{ip_{xt+1}\pi_{t+1}} - 1)}_{\equiv \chi_{imc_{t+1}\pi_{t+1}}} \tau_{i\pi_{t+1}} \quad (204)$$

where I have used the observation that  $\tilde{\alpha}_n + \tilde{\alpha}_x = 1$ , the definition  $P_{t+1} = P_{t-1} + \pi_t + \pi_{t+1}$ , the fact that  $P_{t-1}$  is fixed as it was determined before the shock, and the baseline assumption that there is no cross-learning from  $\pi_{t+1}$  back to  $\pi_t$ .

The scaled average effect of a  $\pi_{t+1}$  signal on  $mc_{t+1}$  expectations can therefore be computed as<sup>40</sup>

$$\int_i \chi_{imc_{t+1}\pi_{t+1}} \frac{\tau_{i\pi_{t+1}}}{\bar{\tau}_{\pi_{t+1}}} di = \int_i (\tilde{\alpha}_n \chi_{iw_{t+1}\pi_{t+1}} + \tilde{\alpha}_x \chi_{ip_{xt+1}\pi_{t+1}} - 1) \frac{\tau_{i\pi_{t+1}}}{\bar{\tau}_{\pi_{t+1}}} di \quad (205)$$

$$= \tilde{\alpha}_n \times \gamma_{1,w_{t+1}} + (1 - \tilde{\alpha}_n) \times \gamma_{1,p_{xt+1}} - 1 \quad (206)$$

$$= -0.498 \quad (207)$$

where for the final equality I have assumed that the labor share of total costs is  $\tilde{\alpha}_n = 0.58$ . As this data is not available for the Euro area, this is taken from the 2023 measurement for the US.<sup>41</sup> On average firms therefore decrease their real marginal cost expectations when they learn of higher future inflation. This is because the positive updating found by [Baumann et al. \(2026\)](#) is over the components of nominal marginal costs, and those updates are dominated by the increase in inflation,

<sup>40</sup>Note that this uses  $\gamma_{1,w_{t+1}} = 0.26$  and  $\gamma_{1,p_{xt+1}} = 0.84$ , both significantly different from 0 at the 1% level. This differs slightly from [Baumann et al. \(2026\)](#) Table 4 Panel D, which reports that using the  $\pi_{t+1}$  instrument yields  $\gamma_{1,w_{t+1}} = 0.29$  and  $\gamma_{1,p_{xt+1}} = 0.90$ . The difference arises because I use the [Baumann et al. \(2026\)](#) replication package applied to the public-release version of the SAFE pilot survey, which differs in small respects from the internal version used by [Baumann et al. \(2026\)](#). The corresponding estimates using the  $\pi_t$  signal are  $-0.03$  and  $-0.78$  (my estimates) or  $-0.06$  and  $-0.47$  ([Baumann et al. \(2026\)](#) Table 4 Panel C), none of which are significant at conventional significance levels, further justifying my assumption of no cross-learning from  $\pi_t$  to future variables.

<sup>41</sup>Available at <https://fred.stlouisfed.org/series/MPU4900141>.



causing real marginal cost expectations to fall.

One could in principle extrapolate from this to cross-learning about marginal costs at longer horizons, but in the absence of estimated effects on cost expectations at other horizons I assume that all cross-learning beyond the horizon one year is zero. Allowing for the cross-learning from  $\pi_{t+1}$  to  $mc_{t+1}$ , the partial equilibrium effects of  $\bar{E}_t(\pi_{t+1})$  are therefore given by

$$\Omega^{PE}(\pi_{t+1}) = \Omega^{PE}(\pi_{t+1} : \chi_{imc_{t+1}\pi_{t+1}} = 0) + (1 - \lambda)(1 - \beta\lambda)\beta\lambda \int_i \chi_{imc_{t+1}\pi_{t+1}} \frac{\tau_{i\pi_{t+1}}}{\tau_{\pi_{t+1}}} di \quad (208)$$

where  $\Omega^{PE}(\pi_{t+1} : \chi_{imc_{t+1}\pi_{t+1}} = 0)$  denotes the partial equilibrium effect calculated in Figure 1a under the assumption of no cross-learning to real marginal costs. The general equilibrium effects can then be calculated from this using equation (16). Since I am assuming no cross-learning from  $\pi_t$  to marginal costs, the  $\Omega^{PE}(\pi_t)$  term in equation (16) is unaffected.

### C.3 Inference

Using Proposition 2 and the calibration assumptions in Appendix C.2, we can rewrite the partial-equilibrium causal effect as a linear combination of RCT estimates. In the case without marginal-cost cross-learning, this is

$$\begin{aligned} \Omega^{PE}(\pi_{t+1} : \chi_{imc_{t+1}\pi_{t+1}} = 0) &= (1 - \lambda) \left[ \beta\lambda + \frac{1}{2}(\beta\lambda)^2 \right] \\ &+ (1 - \lambda) \left[ ((\beta\lambda)^3 + \frac{1}{2}(\beta\lambda)^2 + \frac{1}{2}(\beta\lambda)^4)\hat{\gamma}_{1,\pi_{t+3}} + ((\beta\lambda)^5 + \frac{1}{2}(\beta\lambda)^4)\hat{\gamma}_{1,\pi_{t+5}} \right] \end{aligned} \quad (209)$$

where the terms multiplied by a half stem from the interpolation procedure, through which the RCT estimates for each horizon also have an effect on the calibration for neighboring horizons (where those are not directly estimated).

With marginal cost cross-learning, the expression becomes

$$\begin{aligned} \Omega^{PE}(\pi_{t+1} : \chi_{imc_{t+1}\pi_{t+1}} \neq 0) &= \Omega^{PE}(\pi_{t+1} : \chi_{imc_{t+1}\pi_{t+1}} = 0) \\ &+ (1 - \lambda)(1 - \beta\lambda)\beta\lambda [\tilde{\alpha}_n \hat{\gamma}_{1,w_{t+1}} + (1 - \tilde{\alpha}_n) \hat{\gamma}_{1,px_{t+1}} - 1] \end{aligned} \quad (210)$$

This linearity in estimated coefficients means that for a given  $\lambda$ , the uncertainty around the causal effect is given by  $\text{Var}(\Omega^{PE}(\pi_{t+1}; \lambda)) = \mathbf{a}(\lambda)' \mathbf{V} \mathbf{a}(\lambda)$ , where  $\mathbf{V}$  is the variance-covariance matrix of the vector of RCT estimates  $\gamma$ , and  $\mathbf{a}(\lambda)$  is the vector of corresponding loadings on each estimated coefficient. Importantly, the four regressions producing  $\gamma$  share an estimation sample, an instrument set, and an endogenous regressor. The off-diagonal elements of  $\mathbf{V}$  will therefore be generally non-zero, and the standard errors reported in the equation-by-equation estimation

of [Baumann et al. \(2026\)](#) are insufficient for computing the uncertainty around  $\Omega^{PE}(\pi_{t+1}; \lambda)$ . I therefore re-estimate the equations jointly, in a stacked system, to obtain  $V$ .

In doing this, I keep to the exact procedure as [Baumann et al. \(2026\)](#), using their replication package in all respects apart from the stacking. Standard errors are therefore Huber-robust and clustered as in their paper.

The 90% pointwise confidence interval for the partial-equilibrium effect at each  $\lambda$  is then given by

$$\left[ a_0(\lambda) + \mathbf{a}(\lambda)' \hat{\gamma} \pm 1.645 \sqrt{\mathbf{a}(\lambda)' \mathbf{V} \mathbf{a}(\lambda)} \right]. \quad (211)$$

where  $a_0(\lambda)$  is the constant term in equation (209) or (210) (depending on the specification in use), evaluated at a given  $\lambda$ .

The interval narrows to a point as  $\lambda \rightarrow 0$  in the specification without marginal-cost cross-learning, because every loading is  $O(\lambda^2)$  there: as prices approach full flexibility, only the  $s = 1$  term survives, and that term is known rather than estimated. In the specification with marginal-cost cross-learning the loadings  $a_c$  and  $a_w$  are  $O(\lambda)$ , so the interval retains positive width throughout.

For the causal effects in general equilibrium, it is helpful to note that the assumptions of  $\bar{\tau}_{\pi_t} = 1$  and zero cross-learning from  $\pi_t$  imply that the denominator of equation (16) reduces to  $1 - \Omega^{PE}(\pi_t)$ , and moreover that  $\Omega^{PE}(\pi_t) = 1 - \lambda$ . Therefore, the general-equilibrium causal effect is equal to the partial-equilibrium effect divided by  $\lambda$ . Since there is no uncertainty around  $\lambda$  (the intervals are computed pointwise at each  $\lambda$ ), the confidence interval for  $\Omega^{GE}(\pi_{t+1})$  is simply that for  $\Omega^{PE}(\pi_{t+1})$ , scaled by  $\lambda$ .

Note that inference becomes more complicated for the non-linear functions of RCT estimates considered in Appendices C.4 and C.6, as discussed below.

## C.4 The share of fundamental shock transmission due to expectations

Following the strategy in Proposition 4, we now compute the share of the general equilibrium transmission of a fundamental shock that can be attributed to inflation expectations, for the RCT data and both benchmarks (perfect foresight and rational expectations). Since we are assuming  $\bar{\tau}_{\pi_t} = 1$ , these shares for an i.i.d.  $mc_t$  shock would just equal the partial equilibrium effect of a  $\bar{E}_t(\pi_t)$  shock (equation (21)). We therefore opt instead to consider a news shock: firms learn in period  $t$  of an exogenous increase in  $mc_{t+1}$ . This demonstrates the flexibility of the approach developed here.

With this shock the transmission channels are more numerous. Once period  $t + 1$  arrives, the increased marginal costs will imply inflation above steady state, as described in equation (20). In period  $t$ , firms therefore not only learn that  $mc_{t+1}$  will be above steady state; their signals about  $\pi_{t+1}$  also endogenously increase. Even without cross-learning from marginal cost expectations

to inflation expectations, there is therefore an increase in expected future inflation, which has cross-learning effects on inflation expectations at other horizons. All of these forces contribute to the increase in  $\pi_t$ , which in turn has general equilibrium feedback effects through  $s_{it}(\pi_t)$ .

Formally, we model the news shock by assuming that  $\tau_{imc_{t+1}} = 1$ .<sup>42</sup> We also assume that marginal costs are i.i.d., so there is no cross-learning from the news about future marginal cost to expected marginal costs at any other horizon, or to expected inflation. Differentiating equation (11) with respect to  $mc_{t+1}$  with these assumptions implies

$$\frac{d\pi_t}{dmc_{t+1}} = (1 - \lambda)(1 - \beta\lambda)\beta\lambda + \bar{\tau}_{\pi_{t+1}}\Omega^{PE}(\pi_{t+1})\frac{d\pi_{t+1}}{dmc_{t+1}} + \Omega^{PE}(\pi_t)\frac{d\pi_t}{dmc_{t+1}} \quad (212)$$

where we have also used our recurring assumption that  $\bar{\tau}_{\pi_t} = 1$ .

Next, from equation (20) we have

$$\frac{d\pi_{t+1}}{dmc_{t+1}} = \frac{d\pi_t}{dmc_t} = \frac{(1 - \lambda)(1 - \beta\lambda)}{1 - \Omega^{PE}(\pi_t)} \quad (213)$$

where it is  $\Omega^{PE}(\pi_t)$ , not  $\Omega^{PE}(\pi_{t+1})$  that matters because by the time period  $t + 1$  arrives, the shock will be contemporaneous, so it is the effect of a contemporaneous shock that determines the contemporaneous inflation rate – i.e.  $\pi_{t+1}$ .

Substituting this into equation (212) and rearranging, we find

$$\frac{d\pi_t}{dmc_{t+1}} = \frac{(1 - \lambda)(1 - \beta\lambda)\beta\lambda}{1 - \Omega^{PE}(\pi_t)} + \frac{(1 - \lambda)(1 - \beta\lambda)\bar{\tau}_{\pi_{t+1}}\Omega^{PE}(\pi_{t+1})}{(1 - \Omega^{PE}(\pi_t))^2} \quad (214)$$

We define the share of this overall transmission that is due to expectations as in equation (21). The counterfactual shock transmission is computed assuming firms only observe the  $mc_{t+1}$  signal, with no updating of any other expectations. Thus this calculation gives the share of shock transmission due to expectations of inflation specifically, as marginal cost expectations are the same in the overall effect and the counterfactual.

This counterfactual with no response of inflation expectations is found by setting  $\Omega^{PE}(\pi_t) = \Omega^{PE}(\pi_{t+1}) = 0$ , at which point

$$\left. \frac{d\pi_t}{dmc_{t+1}} \right|_{\sim E} = (1 - \lambda)(1 - \beta\lambda)\beta\lambda \quad (215)$$

Using these expressions, we calculate the share of the shock transmission to  $\pi_t$  that is due to the

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<sup>42</sup>Note that this shuts off cross-learning from other variables to  $mc_{t+1}$  by default (see equation (10)).

response of inflation expectations as

$$\frac{\left. \frac{d\pi_t}{dm c_{t+1}} - \frac{d\pi_t}{dm c_{t+1}} \right|_{\sim E}}{\frac{d\pi_t}{dm c_{t+1}}} = 1 - \frac{\beta\lambda(1 - \Omega^{PE}(\pi_t))^2}{\beta\lambda(1 - \Omega^{PE}(\pi_t)) + \bar{\tau}_{\pi_{t+1}}\Omega^{PE}(\pi_{t+1})} \quad (216)$$

To construct each line in Figure 5, we replace  $\Omega^{PE}(\pi_t)$  and  $\Omega^{PE}(\pi_{t+1})$  with values calculated using equations (12) and (13) and the relevant calibrations of  $\chi_i$  and  $\tau_i$ .

**Inference.** Constructing a confidence interval for Figure 5 differs from Figures 1 and 2 in two respects. First, the share in (216) is a nonlinear function of the estimated RCT coefficients, so the variance cannot be read off a quadratic form directly. Second, the share depends on  $\bar{\tau}_{\pi_{t+1}}$  beyond how it affects the coefficients of the second-stage regressions.

It is convenient to define  $\mathcal{S}(\lambda)$  as the share of shock transmission due to the response of inflation expectations in (216), evaluated at stickiness  $\lambda$ . Moreover, note that with the assumptions of  $\bar{\tau}_{\pi_t} = 1$  and no cross-learning from  $\pi_t$  maintained for the RCT calibration, equation (216) reduces to

$$\mathcal{S}(\lambda) = 1 - \frac{\beta\lambda^3}{\beta\lambda^2 + \bar{\tau}_{\pi_{t+1}}\Omega^{PE}(\pi_{t+1})} \quad (217)$$

That is,  $\mathcal{S}(\lambda)$  in the RCT calibration depends on the estimated parameters only through the scalar

$$g(\lambda) \equiv \bar{\tau}_{\pi_{t+1}}\Omega^{PE}(\pi_{t+1}; \lambda) = \bar{\tau}_{\pi_{t+1}}\left(a_0(\lambda) + \mathbf{a}(\lambda)' \boldsymbol{\gamma}\right), \quad (218)$$

with  $a_0(\lambda)$  and  $\mathbf{a}(\lambda)$  as defined in Appendix C.3. Inference therefore proceeds in two steps: an interval for  $g(\lambda)$  via the delta method, and then a mapping from that interval to one for  $\mathcal{S}$ , which is nonlinear but monotone.

I first consider the interval for  $g(\lambda)$ . As discussed in Section 4,  $\bar{\tau}_{\pi_{t+1}}^{RCT}$  can be obtained from the first-stage regression (31). As with the coefficients used for inference in Figures 1 and 2, this is likely to be correlated with the second-stage coefficients. I therefore stack the first-stage regression with the four second-stage regressions discussed in Appendix C.3, from which I obtain the covariances between  $\hat{\tau}_{\pi_{t+1}}$  and  $\hat{\boldsymbol{\gamma}}$ . Since  $\bar{\tau}_{\pi_{t+1}}^{RCT} = -b_j$ , the variance carries over unchanged but every covariance with  $\boldsymbol{\gamma}$  changes sign.

With this in hand, write  $\boldsymbol{\theta} \equiv (\boldsymbol{\gamma}', \bar{\tau}_{\pi_{t+1}})'$  and let  $\mathbf{V}_{\boldsymbol{\theta}}$  denote its variance-covariance matrix, obtained from the augmented stacked system as above. From (218),

$$\widehat{\text{Var}}(g(\lambda)) \approx \nabla_{\boldsymbol{\theta}} g' \mathbf{V}_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} g, \quad \text{where } \nabla_{\boldsymbol{\theta}} g(\lambda) = \begin{pmatrix} \bar{\tau}_{\pi_{t+1}} \mathbf{a}(\lambda) \\ \Omega^{PE}(\pi_{t+1}; \lambda) \end{pmatrix}. \quad (219)$$

Note that because  $g$  is bilinear rather than linear in  $\theta$ , (219) is a first-order approximation.

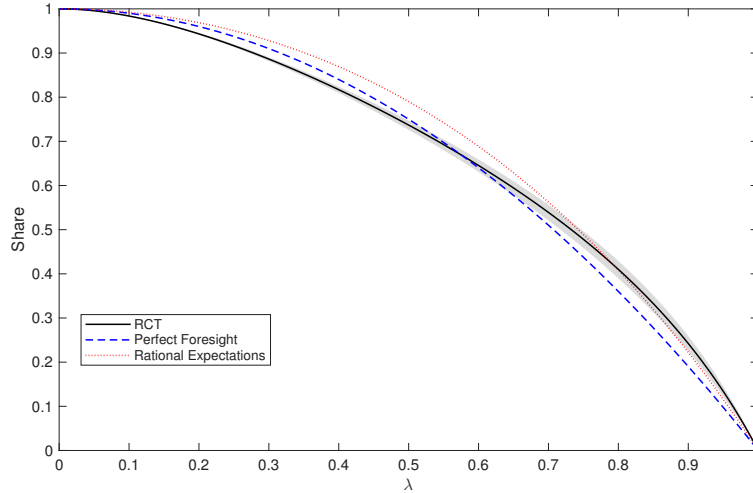
With this in hand, I now turn to the confidence interval for  $\mathcal{S}(\lambda)$ . As long as we are in the region where  $\beta\lambda^2 + g(\lambda) > 0$ , then from (217) we have that  $\mathcal{S}(\lambda)$  is strictly increasing in  $g(\lambda)$ . Therefore, if  $[\hat{g}_{lo}, \hat{g}_{hi}]$  is a 90% confidence interval for  $g(\lambda)$ ,

$$\left[ 1 - \frac{\beta\lambda^3}{\beta\lambda^2 + \hat{g}_{lo}}, 1 - \frac{\beta\lambda^3}{\beta\lambda^2 + \hat{g}_{hi}} \right] \quad (220)$$

is a 90% confidence interval for  $\mathcal{S}(\lambda)$ .

All that is required for this simple approach is that  $\beta\lambda^2 + g(\lambda) > 0$ . At the estimates used here, the lower limit  $\beta\lambda^2 + \hat{g}_{lo}$  remains strictly positive at every  $\lambda$ .

**Figure 5:** Share of transmission from a  $mc_{t+1}$  news shock to  $\pi_t$  due to the adjustment of inflation expectations, comparing RCT estimates to benchmarks



*Note:* Figure plots the share of the transmission of a  $mc_{t+1}$  news shock that is due to the endogenous response of inflation expectations in the Calvo model of Section 2, for calibrations based on perfect foresight, rational expectations, and the RCT results of Baumann et al. (2026) (detail in Appendix C.2). The gray shaded band shows the 90% pointwise confidence interval for the RCT calibration (detail in the text).

**Results.** At  $\lambda = 0.2$ , the share of the shock transmission due to the dynamics of inflation expectations is just over 2.5% (2.55 percentage points) smaller in the RCT calibration than in the rational expectations benchmark. At  $\lambda = 0.5$ , this is 6.7% (5.3 percentage points). However, the dampening of the share under the RCT calibration is not monotonic in  $\lambda$ : for very sticky prices ( $\lambda > 0.79$ ) the effect even changes sign, with marginally greater shares attributable to expectations under the RCT calibration. This sign change occurs because the RCT results imply no cross-learning from  $\pi_t$  to longer horizon expectation, while under rational expectations this is driven by the same AR(1) coefficient as the cross-learning from  $\pi_{t+1}$ . This means that when prices are rather flexible, and short-horizon expectations drive the vast majority of pricing decisions, the  $mc_{t+1}$  shock entails a smaller reaction in the most important expectation horizons than is generated with rational

expectations. Of course, even the limited short-horizon cross-learning in the RCT calibration is stronger than seen under perfect foresight,<sup>43</sup> so the transmission due to expectations is larger than the perfect foresight case for a wider range of price stickiness calibrations.

## C.5 Other models as benchmarks

Recall that the equilibrium New Keynesian Phillips Curve cannot reveal the causal effects of inflation expectations because expectations there simultaneously cause and are caused by dynamics of other variables. The same is true if we attempt to compare different models of expectation formation by studying equilibrium conditions. If, for example, a particular model implies a reduced link between inflation expectations and realized inflation, we cannot tell if that is because expectations have been made less responsive to inflation, or if their causal effect on inflation has been reduced.

The temporary-information equilibrium approach therefore offers a structured way to disentangle why certain models have different implications from one another, and to assess whether they match the equilibrium implications of RCT evidence. I provide an example here, re-computing Figure 1 for diagnostic expectations (Bordalo et al., 2020) and inattentiveness (Gabaix, 2020).

In Gabaix (2020), expectations of a variable  $x_{t+h}$  are given by

$$E_{it}x_{t+h} = m^h E_t^* x_{t+h} \quad (221)$$

where  $E_t^* x_{t+h}$  is the rational expectation of  $x_{t+h}$  formed with information on all variables realized up to period  $t$ , and  $m \in [0, 1]$  is a parameter governing the degree of inattentiveness. The first implication of this setup is that when  $h = 0$ , inattentiveness has no effect on expectations. That is, inattentiveness does not affect the mapping from observed information on a variable  $x_t$  into beliefs about  $x_t$ . It only affects how that information is used to form beliefs about  $x_{t+h}$ . Inattentiveness therefore affects cross-learning from  $x_t$  to  $x_{t+h}$ , but not the information matrix  $\tau_i$ .

Specifically, we will restrict ourselves here to cases with no cross-learning to or from marginal costs. The inattentiveness case can therefore be summarized by

$$\chi_{i\pi_{t+s}\pi_t}^{\text{IA}} = m^s \chi_{i\pi_{t+s}\pi_t}^{\text{RE}} \quad (222)$$

where  $\chi_{i\pi_{t+s}\pi_t}^{\text{RE}}$  is the cross-learning from inflation  $\pi_t$  to inflation  $\pi_{t+s}$  under rational expectations. Assuming as before that  $\chi_{i\pi_{t+s}\pi_t}^{\text{RE}} = \rho^s$ , the benchmark is implemented by setting  $\chi_{i\pi_{t+s}\pi_t}^{\text{IA}} = (m\rho)^s$ .

The second model we consider here is diagnostic expectations, as in Bordalo et al. (2020). In

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<sup>43</sup>Specifically, both the RCT calibration and perfect foresight feature 0 cross-learning from  $\pi_t$  to inflation at future horizons, so they share the same  $\Omega^{PE}(\pi_t)$ ; the difference arises through  $\Omega^{PE}(\pi_{t+1})$ , where the RCT calibration features positive cross-learning from  $\pi_{t+1}$  to inflation at horizons  $t+2$  and beyond (Table 3), which is absent under perfect foresight.

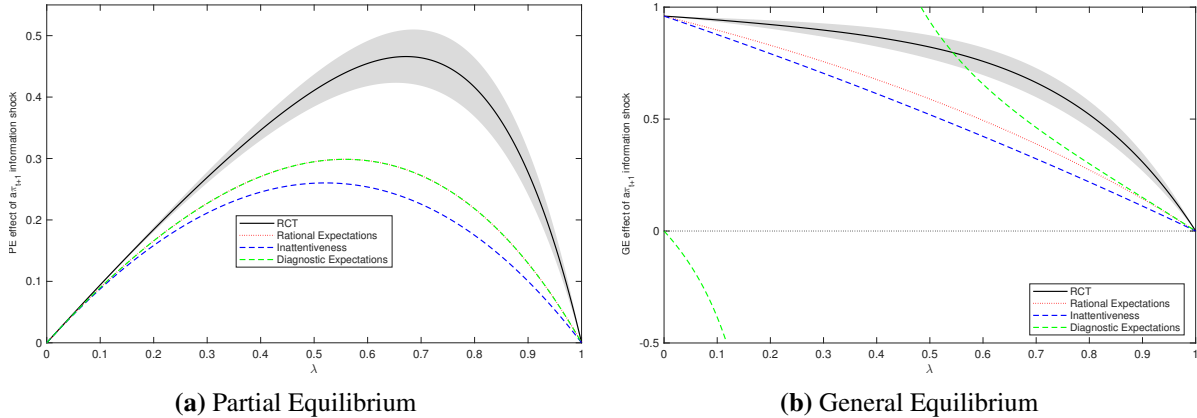
this model, if  $x_t$  is an AR(1) process with persistence  $\rho$ , we have

$$E_{it}x_{t+h} = \rho^h E_{it}x_t = \rho^h \left( (1 + \theta) \frac{\sigma_x^2}{\sigma_x^2 + \sigma_{\varepsilon x}^2} s_{it}(x_t) \right) \quad (223)$$

where  $\theta$  is the degree of diagnosticity,  $s_{it}(x_t)$  is the (noisy) signal observed about  $x_t$ ,  $\sigma_x^2$  is the variance of  $x_t$ , and  $\sigma_{\varepsilon x}^2$  is the variance of the noise in  $s_{it}(x_t)$ . In this case, the departure from rational expectations operates solely on the way in which agents map from direct signals on  $x_t$  into expectations of  $x_t$ . It therefore affects  $\tau_i$  (the direct response to signals) and not  $\chi_i$  (cross-learning across horizons). Specifically, diagnostic expectations imply  $\tau_i$  is multiplied by  $1 + \theta$ , so for  $\theta > 0$  agents overreact to news.

Figure 6 plots the partial and general equilibrium effects of  $\bar{E}_t(\pi_{t+1})$  in each of these models, alongside the effects under rational expectations and the RCT benchmark. For the calibration, I set  $m = 0.8^4$ , converting the quarterly firm-inattention calibration recommended by Gabaix (2020) to annual frequency. I set  $\theta = 0.37$ , the average of the results from the two methods employed by Bordalo et al. (2020) to estimate  $\theta$  in the context of CPI inflation expectations (taken from their Table 4).

**Figure 6:** Partial and general equilibrium effects of a shock to  $s_{\pi_{t+1}}$ , comparing RCT estimates to rational expectations, inattentiveness, and diagnostic expectations.



*Note:* Figure plots the partial- and general-equilibrium causal effects of an increase in  $\bar{E}_t\pi_{t+1}$  in the Calvo model, following the formulae in Propositions 2 and 3, and a calibration derived from the estimates in Baumann et al. (2026) (detail in Appendix C.2). This RCT calibration is compared to calibrations based on perfect foresight, rational expectations, inattentiveness as in Gabaix (2020), and diagnostic expectations as in Bordalo et al. (2020). Gray shaded bands show 90% pointwise confidence intervals (detail in Appendix C.3).

Inattentiveness generates similar results to rational expectations, with a small degree of dampening relative to that benchmark in both partial and general equilibrium. The dampening is proportionately larger at higher  $\lambda$ , when longer-horizon expectations are more important for current pricing.

Diagnostic expectations are identical to rational expectations in partial equilibrium, because the only difference to rational expectations is in  $\bar{\tau}_{\pi_{t+1}}$ . When we scale the shock to ensure a unit

response of  $\bar{E}_t(\pi_{t+1})$ , we therefore cancel out any effects of diagnosticity.

However, in general equilibrium diagnostic expectations deliver radically different results from the other models and from the RCT results. The over-reaction to news generated by diagnostic expectations makes general equilibrium feedback effects more powerful, as any increase in  $\pi_t$  is met with a further round of over-reaction, entailing even more  $\pi_t$  response. This creates powerful strategic complementarities in price-setting. As price stickiness falls from  $\lambda = 1$ , general equilibrium effects grow in all models. The extra complementarity induced by diagnostic expectations means that as  $\lambda$  falls the causal effects of expectations grow more rapidly with diagnostic expectations than they do with rational expectations. At sufficiently low  $\lambda$ , the strategic complementarities become so strong that an increase in  $\bar{E}_t(\pi_t)$  generates a greater than one-for-one response in  $\pi_t$  (i.e.  $\bar{\tau}_{\pi_t} \Omega^{PE}(\pi_t) > 1$  in the denominator of Proposition 3), implying that in equilibrium greater  $\bar{E}_t(\pi_{t+1})$  must in fact be met with a decrease in  $\pi_t$ .

## C.6 Allowing for $\pi_t$ cross-learning

The exercises so far have assumed that  $\chi_{iY\pi_t} = 0$  for all  $Y \neq \pi_t$ , motivated by the observation that the coefficient estimates from the second-stage regression (32) are insignificantly different from 0 when the  $\pi_t$  treatment is the only included source of exogenous variation. For the partial-equilibrium causal effects (Figures 1a, 2a, and 6a) this is without loss of generality, as signals on  $\pi_t$  are held constant by assumption. For general-equilibrium causal effects (Figures 1b, 2b, and 6b), however, it does make a difference. In this appendix I add  $\pi_t$  cross-learning back in, using the point estimates and their (wide) confidence intervals.

To do this, the first thing to note is that Proposition 8 needs adjusting, as that was proved for the case in which the treatment signal is about the same variable that is being instrumented. Since priors and posteriors on  $\pi_t$  are not collected in the SAFE survey, this is not possible here. Rather, I will use the strategy behind Baumann et al. (2026) Table 4 Panel C, in which they instrument  $E_{it}^{post} \pi_{t+1}$  with instruments derived from the  $\pi_t$  treatment. Future survey design could elicit priors and posteriors on  $\pi_t$  to make these steps (and the associated approximations) unnecessary. The results here should be interpreted as the best available with the existing survey.

**What the second-stage coefficients identify.** Let  $D_{ij,\pi_t}$  be an indicator for treatment using the  $\pi_t$  signal. Using the same normalization conventions as the proof of Proposition 8, the Wald estimand in equation (167) becomes:

$$\gamma_{1,j}^{Wald} = \frac{E_I[E_{it}^{post}(z_j) \mid D_{ij,\pi_t} = 1] - E_I[E_{it}^{post}(z_j) \mid D_{ij,\pi_t} = 0]}{E_I[E_{it}^{post}(\pi_{t+1}) \mid D_{ij,\pi_t} = 1] - E_I[E_{it}^{post}(\pi_{t+1}) \mid D_{ij,\pi_t} = 0]} = \frac{E_I[\Delta_{ij}]}{E_I[\Delta_{in}]} \quad (224)$$



which simplifies, using (165), to:

$$\gamma_{1,j}^{Wald} = \frac{\int_i \chi_{ij\pi_t} \tau_{i\pi_t}^{RCT} di}{\int_i \chi_{i\pi_{t+1}\pi_t} \tau_{i\pi_t}^{RCT} di} = \frac{\int_i \chi_{ij\pi_t} \frac{\tau_{i\pi_t}^{RCT}}{\bar{\tau}_{\pi_t}^{RCT}} di}{\int_i \chi_{i\pi_{t+1}\pi_t} \frac{\tau_{i\pi_t}^{RCT}}{\bar{\tau}_{\pi_t}^{RCT}} di} \quad (225)$$

The coefficient therefore identifies the *ratio* of the desired weighted-average cross-learning ( $\int_i \chi_{ij\pi_t} \frac{\tau_{i\pi_t}^{RCT}}{\bar{\tau}_{\pi_t}^{RCT}} di$ ) and the weighted-average cross-learning from  $\pi_t$  to  $\pi_{t+1}$ . Given the already substantial complexity of the exercise below, I leave the derivation for the 2SLS case including  $D_{ij,\pi_t} \times E_{it}^{prior} \pi_{t+1}$  as a second instrument for future work, and continue with this as an approximate guide to interpreting the coefficients from the  $\pi_t$  treatment. Eliciting  $\pi_t$  priors and posteriors in future experiments would remove the need for this, as Proposition 8 could then be applied directly.

One final point to note is that this derivation assumed, like Proposition 8, that the treatment variable (here  $\pi_t$ ) is ordered last in  $M_i$ . This contradicts the assumptions required for the interpretation of the estimates using the  $\pi_{t+1}$  treatment, in which  $\pi_{t+1}$  is ordered last. The two sets of estimates are therefore best read as characterizing cross-learning from each treated variable under the ordering appropriate to that treatment.

**Calibrating  $\pi_t$  cross-learning.** Define  $\tilde{\gamma}_{\pi_{t+1}\pi_t} \equiv \int_i \chi_{i\pi_{t+1}\pi_t} \frac{\tau_{i\pi_t}^{RCT}}{\bar{\tau}_{\pi_t}^{RCT}} di$  as the weighted-average cross-learning from  $\pi_t$  to  $\pi_{t+1}$ . The desired vector of cross-learning parameters for use in calibrating the model, equivalent to the ‘ $\pi_t$  signal’ column in Table 3, has the form  $(1, \tilde{\gamma}_{\pi_{t+1}\pi_t}, \tilde{\gamma}_{\pi_{t+1}\pi_t} \hat{\gamma}_{1,\pi_{t+2}}, \tilde{\gamma}_{\pi_{t+1}\pi_t} \hat{\gamma}_{1,\pi_{t+3}}, \tilde{\gamma}_{\pi_{t+1}\pi_t} \hat{\gamma}_{1,\pi_{t+4}}, \tilde{\gamma}_{\pi_{t+1}\pi_t} \hat{\gamma}_{1,\pi_{t+5}})$ , where  $\hat{\gamma}_{1,\pi_{t+h}}$  is the estimated coefficient from the second-stage regression using the  $\pi_t$  treatment and  $\pi_{t+h}$  as the dependent variable.  $\hat{\gamma}_{1,\pi_{t+3}}$  and  $\hat{\gamma}_{1,\pi_{t+5}}$  can be estimated using the same procedure as Baumann et al. (2026) Table 4 Panel C. The values are 0.194 and 0.137 respectively, neither significantly different from 0 at conventional levels.

I then calibrate  $\hat{\gamma}_{1,\pi_{t+4}}$  using the same linear interpolation strategy as in Appendix C.2, which implies  $\hat{\gamma}_{1,\pi_{t+4}} = \frac{1}{2}(\hat{\gamma}_{1,\pi_{t+3}} + \hat{\gamma}_{1,\pi_{t+5}}) = 0.165$ . That leaves two parameters to calibrate,  $\tilde{\gamma}_{\pi_{t+1}\pi_t}$  and  $\hat{\gamma}_{1,\pi_{t+2}}$ . Again, I opt for a simple linear interpolation, assuming a linear decay between the first four elements of the parameter vector, which implies:

$$\hat{\gamma}_{1,\pi_{t+2}} = \frac{1 + \hat{\gamma}_{1,\pi_{t+3}}}{2} = 0.597, \quad \tilde{\gamma}_{\pi_{t+1}\pi_t} = \frac{1}{2 - \hat{\gamma}_{1,\pi_{t+2}}} = 0.713 \quad (226)$$

The full parameter vector for cross-learning from  $\pi_t$  to  $\pi_{t+h}$  is therefore  $(1, 0.713, 0.425, 0.138, 0.118, 0.098)$ . In Figure 7 below, I compute causal effects with and without cross-learning from  $\pi_t$  to  $m_{t+1}$ . For the latter case, the relevant coefficients from estimating (32) with the current-inflation treatment are  $\hat{\gamma}_{1,w_{t+1}} = -0.03$ ,  $\hat{\gamma}_{1,p_{xt+1}} = -0.78$  (both

insignificantly different from 0), implying:

$$\int_i \chi_{imc_{t+1}\pi_t} \frac{\tau_{i\pi_t}}{\bar{\tau}_{\pi_t}} di = \int_i (\tilde{\alpha}_n \chi_{iw_{t+1}\pi_t} + \tilde{\alpha}_x \chi_{ip_{xt+1}\pi_t} - \chi_{i\pi_t\pi_t} - \chi_{i\pi_{t+1}\pi_t}) \frac{\tau_{i\pi_t}}{\bar{\tau}_{\pi_t}} di \quad (227)$$

$$= \tilde{\alpha}_n \times (\hat{\gamma}_{1,w_{t+1}} \tilde{\gamma}_{\pi_{t+1}\pi_t}) + (1 - \tilde{\alpha}_n) \times \hat{\gamma}_{1,p_{xt+1}} \tilde{\gamma}_{\pi_{t+1}\pi_t} - 1 - \tilde{\gamma}_{\pi_{t+1}\pi_t} \quad (228)$$

$$= -1.962 \quad (229)$$

**Inference.** Inference here differs from Appendix C.3 in two respects. First, the general-equilibrium causal effect now depends on estimates from both  $\pi_{t+1}$  and  $\pi_t$  treatment arms. Second, the object of interest is a ratio whose denominator is itself estimated, and which depends on the estimates nonlinearly. I therefore extend the stacked system used there, and apply the delta method to the ratio. To minimize notation I write  $\tilde{\gamma} \equiv \tilde{\gamma}_{\pi_{t+1}\pi_t}$  throughout, and let  $\mathbb{1}(mc) \in \{0, 1\}$  indicate whether cross-learning to  $mc_{t+1}$  is included.

Let

$$\hat{\theta} \equiv (\hat{\gamma}_{1,\pi_t+3}^{\pi_t}, \hat{\gamma}_{1,\pi_t+5}^{\pi_t}, \hat{\gamma}_{1,w_{t+1}}^{\pi_t}, \hat{\gamma}_{1,p_{xt+1}}^{\pi_t}, \hat{\gamma}_{1,\pi_t+3}^{\pi_{t+1}}, \hat{\gamma}_{1,\pi_t+5}^{\pi_{t+1}}, \hat{\gamma}_{1,w_{t+1}}^{\pi_{t+1}}, \hat{\gamma}_{1,p_{xt+1}}^{\pi_{t+1}})' \quad (230)$$

where the superscript denotes the treatment used to instrument  $E_{it}^{post} \pi_{t+1}$  in equation (32). Denote the variance-covariance matrix of  $\hat{\theta}$  as  $V$ . As in Appendix C.3, I obtain  $V$  by stacking all relevant regressions and estimating jointly.

Let  $a_0(\lambda)$  and  $a(\lambda)$  be the constant and the vector of loadings defined in Appendix C.3. Comparing the calibrated cross-learning vector for the  $\pi_t$  signal,  $(1, \tilde{\gamma}, \tilde{\gamma} \hat{\gamma}_{1,\pi_t+2}^{\pi_t}, \dots, \tilde{\gamma} \hat{\gamma}_{1,\pi_t+5}^{\pi_t})$ , with the corresponding vector for the  $\pi_{t+1}$  signal in Table 3, the former is  $\tilde{\gamma}$  times the latter, except for  $\chi_{i\pi_t\pi_t}$ , which is set to 1 by construction. The same is true of the marginal-cost term: equation (228) can be written as  $\tilde{\gamma} [\tilde{\alpha}_n \hat{\gamma}_{1,w_{t+1}}^{\pi_t} + (1 - \tilde{\alpha}_n) \hat{\gamma}_{1,p_{xt+1}}^{\pi_t} - 1] - 1$ , where  $\hat{\gamma}^{\pi_t}$  contains the first four elements of  $\hat{\theta}$ . This is  $\tilde{\gamma}$  times the  $\pi_{t+1}$ -treatment expression in (206), minus one. Since the interpolation scheme is common to the two arms, the loadings are therefore identical, and

$$\Omega^{PE}(\pi_t) = (1 - \lambda) - \mathbb{1}(mc) m(\lambda) + \tilde{\gamma} [a_0(\lambda) + a(\lambda)' \hat{\gamma}^{\pi_t}] \quad (231)$$

where  $m(\lambda) \equiv (1 - \lambda)(1 - \beta\lambda)\beta\lambda$ .

Substituting (231) into (16), and using  $\bar{\tau}_{\pi_t} = 1$ , we obtain  $\Omega^{GE}(\pi_{t+1}; \lambda) = N(\lambda; \theta)/D(\lambda; \theta)$  with

$$N = a_0(\lambda) + a(\lambda)' \hat{\gamma}^{\pi_{t+1}}, \quad D = \lambda + \mathbb{1}(mc) m(\lambda) + \tilde{\gamma} \underbrace{a(\lambda)' (\hat{\gamma}^{\pi_{t+1}} - \hat{\gamma}^{\pi_t})}_{\equiv Q(\lambda)} \quad (232)$$

$N$  is linear in  $\hat{\theta}$  and  $D$  is smooth in it, with gradients

$$\nabla_{\theta} N = \begin{pmatrix} 0 \\ a(\lambda) \end{pmatrix}, \quad \nabla_{\theta} D = \begin{pmatrix} \frac{\tilde{\gamma}^2}{2} Q(\lambda) e_1 - \tilde{\gamma} a(\lambda) \\ \tilde{\gamma} a(\lambda) \end{pmatrix} \quad (233)$$

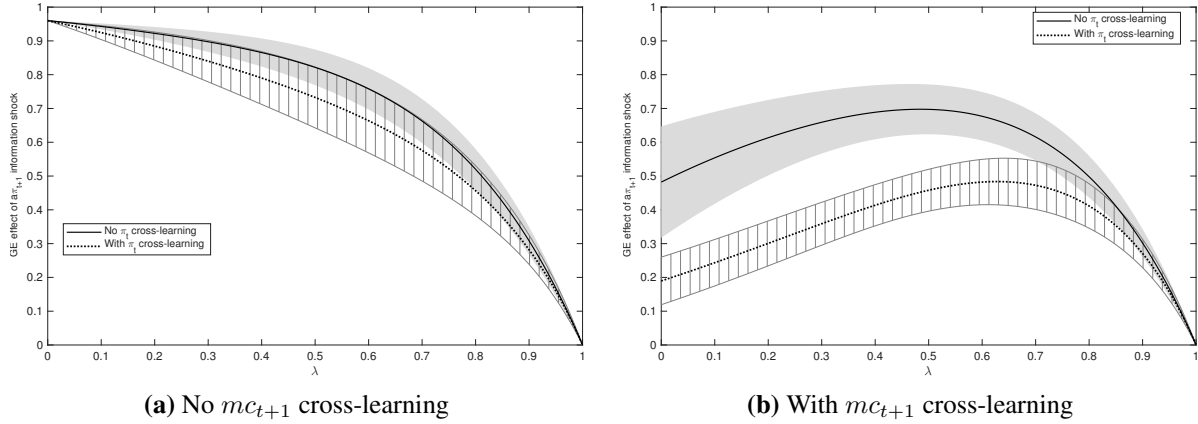
where  $e_1$  is the first unit vector, picking out  $\hat{\gamma}_{1,\pi_{t+3}}^{\pi_t}$ . This expression uses the fact that combining the restrictions in (226) gives  $\tilde{\gamma} = 2/(3 - \hat{\gamma}_{1,\pi_{t+3}}^{\pi_t})$ , and hence  $\partial \tilde{\gamma} / \partial \hat{\gamma}_{1,\pi_{t+3}}^{\pi_t} = \tilde{\gamma}^2/2$ . This is the only channel through which the estimates enter nonlinearly. The delta method then gives

$$\widehat{\text{Var}}(\hat{r}(\lambda)) = \frac{1}{\hat{D}^2} (\nabla_{\theta} N - \hat{r} \nabla_{\theta} D)' V (\nabla_{\theta} N - \hat{r} \nabla_{\theta} D), \quad \hat{r} \equiv \hat{N} / \hat{D} \quad (234)$$

and the 90% pointwise confidence interval plotted in Figure 7 is  $\hat{r}(\lambda) \pm 1.645 \sqrt{\widehat{\text{Var}}(\hat{r}(\lambda))}$ .

**Results.** Figure 7 plots  $\Omega^{GE}(\pi_{t+1})$  with and without cross-learning from  $\pi_t$ , without cross-learning into marginal costs (panel (a)) and with it (panel (b)).

**Figure 7:** General equilibrium effects of a shock to  $s_{\pi_{t+1}}$ , comparing baseline RCT estimates to an alternative including  $\pi_t$  cross-learning.



*Note:* In each panel, the figure plots (in black) the general-equilibrium causal effects of an increase in  $\bar{E}_t \pi_{t+1}$  in the Calvo model, following the formulae in Propositions 2 and 3, and a calibration derived from the estimates in Baumann et al. (2026). The dotted lines plot the equivalent for an alternative model allowing for cross-learning from  $\pi_t$  to other expectations, using the insignificant point estimates from the relevant regressions in Baumann et al. (2026). Panel (a) excludes cross-learning from  $\pi_t$  and  $\pi_{t+1}$  to marginal costs, panel (b) includes these forces. Gray shaded bands show 90% pointwise confidence intervals without cross-learning from  $\pi_t$ , hatched regions show them with cross-learning from  $\pi_t$  (detail in the text).

Without marginal cost cross-learning, including the response of longer-term inflation expectations to perceptions of  $\pi_t$  has a relatively small dampening effect on the causal effect of  $\bar{E}_t \pi_{t+1}$  in general equilibrium. The dampening occurs because the only sizeable cross-learning from  $\pi_t$  is to expectations of  $\pi_{t+1}$ , and this causes a rescaling to ensure the aggregate  $\bar{E}_t \pi_{t+1}$  rises by 1 unit. With marginal cost cross-learning, the dampening is more substantial, because the estimated response-weighted effect of the  $\pi_t$  signal on expected future wages and (especially) other

costs is negative. Increased perceptions of  $\pi_t$  therefore on average cause a more than one-for-one decline in expected real marginal costs, which reduces firms' desired price increases. However, these negative effects are imprecisely estimated, and not significant at conventional significance levels, which is why I abstract from them in the main analysis above.

With this final observation, it is worth noting however that imprecision in the estimated RCT coefficients does not translate into substantially wider confidence intervals around the causal effects in either panel, because of the nonlinear way the coefficient estimates map into  $\Omega^{GE}(\pi_{t+1})$ . In particular, if we changed coefficient estimates such that  $\Omega^{PE}(\pi_{t+1})$  rose, that would imply a greater rise in  $\pi_t$ , and thus (through cross-learning from  $\pi_t$ ) a greater rise in  $\bar{E}_t\pi_{t+1}$ . This would imply a greater rescaling to keep the expectation rise at unity. Therefore, switching on cross-learning from  $\pi_t$  limits the extent to which greater coefficient estimates can affect  $\Omega^{GE}(\pi_{t+1})$ , and this force narrows the confidence intervals in the  $\pi_t$  cross-learning case. In panel (b), an extra effect arises because the cross-learning from  $\pi_t$  to  $mc_{t+1}$  is estimated to be strongly negative. Switching on this cross-learning therefore reduces  $\Omega^{PE}(\pi_t)$ , which increases the denominator in (16), and thus again reduces the influence of noise in  $\Omega^{PE}(\pi_{t+1})$ .

## D Empirical Appendix for Section 5

### D.1 Defining the direct information indicator in the IAS

The full set of questions used to construct the information dummy is set out below, along with the dates at which each was asked and how the answers are mapped into the information indicator used above. Note that my question numbering differs from the labels in the IAS microdata, to aid the logical organization of the paper. All of the questions were only asked in the first quarter of the year(s) indicated. In the main exercises I exclude questions 2e and 2g from the total information variable, to ensure that there are no periods in which two questions are asked. I remove these rather than the short run questions in those periods to keep the majority of questions as short run expectations. The results are robust to including these extra questions. See Appendix D.3 for this, and robustness checks with other variations in the definition of the information indicator.

**Question 2a** *What were the most important factors in getting to your expectation for how prices in the shops would change over the next 12 months?*

*Please select up to 4.*

1. *How prices have changed in the shops recently, over the last 12 months*
2. *How prices have changed in the shops, on average, over the longer term i.e the last few years*
3. *Reports of current inflation in the media*
4. *Discussion of the prospects for inflation in the media*

5. *The level of interest rates*
6. *The inflation target set by the government*
7. *The current strength of the UK economy*
8. *Expectations about how economic conditions in the UK are likely to evolve*
9. *Other factors*
10. *None*

We can divide the possible answers into four categories. First, options 1 and 2 concern past experienced price rises. Options 3 and 4 are direct information about inflation. Options 5-8 concern other macroeconomic variables, either current or expected, and options 9 and 10 are extras. A rational household may well use the information sources in options 1,2 and 5-9 to forecast inflation, but in the decomposition in Proposition 6 this would represent cross-learning from information about other variables. To use the level of interest rates (5) to forecast inflation, for example, a household must employ a model of how interest rates relate to inflation. Similarly, to use past experienced price changes (1-2), households need a model of the persistence of inflation.<sup>44</sup> The only answers that represent the use of direct information about inflation are options 3 and 4.

**Question 2b** *What were the most important factors that led you [to change (insert their response to how expectation has changed)] your expectation of prices in the shops over the next 12 months?*

*Please select up to 4:*

1. *How prices have changed in the shops recently, over the last 12 months*
2. *How prices have changed in the shops, on average, over the longer term i.e the last few years*
3. *Reports of current inflation in the media*
4. *Discussion of the prospects for inflation in the media*
5. *The level of interest rates*
6. *The inflation target set by the government*
7. *The current strength of the UK economy*
8. *Expectations about how economic conditions in the UK are likely to evolve*
9. *The level of the exchange rate (the value of sterling)*
10. *Other factors*
11. *None*

Asked: 2017

Information indicator: =1 if items 3 or 4 selected, =0 otherwise.

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<sup>44</sup>Macaulay and Moberly (2026) find this perceived persistence is very heterogeneous across households. Note that strictly, option 3 also concerns past price changes, so the assumption here is that media reports of inflation tend to discuss both current and future inflation simultaneously, or (equivalently) that households interpret the “current inflation” mentioned in option 3 to mean what is happening at the present moment, not past changes over the previous year as economists may strictly mean it. Appendix D.3 shows that the results below are robust to various small changes to this definition of the information indicator.

**Question 2c** *What were the most important factors that led you to change/not change your expectation of prices in the shops in the longer term?*

1. *How prices have changed in the shops recently, over the last 12 months*
2. *How prices have changed in the shops, on average, over the longer term i.e the last few years*
3. *Reports of current inflation in the media*
4. *Discussion of the prospects for inflation in the media*
5. *The level of interest rates*
6. *The inflation target set by the government*
7. *The current strength of the UK economy*
8. *Expectations about how economic conditions in the UK are likely to evolve*
9. *The level of the exchange rate (the value of sterling)*
10. *Other factors*
11. *None*

Asked: 2018, 2019

Information indicator: =1 if items 3 or 4 selected, =0 otherwise.

**Question 2d** *When you said prices would go up in the next 12 months, how important were the following things in getting to that answer?*

*For each option, possible answers are:*

- *Very important*
- *Fairly important*
- *Not very important*
- *Not at all important*
- *Don't know*
- *Refused*

*Options:*

1. *How prices have changed in the shops in your most recent visits (i.e. the last 1 to 6 months).*
2. *How prices have changed in the shops over the longer term (i.e. the last 12 months or more)*
3. *The current level of interest rates.*
4. *The current strength of the British Economy.*
5. *The inflation target set by the government.*
6. *Reports on inflation outlook in the media.*
7. *Reports of VAT changes in the media.*
8. *Other factor(s).*

Asked: 2009, 2010, 2011, 2013

Information indicator: =1 if 'very important' selected for option 6, =0 otherwise.

**Question 2e** *And which, if any, of the same factors were important in getting to your expectation of how prices will change over the longer term (say in 5 years time)?*

1. *How prices have changed in the shops in your most recent visits (i.e. the last 1 to 6 months).*
2. *How prices have changed in the shops over the longer term (i.e. the last 12 months or more)*
3. *The current level of interest rates.*
4. *The current strength of the British Economy.*
5. *The inflation target set by the government.*
6. *Reports on inflation outlook in the media.*
7. *Reports of VAT changes in the media.*
8. *Other factor(s).*

Asked: 2011, immediately after Question 2d

Information indicator: =1 if item 6 selected, =0 otherwise.

**Question 2f** *What were the most important factors in getting to your expectation for how prices in the shops would change over the next 12 months?*

*Please select up to 4:*

1. *How prices have changed in the shops recently, over the last 12 months*
2. *How prices have changed in the shops, on average, over the longer term i.e the last few years*
3. *Reports of current inflation in the media*
4. *Discussion of the prospects for inflation in the media*
5. *The level of interest rates*
6. *The inflation target set by the government*
7. *The current strength of the UK economy*
8. *Expectations about how economic conditions in the UK are likely to evolve*
9. *Other factors*
10. *None*

Asked: 2016

Information indicator: =1 if items 3 or 4 selected, =0 otherwise.

**Question 2g** *And what were the most important factors in getting to your expectation for how prices in the shops would change over the longer term (say in 5 years' time)?*

*Please select up to 4:*

1. *How prices have changed in the shops recently, over the last 12 months*
2. *How prices have changed in the shops, on average, over the longer term i.e the last few years*
3. *Reports of current inflation in the media*
4. *Discussion of the prospects for inflation in the media*

5. *The level of interest rates*
6. *The inflation target set by the government*
7. *The current strength of the UK economy*
8. *Expectations about how economic conditions in the UK are likely to evolve*
9. *Other factors*
10. *None*

Asked: 2016

Information indicator: =1 if items 3 or 4 selected, =0 otherwise.

## D.2 The relationship of planned consumption with measured information and subjective models

To confirm that the survey measures of information and subjective models uncover meaningful aspects of household beliefs, I consider how they correlate with planned consumption behavior. To this end, I use the following survey question:

**Question 3** *Which, if any, of the following actions are you taking, or planning to take, in the light of your expectations of price changes over the next twelve months?*

- *Cut back spending and save more.*

Crucially, this asks about consumption choices which are explicitly driven by expected inflation.<sup>45</sup> A household answering ‘yes’ to this question, and who reports elsewhere in the survey that they expect prices to rise in the next year, is therefore indicating that  $dc_{it}/dE_{it}p_{t+1} < 0$ . A question that only asked about consumption or consumption changes, without reference to the cause of the behavior, would conflate this with reactions to expectations of other variables, which might also be influenced by the same shocks as expected inflation, either directly or through cross-learning. Question 3 is therefore informative about the sign of  $\frac{dc_{it}}{dE_{it}p_{t+1}}$ . If current prices are taken as given by the household, then this is the same as the sign of  $\tilde{\chi}_i = \frac{dc_{it}}{dE_{it}\pi_{t+1}}$ .

The vast majority of respondents (98%) expect positive inflation over the next 12 months.<sup>46</sup> For

<sup>45</sup>Another question in the survey asks if the respondent will “bring forward major purchases such as furniture or electrical goods” as a result of expected inflation. I do not use this for two reasons. First, as Nunes and Park (2020) note, the question refers specifically to durable goods, which may not respond to prices in the same way as aggregate consumption, the object of interest. Second, it is very rarely chosen: just 6% of respondents said they would bring forward major purchases. In contrast, 40% report that they will cut back spending and save more. Any estimation on this variable will therefore be heavily influenced by a small subset of agents.

<sup>46</sup>The analysis in this section excludes any households who report expecting zero inflation over the next 12 months, or who do not answer the inflation expectation question, as Question 3 is difficult to interpret for these households. I discuss the appropriate counterfactual implicit in the question below. Including these people, 79% of respondents to Question 3 expect positive inflation, 7% expect zero inflation, 2% expect deflation, and 12% do not answer.



these households, yes and no responses to Question 3 respectively indicate that:

$$\frac{dc_{it}}{dE_{it}p_{t+1}} \begin{cases} < 0 & \text{if answer yes} \\ \geq 0 & \text{if answer no} \end{cases} \quad (235)$$

For the minority who expect deflation, these inequalities are reversed: responding with ‘yes’ indicates consumption is being cut because of an expected fall in prices. I therefore define the following indicator:

$$\frac{\widetilde{dc}_{it}}{dE_{it}p_{t+1}} = \begin{cases} 1 & \text{if Q3='no' and } E_{it}\pi_{t+1} > 0 \\ 0 & \text{if Q3='yes' and } E_{it}\pi_{t+1} > 0 \\ 1 & \text{if Q3='yes' and } E_{it}\pi_{t+1} < 0 \\ 0 & \text{if Q3='no' and } E_{it}\pi_{t+1} < 0 \end{cases} \quad (236)$$

For those who expect inflation, this is equal to 1 if  $\frac{dc_{it}}{dE_{it}p_{t+1}} \geq 0$ , and equal to 0 if the reaction to expected price rises is strictly negative. The same is true of the minority who expect deflation, except that any household with  $\frac{dc_{it}}{dE_{it}p_{t+1}} = 0$  would respond ‘no’ to Question 3, and so is counted as if their response to expected price rises is strictly negative. The mislabeling is not a large issue, as less than 1% of respondents to Question 3 both expect deflation and answer ‘no’. The results below are robust to removing the few households who expect deflation (see Table 4 column 2).

Table 4 shows how this is related to the information indicator and the subjective models (responses to Question 1). Column 1 shows the results from estimating a probit regression of  $\frac{\widetilde{dc}_{it}}{dE_{it}p_{t+1}}$  on the information indicator interacted with subjective models (Question 1), plus the standard household controls and time fixed effects used above. The coefficient on information is significantly negative for those with negative subjective models of inflation, despite the fact that substitution effects imply  $\frac{dc_{it}}{dE_{it}p_{t+1}} \geq 0$  in many standard models. Being informed is therefore associated with a *lower* probability of responding positively to expected inflation for these households.

However, for those who believe inflation makes the economy stronger, being informed is associated with a significantly higher  $\Pr(\frac{dc_{it}}{dE_{it}\pi_{t+1}} \geq 0)$ . For those who believe inflation makes no difference, the average value of  $\Pr(\frac{dc_{it}}{dE_{it}\pi_{t+1}} \geq 0)$  is greater than for those with negative subjective models. This is true with and without information, consistent with the interpretation of these variables given above, since  $\frac{\widetilde{dc}_{it}}{dE_{it}p_{t+1}} = 1$  includes the case where  $\frac{dc_{it}}{dE_{it}\pi_{t+1}} = 0$ .

**Table 4:** Consumption response to inflation correlates with information, by subjective model

|                                                            | (1)<br>c response to $E\pi$ | (2)<br>c response to $E\pi$ |
|------------------------------------------------------------|-----------------------------|-----------------------------|
| information<br>indicator=1                                 | -0.213***<br>(0.0611)       | -0.224***<br>(0.0613)       |
| end up stronger                                            | 0.0108<br>(0.0891)          | 0.0392<br>(0.0906)          |
| information<br>indicator=1 $\times$ end up stronger        | 0.348*<br>(0.185)           | 0.313*<br>(0.186)           |
| make little<br>difference                                  | 0.130**<br>(0.0594)         | 0.157***<br>(0.0600)        |
| information<br>indicator=1 $\times$ make little difference | 0.0240<br>(0.126)           | -0.0149<br>(0.128)          |
| dont know                                                  | 0.0958<br>(0.0833)          | 0.0978<br>(0.0846)          |
| information<br>indicator=1 $\times$ dont know              | -0.0158<br>(0.186)          | -0.0342<br>(0.187)          |
| Expected Inflation                                         | All                         | Exclude Deflation           |
| Controls                                                   | All                         | All                         |
| Time FE                                                    | Yes                         | Yes                         |
| Observations                                               | 4940                        | 4871                        |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ 

Note: The table reports the results of probit regressions of the  $\frac{\widehat{dc}_{it}}{dE_{it}\pi_{t+1}}$  indicator on the information indicator, interacted with responses to Question 1. The omitted category is a household with information indicator=0 who holds the belief that inflation makes the economy weaker. All regressions are weighted using the survey weights provided in the IAS.

This is consistent with individuals filtering information through their subjective models of the economy. If a household who believes inflation weakens the economy gets more information about future positive inflation, their subjective model implies that they should cut consumption, because bad times lie ahead. If instead a household believes inflation strengthens the economy, then they will react in the opposite way to the same inflation. The overall correlation of information and consumption response is negative because the majority of households believe inflation makes the economy weaker. This therefore supports the claim that the information indicator and answers to Question 1 reflect the information and subjective models used by households in making their consumption decisions.

The analysis here assumes that when asked whether they will cut back consumption and save

more, households are comparing their actions to a counterfactual in which there are no price rises over the next 12 months. An alternative possibility is that they are comparing with a consumption plan made in the past, in which case the relevant counterfactual is where expected inflation is unchanged from the level expected when the plan was made. I consider this in two ways, and find that the qualitative patterns in reported consumption responses to inflation are the same for households expecting inflation to increase or decrease relative to the previous year. It does not therefore appear that past inflation is the relevant counterfactual for most respondents.

**Table 5:** Consumption response to inflation correlates with information, by subjective model and sign of perceived  $E\pi$  change.

|                                           | (1)<br>$E\Delta\pi < 0$ | (2)<br>$E\Delta\pi = 0$ | (3)<br>$E\Delta\pi > 0$ |
|-------------------------------------------|-------------------------|-------------------------|-------------------------|
| Information=1                             | -0.140<br>(0.116)       | -0.305***<br>(0.101)    | -0.257**<br>(0.107)     |
| end up stronger                           | 0.0668<br>(0.164)       | -0.178<br>(0.151)       | 0.195<br>(0.165)        |
| Information=1<br>× end up stronger        | 0.586<br>(0.441)        | 0.349<br>(0.293)        | 0.397<br>(0.307)        |
| make little<br>difference                 | 0.165<br>(0.111)        | 0.136<br>(0.0957)       | 0.181<br>(0.112)        |
| Information=1<br>× make little difference | 0.129<br>(0.241)        | -0.300<br>(0.211)       | 0.113<br>(0.216)        |
| dont know                                 | 0.156<br>(0.176)        | 0.0293<br>(0.128)       | 0.0264<br>(0.167)       |
| Information=1<br>× dont know              | -0.141<br>(0.354)       | 0.469<br>(0.359)        | 0.117<br>(0.325)        |
| Controls                                  | All                     | All                     | All                     |
| Time FE                                   | Yes                     | Yes                     | Yes                     |
| Observations                              | 1384                    | 1876                    | 1463                    |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

*Note:* The table reports the results of probit regressions of the  $\frac{dc_{it}}{dE_{it}\pi_{t+1}}$  indicator on the information indicator, interacted with responses to Question 1, split by the sign of the respondent's inflation expectations. The omitted category in all cases is a household with information indicator=0 who holds the belief that inflation makes the economy weaker. All regressions are weighted using the survey weights provided in the IAS.

First, column 2 of Table 4 re-runs the regression in column 1, excluding any respondent who reports expecting prices to fall over the next year. All results are qualitatively the same as over the

full sample, showing that the few respondents expecting deflation are not driving the results.

Second, I split the sample by the sign of the respondent's expected change in inflation, computed as the sign of the difference between 12-month ahead inflation forecast and their perception of inflation over the previous 12 months. The results are in Table 5. The sample sizes in each group are substantially smaller than over the full sample, so some significance is lost, but importantly the signs of the key coefficients remain the same. In each group, households who believe inflation makes the economy weaker are less likely to have  $\frac{dc_{it}}{dE_{it}\pi_{t+1}} \geq 0$  when they get inflation information. For households who believe inflation makes the economy stronger, this effect is reversed. The similarity of these patterns suggests that most respondents use 'no price change' as the counterfactual when answering Question 3, not 'no inflation change'. If the latter was used, we would expect to see changes of sign across the columns in Table 5, as a household expecting a fall in inflation would be reporting  $-1 \times \frac{dc_{it}}{dE_{it}\pi_{t+1}}$ , while one expecting a rise in inflation would report  $\frac{dc_{it}}{dE_{it}\pi_{t+1}}$ .

### D.3 Cross-sectional patterns in information on inflation

**Table 6:** Information correlates with subjective models, split by information question type

|                           | (1)<br>Revision     | (2)<br>Point           | (3)<br>Short horz.     | (4)<br>Extra Qs        | (5)<br>Q2d wider       | (6)<br>+Other          |
|---------------------------|---------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| end up<br>stronger        | 0.0575<br>(0.0380)  | -0.0335<br>(0.0218)    | -0.0123<br>(0.0206)    | 0.00114<br>(0.0196)    | -0.00126<br>(0.0196)   | -0.00779<br>(0.0205)   |
| make little<br>difference | -0.0191<br>(0.0233) | -0.0331**<br>(0.0155)  | -0.0392***<br>(0.0141) | -0.0310**<br>(0.0132)  | -0.0312**<br>(0.0131)  | -0.0429***<br>(0.0139) |
| dont know                 | -0.0408<br>(0.0297) | -0.0715***<br>(0.0206) | -0.0622***<br>(0.0192) | -0.0663***<br>(0.0174) | -0.0472***<br>(0.0180) | -0.0663***<br>(0.0191) |
| Controls                  | All                 | All                    | All                    | All                    | All                    | All                    |
| Time FE                   | Yes                 | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations              | 2364                | 5906                   | 6848                   | 8306                   | 8270                   | 8270                   |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

*Note:* The table reports the average marginal effects from estimating probit regressions of the information indicators constructed from subsets of the questions listed in Appendix D.1 on the responses to Question 1. The omitted category is the belief that inflation makes the economy weaker. All regressions are weighted using the survey weights provided in the IAS.

Columns 1-3 of Table 6 show the results of probit regressions of the information indicator on subjective models, controls, and period fixed-effects, for three subsamples. The first only uses questions about the information used to arrive at the respondent's *change* in expected inflation, and the second uses only questions about information used to form point forecasts. The third column excludes questions relating to forecast horizons longer than 12 months. The signs of the marginal

effects are the same as in the main exercise in Table 1, though they are not significant in the case of the revisions questions, as the sample size is small.

**Table 7:** Information correlates with subjective models, with selection correction

|                                 | (1)<br>Information     | (2)<br>Information     |
|---------------------------------|------------------------|------------------------|
| end up stronger                 | -0.0178<br>(0.0172)    | -0.0183<br>(0.0172)    |
| make little<br>difference       | -0.0306**<br>(0.0120)  | -0.0315***<br>(0.0121) |
| dont know                       | -0.0575***<br>(0.0172) | -0.0579***<br>(0.0173) |
| Inverse Mills ratio             | -0.282***<br>(0.0820)  | -0.0696**<br>(0.0355)  |
| <i>Selection stage</i>          |                        |                        |
| Economic Literacy               | 0.205***<br>(0.0226)   |                        |
| HH does not know past $\pi$     |                        | -0.876***<br>(0.0365)  |
| $r$ affects $\pi$ in 1-2 months |                        | 0.0334<br>(0.0236)     |
| $r$ affects $\pi$ in 1-2 yrs    |                        | 0.0882***<br>(0.0231)  |
| Pseudo- $R^2$ (selection)       | 0.103                  | 0.127                  |
| Controls                        | All                    | All                    |
| Time FE                         | Yes                    | Yes                    |
| Observations                    | 18026                  | 18026                  |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

*Note:* The table reports the coefficients from estimating a linear regression of the information indicator defined in Section 5.1 on the responses to Question 1, augmented with the inverse Mills ratio from a first-stage probit regression of whether the information indicator is observed on measures of economic literacy defined above. The omitted category is the belief that inflation makes the economy weaker. The selection stage is only run for quarters in which the information questions were asked. The model is run using the 2-step limited information method in Heckman (1979). Time fixed effects and controls as in Table 1 are included in both stages.

The remaining columns of Table 6 repeat the regression for broader definitions of the information dummy than that used in Table 1. In the fourth column, the information indicator includes Questions 2e and 2g. In the fifth column, I extend the criteria for setting the information indicator equal to 1 in Question 2d to account for the fact that some people may be unwilling to select the highest importance box for any information source. I therefore set the information indicator to 1 if in answer to Question 2d, the respondent selects ‘very important’ for direct inflation information (as before), or if they do not select ‘very important’ for any option, but do respond that four or fewer options

were ‘fairly important’, and direct inflation information is among them. In the final column, I set the information indicator =1 if the household chooses a direct information source or ‘Other’, in case this includes direct information sources (e.g. checking the Bank of England published forecasts). In all of these, the results are robust.

To account for possible selection bias from missing observations, I estimate a version of Table 1 amended for selection as in Heckman (1979). As in Michelacci and Paciello (2024), I predict observing the relevant survey response using a measure of economic literacy. Here the relevant response is only the information indicator, as there are no missing values for Question 1. Following Michelacci and Paciello (2024), economic literacy is measured with three indicators: the household reports a value for perceived current inflation, and they answer ‘agree’ or ‘strongly agree’ to the statements “a rise in interest rates makes prices in the high street rise more slowly in the short term (say a month or two)” and “a rise in interest rates makes prices rise more slowly in the medium term (say a year or two)”. I estimate versions of the model with this as an aggregate index (=1 if and only if the household scores on all components), and with the components disaggregated. The results are in Table 7. The predictors used in the first stage are strongly significant. Qualitatively the second-stage results are unchanged from Table 1, and the quantitative differences are small.

## D.4 Time series patterns in subjective models of inflation

Bhandari et al. (2024) also study the time series of responses to Question 1, and conclude that households are more pessimistic about inflation when output growth is low. To explore this, I regress the proportion of households responding ‘end up weaker’ on realized annual CPI inflation and quarterly GDP growth. The results are in column 2 of Table 8. Consistent with Bhandari et al. (2024), the coefficient on GDP growth is significantly negative. However, the  $R^2$  is only slightly higher than that of a regression on inflation only (column 1), so GDP growth does not account for much of the variation in survey answers. Indeed, GDP growth does not have any significant relationship with the proportion of households with a negative view of inflation outside of the four worst months of the Great Recession (column 3).

To explore which measure of inflation affects subjective models, Table 9 reports the results of regressing an indicator variable for if the respondent reports a negative subjective model on a variety of inflation measures. The first column uses CPI inflation, so is very similar to the time-series regression in Table 8. Columns 2-4 use more granular measures of the inflation rate experienced by different households, split by whether they are above retirement age (65), above median income, and by their housing tenure. Inflation rates split by these characteristics are provided by the ONS.<sup>47</sup> Column 5 uses perceived current inflation. The different realized inflation measures are strongly

<sup>47</sup>Finer decompositions of inflation by household characteristics are not reliable, given the data available for the UK (see e.g. Dawber et al., 2022).

correlated, so cannot be included jointly. Although the coefficient sizes vary as the different inflation rates have different levels of volatility, in all cases higher inflation is associated with a significantly greater probability of reporting a negative subjective model. The  $R^2$  is highest for perceived inflation, supporting the view that inflation information affects subjective models.

**Table 8:** Regressions of the proportion of households answering weaker to Question 1 on aggregate variables.

|                  | (1)                    | (2)                     | (3)                    |
|------------------|------------------------|-------------------------|------------------------|
|                  | Proportion weaker      | Proportion weaker       | Proportion weaker      |
| Inflation        | 0.0568***<br>(0.00489) | 0.0517***<br>(0.00479)  | 0.0501***<br>(0.00469) |
| GDP growth       |                        | -0.0261***<br>(0.00869) | -0.0110<br>(0.0180)    |
| Constant         | 0.466***<br>(0.0109)   | 0.487***<br>(0.0123)    | 0.482***<br>(0.0152)   |
| Omitted quarters | None                   | None                    | 2008Q2-2009Q1          |
| R-squared        | 0.615                  | 0.647                   | 0.554                  |
| Observations     | 70                     | 70                      | 66                     |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

*Note:* The table reports the results of regressing the proportion of households answering Question 1 that inflation makes the economy weaker on annual CPI inflation and quarter-on-quarter real GDP growth. Proportions are computed using survey weights.

**Table 9:** Probability of reporting negative subjective model by experienced and perceived inflation

|                   | (1)                    | (2)                    | (3)                    | (4)                    | (5)                     |
|-------------------|------------------------|------------------------|------------------------|------------------------|-------------------------|
|                   | Weaker                 | Weaker                 | Weaker                 | Weaker                 | Weaker                  |
| Inflation         | 0.0510***<br>(0.00177) | 0.0463***<br>(0.00170) | 0.0457***<br>(0.00165) | 0.0292***<br>(0.00137) | 0.0254***<br>(0.000720) |
| Inflation measure | CPI                    | by retirement          | by income              | by housing             | perceived               |
| Controls          | All                    | All                    | All                    | All                    | All                     |
| R-squared         | 0.0303                 | 0.0286                 | 0.0292                 | 0.0237                 | 0.0371                  |
| Observations      | 68269                  | 68269                  | 68269                  | 68269                  | 68269                   |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

*Note:* The table reports the results of estimating a linear probability model of whether a respondent reports that inflation makes the economy weaker in response to Question 1 on various measures of inflation. These are annual CPI inflation, inflation split by whether the respondent is of retirement age, split by whether the respondent has above or below median income, split by the respondent's housing tenure, and finally the respondent's perceived current inflation. Sample begins in 2006 Q1, as this is when the ONS sub-group inflation data is available from. Households not reporting a perceived rate of inflation are dropped in all regressions. All regressions are weighted using the survey weights provided in the IAS.

**Table 10:** Regressions of the proportion of households giving each answer to Question 1 on aggregate variables.

|                      | (1)                     | (2)                     | (3)                     |
|----------------------|-------------------------|-------------------------|-------------------------|
|                      | Proportion              | Proportion              | Proportion              |
| <i>Stronger</i>      |                         |                         |                         |
| Inflation            | -0.0123***<br>(0.00193) | -0.0116***<br>(0.00215) | -0.0108***<br>(0.00221) |
| GDP growth           |                         | 0.00346<br>(0.00363)    | -0.00392<br>(0.00646)   |
| Constant             | 0.104***<br>(0.00431)   | 0.102***<br>(0.00550)   | 0.104***<br>(0.00638)   |
| <i>No difference</i> |                         |                         |                         |
| Inflation            | -0.0292***<br>(0.00303) | -0.0262***<br>(0.00313) | -0.0257***<br>(0.00314) |
| GDP growth           |                         | 0.0150***<br>(0.00473)  | 0.0106<br>(0.0107)      |
| Constant             | 0.277***<br>(0.00772)   | 0.264***<br>(0.00883)   | 0.266***<br>(0.0103)    |
| <i>Don't know</i>    |                         |                         |                         |
| Inflation            | -0.0154***<br>(0.00249) | -0.0139***<br>(0.00262) | -0.0135***<br>(0.00267) |
| GDP growth           |                         | 0.00762*<br>(0.00423)   | 0.00428<br>(0.00987)    |
| Constant             | 0.153***<br>(0.00687)   | 0.147***<br>(0.00757)   | 0.148***<br>(0.00884)   |
| Omitted quarters     | None                    | None                    | 2008Q2-2009Q1           |
| Observations         | 70                      | 70                      | 66                      |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

*Note:* The table reports the results of regressing the proportion of households giving each answer to Question 1 (except 'weaker') on annual CPI inflation and quarter-on-quarter real GDP growth. Proportions are computed using survey weights. The  $R^2$ s of the core regressions in column 1 are 0.388 (stronger), 0.534 (no difference), and 0.355 (don't know).

Similar patterns in reverse are observed for the other answers. Table 10 repeats the regressions of Table 8, replacing the dependent variable with the proportion of respondents choosing each of the other answers to Question 1. In all cases, inflation accounts for a large share of the variation in survey answers, and higher inflation is associated with significantly lower proportions giving each answer. Higher GDP growth is associated with higher proportions on these other answers, but that



relationship is not significantly different from zero for any answer when excluding the worst of the Great Recession.

To test if the distribution of beliefs about inflation shifts when the economy reaches the ZLB, I estimate an ordered probit regression of subjective models of inflation on an indicator for the zero lower bound period, and a variety of controls.<sup>48</sup> A response that inflation makes the economy stronger is coded as the highest value, and inflation makes the economy weaker is the lowest value (I exclude the ‘don’t know’ answers). A positive coefficient on the zero lower bound period would therefore imply a shift towards believing inflation makes the economy stronger, as we would expect if households follow a standard New Keynesian model. This is not what the results in Table 11 show: there is no significant shift towards a positive view of inflation in the ZLB period.

**Table 11:** Ordered probit regressions of subjective models of inflation on whether the economy is at the zero lower bound on nominal interest rates.

|              | (1)                   | (2)                   | (3)                   |
|--------------|-----------------------|-----------------------|-----------------------|
|              | Subjective model      | Subjective model      | Subjective model      |
| ZLB          | -0.00801<br>(0.00937) | -0.00785<br>(0.00962) | -0.00513<br>(0.00972) |
| Controls     | None                  | Household             | Household + macro     |
| Observations | 83526                 | 83526                 | 83526                 |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

*Note:* The table reports the results of an ordered probit regression of answers to Question 1 on an indicator for whether the UK economy was at the zero lower bound, defined as the period from 2009Q2 to the end of 2019 (end of the sample). The ordering is: “stronger”, “no difference”, “weaker”. Those answering “don’t know” are omitted. All regressions are weighted using the survey weights provided in the IAS.

## D.5 Inflation perceptions, expectations, and subjective models

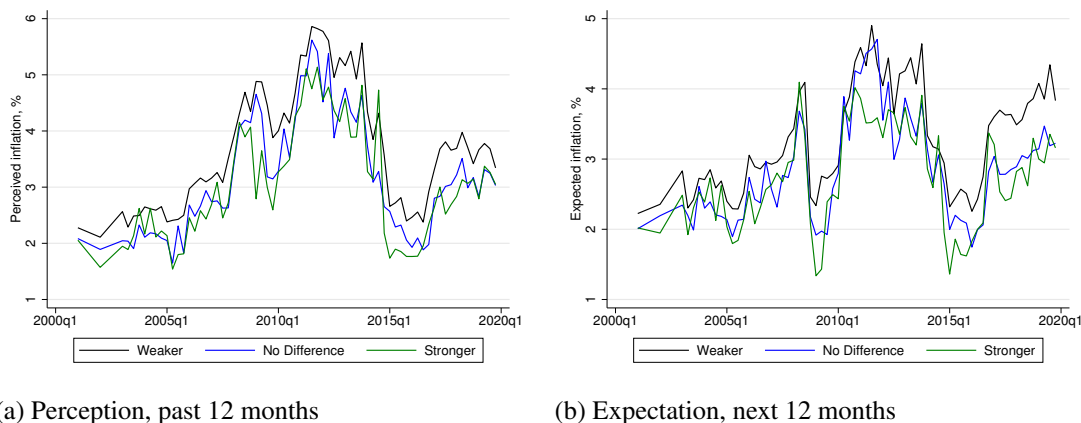
I now compare perceived and expected inflation across households with different subjective model beliefs. The IAS asks households to give point estimates for current and one-year-ahead annual inflation. For each question, responses consist of a 1 percentage-point bin between -5% and +10%, or end ranges  $\leq -5\%$ ,  $\geq 10\%$ . I code perceptions and expectations at the midpoint of the selected bin, with the lowest and highest bins coded as -5.5% and 10.5% respectively.

Figure 8 shows the time series of mean perceived and expected inflation by qualitative subjective model of inflation. Households who believe inflation weakens the economy perceive higher current inflation, and expect higher future inflation, than those who believe inflation makes no difference to the economy. They, in turn, perceive and expect higher inflation than those with positive subjective models of inflation.

<sup>48</sup>The first column of Table 11 has no controls, the second includes the set of household-level covariates used throughout the paper, and the third adds inflation and GDP growth.

The differences are large: Table 12 shows that even after controlling for the full set of available household characteristics, those with a negative model of inflation perceive that inflation has been 54 basis points higher than those with a neutral model, and 70 basis points higher than those with a positive model.<sup>49</sup> The gaps are similarly large and strongly significant for expected inflation.<sup>50</sup> These results are not driven by selection bias from missing observations for inflation perceptions and expectations.

**Figure 8:** Inflation perceptions and expectations by subjective model.



*Note:* Perceived inflation refers to beliefs about what inflation has been over the past 12 months, and expected inflation refers to expectations for the next 12 months. Averages for each variable are calculated using the survey weights provided in the IAS. Average perceptions and expectations among respondents answering ‘Don’t know’ to the subjective model question (Question 1) are omitted for figure clarity.

This is not driven by the households using different kinds of information to arrive at their perceptions and expectations: Table 1 shows that the households with positive subjective models use similar information sources to those with negative models. It is, however, consistent with information about high inflation causing households to update their subjective models towards more negative views. Although the exercises here do not identify the direction of causation, such a mechanism can parsimoniously account for Result 2 and Table 12.

**Robustness.** To account for possible selection bias from missing observations, I estimate a version of Table 12 amended for selection as in Heckman (1979). As in Appendix D.3, I predict observing perceived and expected inflation using the components of the economic literacy indicator in Michelacci and Paciello (2024), this time removing the component concerning whether perceived inflation is reported as this is closely related to the dependent variables of the regressions. The

<sup>49</sup>If anything, these gaps are larger in the more recent waves of the survey omitted from the main analysis. At the peak of the UK’s inflation in 2022Q4, the mean difference in inflation perceptions between those with negative and neutral models of inflation was 374 basis points. For 1-year ahead inflation expectations it was 280 basis points.

<sup>50</sup>Dräger et al. (2022) similarly find for German households that inflation expectations are higher among those reporting that they would prefer inflation to be lower.

**Table 12:** Perceived and expected inflation are higher for those with more negative subjective models.

|                           | (1)<br>Perceived inflation | (2)<br>Expected inflation |
|---------------------------|----------------------------|---------------------------|
| end up stronger           | -0.696***<br>(0.0371)      | -0.565***<br>(0.0353)     |
| make little<br>difference | -0.543***<br>(0.0226)      | -0.466***<br>(0.0207)     |
| dont know                 | -0.462***<br>(0.0315)      | -0.413***<br>(0.0294)     |
| Controls                  | Yes                        | Yes                       |
| Time FE                   | Yes                        | Yes                       |
| R-squared                 | 0.179                      | 0.113                     |
| Observations              | 85803                      | 85201                     |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ 

*Note:* The table reports the results of regressing perceived and expected inflation on respondent subjective models (responses to Question 1). The omitted category is the belief that inflation makes the economy weaker. All regressions are weighted using the survey weights provided in the IAS.

results are in Table 13. The predictors used in the first stage are strongly significant. Qualitatively the second-stage results are unchanged from Table 12, and the quantitative differences are small.

**Table 13:** Perceived and expected inflation are higher for those with more negative subjective models, with selection correction

|                                 | (1)                   | (2)                   |
|---------------------------------|-----------------------|-----------------------|
|                                 | Perceived inflation   | Expected inflation    |
| end up stronger                 | -0.724***<br>(0.0319) | -0.607***<br>(0.0301) |
| make little<br>difference       | -0.548***<br>(0.0213) | -0.478***<br>(0.0201) |
| dont know                       | -0.452***<br>(0.0292) | -0.407***<br>(0.0280) |
| Inverse Mills ratio             | 0.714***<br>(0.248)   | 0.168<br>(0.191)      |
| <i>Selection stage</i>          |                       |                       |
| $r$ affects $\pi$ in 1-2 months | 0.133***<br>(0.0209)  | 0.192***<br>(0.0209)  |
| $r$ affects $\pi$ in 1-2 yrs    | 0.278***<br>(0.0205)  | 0.310***<br>(0.0205)  |
| Pseudo- $R^2$ (selection)       | 0.0477                | 0.0536                |
| Controls                        | All                   | All                   |
| Time FE                         | Yes                   | Yes                   |
| Observations                    | 95339                 | 95339                 |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

*Note:* The table reports the coefficients from estimating a linear regression of perceived and expected inflation on the responses to Question 1, augmented with the inverse Mills ratio from a first-stage probit regression of whether the dependent variable is observed on measures of economic literacy. The omitted category is the belief that inflation makes the economy weaker. The model is run using the 2-step limited information method in Heckman (1979). Time fixed effects and controls as in Table 1 are included in both stages.

## E New Keynesian model further details

### E.1 Proof of Proposition 9

The household maximizes (35) subject to the budget constraint (36). With  $Y = B = 1$  normalized in steady state, goods market clearing  $Y_t = C_t$  implies  $C = 1$  in steady state as well.

The first-order conditions imply

$$C_t^{-\frac{1}{\sigma}} = \beta \bar{E}_t \left[ \frac{R_t}{\Pi_{t+1}} C_{t+1}^{-\frac{1}{\sigma}} \right] + \xi B_t^{-\eta}. \quad (237)$$

In steady state ( $C = B = 1, \Pi = 1$ ), this yields

$$1 = \beta R + \xi, \quad \text{so} \quad R = \frac{1 - \xi}{\beta}. \quad (238)$$

Log-linearizing (237) around the steady state gives

$$c_t = (1 - \xi) [\bar{E}_t c_{t+1} - \sigma(r_t - \bar{E}_t \pi_{t+1})] + \sigma \eta \xi b_t. \quad (239)$$

Log-linearizing the budget constraint (36) around  $C = B = Y = 1$  gives

$$c_t + b_t = R(r_{t-1} - \pi_t + b_{t-1}) + y_t - (R - 1)t_t, \quad (240)$$

where  $y_t$  denotes log-deviation of labor and profit income (equal to output in equilibrium) from steady state. Since fiscal policy is exogenous to the individual household's actions, we log-linearize the tax rule presented below (268) and use this to substitute out for taxes  $t_t$ , giving

$$c_t + b_t = R b_{t-1} + y_t. \quad (241)$$

**Recasting the system.** Using (241) to substitute out for  $b_t$  in (239) and rearranging, we obtain:

$$\bar{E}_t c_{t+1} = \frac{1 + \sigma \eta \xi}{1 - \xi} c_t - \frac{\sigma \eta \xi R}{1 - \xi} b_{t-1} + \sigma(r_t - \bar{E}_t \pi_{t+1}) - \frac{\sigma \eta \xi}{1 - \xi} y_t. \quad (242)$$

Together with the budget constraint, this yields the system

$$\bar{E}_t \begin{pmatrix} c_{t+1} \\ b_t \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{1 + \sigma \eta \xi}{1 - \xi} & -\frac{\sigma \eta \xi R}{1 - \xi} \\ -1 & R \end{pmatrix}}_{\equiv \mathbf{A}} \begin{pmatrix} c_t \\ b_{t-1} \end{pmatrix} + \begin{pmatrix} f_{1,t} \\ f_{2,t} \end{pmatrix}, \quad (243)$$

where  $f_{1,t} = \sigma(r_t - \bar{E}_t \pi_{t+1}) - \frac{\sigma \eta \xi}{1 - \xi} y_t$  and  $f_{2,t} = y_t$ . This system has one forward-looking (jump) variable,  $c_t$ , and one predetermined (state) variable,  $b_{t-1}$ . The determinant and trace of  $\mathbf{A}$  are

$$\det(\mathbf{A}) = \frac{R}{1 - \xi} = \frac{1}{\beta}, \quad \text{tr}(\mathbf{A}) = \frac{1 + \sigma \eta \xi}{1 - \xi} + R. \quad (244)$$

Let  $\lambda_1, \lambda_2$  denote the eigenvalues, satisfying

$$\lambda^2 - \text{tr}(\mathbf{A}) \lambda + \frac{1}{\beta} = 0. \quad (245)$$

For a unique saddle-path solution we require one eigenvalue inside and one outside the unit circle

(Blanchard and Kahn, 1980). Evaluating the characteristic polynomial at  $\lambda = 1$ :

$$p(1) = 1 - \text{tr}(\mathbf{A}) + \frac{1}{\beta} = \xi \left( \frac{1}{\beta} - \frac{1 + \sigma\eta}{1 - \xi} \right). \quad (246)$$

Saddle-path stability requires  $p(1) < 0$ , i.e.

$$\sigma\eta > R - 1 = \frac{1 - \xi - \beta}{\beta}, \quad (247)$$

which is satisfied for all calibrations in this paper. Under this condition,  $0 < \lambda_1 < 1 < \lambda_2$ .

**Saddle-path solution.** Conjecture that the solution takes the form  $c_t = \mu b_{t-1} + g_t$ , where  $\mu$  is a constant and  $g_t$  depends only on current and expected future exogenous variables.

Substituting the conjecture into the budget constraint gives  $b_t = (R - \mu) b_{t-1} + (y_t - g_t)$ , so the stable eigenvalue governs the dynamics of  $b_t$  on the saddle path:

$$\mu = R - \lambda_1. \quad (248)$$

Substituting the conjecture into (242) and matching coefficients on  $b_{t-1}$  yields the quadratic

$$(1 - \xi) \mu^2 - [(1 - \xi)R - 1 - \sigma\eta\xi] \mu - \sigma\eta\xi R = 0, \quad (249)$$

and the saddle-path root is the one satisfying  $R - \mu \in (0, 1)$ , i.e.  $\mu = R - \lambda_1$  where  $\lambda_1$  is the smaller root of (245).

Matching the remaining (non- $b_{t-1}$ ) terms gives

$$\bar{E}_t g_{t+1} = \lambda_2 g_t + f_{1,t} - \mu y_t, \quad (250)$$

where the coefficient on  $g_t$  equals  $\lambda_2$  (verified by  $\frac{1 + \sigma\eta\xi}{1 - \xi} + \mu = \text{tr}(\mathbf{A}) - R + R - \lambda_1 = \lambda_2$ ). Since  $\lambda_2 > 1$ , the unique bounded solution is obtained by solving forward:

$$g_t = \sum_{s=0}^{\infty} \lambda_2^{-(s+1)} \bar{E}_t [\mu y_{t+s} - f_{1,t+s}]. \quad (251)$$

**Expanding notation.** Define

$$\zeta \equiv \mu + \frac{\sigma\eta\xi}{1 - \xi}. \quad (252)$$

Then

$$\mu y_{t+s} - f_{1,t+s} = \zeta y_{t+s} - \sigma(r_{t+s} - \bar{E}_{t+s} \pi_{t+s+1}). \quad (253)$$

Define  $\Lambda \equiv \lambda_2^{-1} < 1$  as the effective discount factor for the consumption function. Substituting this into (251) gives:

$$g_t = \sum_{s=0}^{\infty} \Lambda^{s+1} \bar{E}_t \left[ \zeta y_{t+s} - \sigma(r_{t+s} - \pi_{t+s+1}) \right]. \quad (254)$$

Since  $c_t = \mu b_{t-1} + g_t$ , this implies:

$$c_t = \mu b_{t-1} + \Lambda \sum_{s=0}^{\infty} \Lambda^s \left[ \zeta \bar{E}_t y_{t+s} - \sigma(\bar{E}_t r_{t+s} - \bar{E}_t \pi_{t+s+1}) \right], \quad (255)$$

which is equation (37) in the main text.

## E.2 Model description

**Households.** A representative household chooses consumption  $C_t$ , labor supply  $N_t$ , and real one-period bond holdings  $B_t$  to maximize equation (35) subject to (36). Combining first-order conditions and log-linearizing around the steady state with  $Y = C = B = \Pi = 1$  yields:

$$c_t = (1 - \xi) \left[ \bar{E}_t c_{t+1} - \sigma(r_t - \bar{E}_t \pi_{t+1}) \right] + \sigma \eta \xi b_t \quad (256)$$

$$\psi n_t = w_t - \frac{1}{\sigma} c_t \quad (257)$$

where lower-case letters denote log-deviations from steady state. The derivation of the resulting consumption function (equation (37) in the main text) is presented in Appendix E.1.

**Firms.** A perfectly competitive final goods producer combines a continuum of intermediate goods varieties  $j \in [0, 1]$  using a CES production function:

$$Y_t = \left( \int_0^1 Y_{jt}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}} \quad (258)$$

where  $\varepsilon > 1$  is the elasticity of substitution. Cost minimization yields the demand function for variety  $j$ :

$$Y_{jt} = \left( \frac{P_{jt}}{P_t} \right)^{-\varepsilon} Y_t \quad (259)$$

and the aggregate price index  $P_t = \left( \int_0^1 P_{jt}^{1-\varepsilon} dj \right)^{\frac{1}{1-\varepsilon}}$ .

Each intermediate goods firm  $j$  produces using a linear production function:

$$Y_{jt} = A N_{jt}, \quad (260)$$

where aggregate productivity  $A$  is constant and common across firms. Cost minimization implies that real marginal cost is equal to the real wage for all firms:

$$MC_t = \frac{W_t}{A}. \quad (261)$$

Firms set prices subject to quadratic **Rotemberg (1982)** adjustment costs. Following **Bilbiie (2025)**, the adjustment cost is paid relative to last period's *aggregate* price level  $P_{t-1}$ , rather than the firm's own lagged price. This simplification makes the pricing problem static, allowing us to focus on the core demand mechanisms of interest.

Specifically, firm  $j$  chooses  $P_{jt}$  to maximize current-period profits net of adjustment costs:

$$\frac{P_{jt}}{P_t} Y_{jt} - W_t N_{jt} - \frac{\theta}{2} \left( \frac{P_{jt}}{P_{t-1}} - 1 \right)^2 Y_t \quad (262)$$

subject to the demand function (259) and the production function (260). Here  $\theta > 0$  governs the magnitude of the price adjustment cost. The first-order condition for  $P_{jt}$  is:

$$(1 - \varepsilon) \left( \frac{P_{jt}}{P_t} \right)^{-\varepsilon} \frac{Y_t}{P_t} + \varepsilon \frac{W_t}{A} \left( \frac{P_{jt}}{P_t} \right)^{-\varepsilon-1} \frac{Y_t}{P_t} - \theta \left( \frac{P_{jt}}{P_{t-1}} - 1 \right) \frac{Y_t}{P_{t-1}} = 0. \quad (263)$$

In a symmetric equilibrium ( $P_{jt} = P_t$  for all  $j$ ), this simplifies to:

$$\Pi_t(\Pi_t - 1) = \frac{\varepsilon}{\theta} (MC_t - \mathcal{M}^{-1}) \quad (264)$$

where  $\mathcal{M} \equiv \varepsilon/(\varepsilon - 1)$  denotes the flexible-price markup.

Log-linearizing we obtain:

$$\pi_t = \frac{\varepsilon - 1}{\theta} mc_t \quad (265)$$

From (261), (257), (260), and (40), real marginal costs in terms of output are:

$$mc_t = w_t = \psi n_t + \frac{1}{\sigma} c_t = \left( \psi + \frac{1}{\sigma} \right) y_t \quad (266)$$

Substituting into (265) yields the Phillips curve in the main text (38), with

$$\kappa \equiv \frac{\varepsilon - 1}{\theta} \left( \psi + \frac{1}{\sigma} \right). \quad (267)$$

**Monetary policy.** The central bank sets the nominal interest rate according to the standard Taylor rule (39), where  $\nu_{rt}$  is an AR(1) monetary policy shock:  $\nu_{rt} = \rho_r \nu_{r,t-1} + \epsilon_{rt}$ , with  $\epsilon_{rt} \sim \mathcal{N}(0, \sigma_r^2)$ .



**Fiscal policy.** The fiscal authority issues a constant real supply of bonds  $B = 1$  and levies lump-sum taxes to service its debt. The government budget constraint in levels is:

$$T_t = \frac{R_{t-1}}{\Pi_t} B - B = \frac{R_{t-1}}{\Pi_t} - 1 \quad (268)$$

where the second equality uses  $B = 1$ . That is, the government levies taxes equal to the net real interest payments on its outstanding debt. Since bond supply is constant, in equilibrium  $b_t = 0$  for all  $t$ , so we do not need to track  $b_t$  as a state variable.<sup>51</sup> Log-linearizing (268) yields:

$$t_t = \frac{R}{R-1}(r_{t-1} - \pi_t) \quad (269)$$

**Market clearing.** The goods market clears when aggregate output equals consumption (40). The labor market clears when aggregate labor demand from firms  $\int_0^1 N_{jt} dj$  equals labor supply  $N_t$ . In symmetric equilibrium with linear production, this is equivalent to  $Y_t = AN_t$ , i.e.  $y_t = n_t$ . The bond market clears by Walras' law given the household and government budget constraints.

### E.3 Temporary-information equilibrium

**Equilibrium definition.** For given distributions of  $\tau_{i\pi_t}$ ,  $\tilde{\chi}_i$ , and given exogenous shocks  $v_{rt}, \varepsilon_{\pi t}$ , the temporary-information equilibrium consists of  $\{c_t, r_t, \pi_t, y_t\}$  and signals  $s_{it}$  such that:

1. *Households:* households choose  $c_t$  to maximize expected lifetime utility (following (37)).
2. *Firms:* firms set prices to maximize expected lifetime profits, implying  $\pi_t$  follows (38).
3. *Monetary Policy:* policymakers choose the nominal interest rate  $r_t$  according to (39).
4. *Market Clearing:* the goods and labor markets clear ( $y_t$  satisfies equation (40), and  $y_t = n_t$ ).
5. *Signals:* inflation signals are determined as a function of realized inflation and idiosyncratic and aggregate noise according to (41).

**Rewriting the consumption function.** Using the consumption function (37), and substituting in the assumptions on agent information and subjective models described in Section 6.3, then imposing market clearing, we obtain

$$y_t = -\frac{\Lambda\sigma}{1-\zeta\Lambda}(r_t - \bar{E}_t\pi_{t+1}) + \Lambda \sum_{s=1}^{\infty} \Lambda^s \int \tilde{\chi}_i E_{it}\pi_{t+s} di \quad (270)$$

<sup>51</sup>In the log-linearized consumption function (37), the  $\mu b_{t-1}$  term therefore drops out in equilibrium, though the coefficients  $\mu$ ,  $\zeta$ , and  $\Lambda$  derived in Appendix E.1 still enter the other terms.

where

$$\tilde{\chi}_i = \frac{(\zeta - \sigma\phi_y)\chi_{iy\pi} + \sigma(\rho - \phi_\pi)}{1 - \zeta\Lambda} \quad (271)$$

summarizes the passthrough from  $E_{it}\pi_{t+s}$  to  $c_t$  before applying appropriate discounting.

Using equation (43), we evaluate the summation term to give

$$y_t = -\frac{\Lambda\sigma}{1 - \zeta\Lambda}(r_t - \bar{E}_t\pi_{t+1}) + \frac{\Lambda^2}{1 - \Lambda\rho} \int \tilde{\chi}_i E_{it}\pi_{t+1} di \quad (272)$$

**Causal effects of  $\bar{E}_t\pi_t$ .** For these, we use (43) once more to obtain

$$y_t = -\frac{\Lambda\sigma}{1 - \zeta\Lambda}(r_t - \rho\bar{E}_t\pi_t) + \frac{\Lambda^2\rho}{1 - \Lambda\rho} \int \tilde{\chi}_i E_{it}\pi_t di \quad (273)$$

Given the Gaussian signal structure (41) and AR(1) perceived law of motion for inflation (43), the inflation perception  $E_{it}\pi_t$  is formed according to

$$E_{it}\pi_t = (1 - \tau_{i\pi_t})\rho E_{it-1}\pi_{t-1} + \tau_{i\pi_t}s_{it}(\pi_t) \quad (274)$$

where  $\tau_{i\pi_t}$  is the Kalman gain of agent  $i$ .

To find the causal effects of inflation expectations on  $y_t$  in partial equilibrium, we hold  $\pi_t$ ,  $r_t$ , and signals fixed, except for the public noise in  $s_{it}(\pi_t)$ . This isolates the pure demand impulse from a change in perceived inflation. We then find:

$$\Omega_y^{PE}(\pi_t) = \frac{1}{\bar{\tau}_{\pi_t}} \frac{dy_t}{d\varepsilon_{\pi_t}} = \frac{\Lambda\rho\sigma}{1 - \zeta\Lambda} + \int \frac{\Lambda^2\rho\tilde{\chi}_i\tau_{i\pi_t}}{(1 - \Lambda\rho)\bar{\tau}_{\pi_t}} di \quad (275)$$

where following the notation of Section 3 I use  $\Omega_y(\pi_t)$  to denote the causal effect of expectations of  $\pi_t$  on  $y_t$ , rescaled such that aggregate expectations move by 1 unit.

In general equilibrium,  $\pi_t$  and  $r_t$  respond endogenously, and the change in  $\pi_t$  also feeds back into signals.<sup>52</sup>

$$\Omega_y^{GE}(\pi_t) = \Omega_y^{PE}(\pi_t) \cdot \frac{1 - \zeta\Lambda}{1 - \zeta\Lambda + \Lambda\sigma(\phi_\pi\kappa + \phi_y)} \quad (276)$$

**Causal effects of  $\bar{E}_t\pi_{t+1}$ .** We now return to equation (272). Given signals about both  $\pi_t$  and  $\pi_{t+1}$ , and the imposed degree of cross-learning from  $E_{it}\pi_t$  to  $E_{it}\pi_{t+1}$ , the expectation in the form of (7)

<sup>52</sup>In this case, this is the same as the effect that would be obtained if signals were held constant, because the rescaling to give a unit change in  $\bar{E}_t\pi_t$  exactly offsets the signal-feedback channel, as in the causal effects of inflation perceptions in the Calvo model in Section 2.

is<sup>53</sup>

$$E_{it}\pi_{t+1} = (1 - \tau_{i\pi_{t+1}})\rho E_{it}\pi_t + \tau_{i\pi_{t+1}}s_{it}(\pi_{t+1}). \quad (277)$$

I impose throughout that there is no cross-learning from  $E_{it}\pi_{t+1}$  to  $E_{it}\pi_t$ . This reflects the focus in existing empirical literature on the pass-through of short to longer-horizon inflation expectations, but not the reverse. Substituting (277) into (272) and differentiating with respect to the public noise in  $s_{it}(\pi_{t+1})$ , holding  $r_t, \pi_t$  constant, and rescaling, gives the PE causal effect of  $\bar{E}_t\pi_{t+1}$ .

$$\Omega_y^{PE}(\pi_{t+1}) = \frac{1}{\bar{\tau}_{\pi_{t+1}}} \frac{dy_t}{d\varepsilon_{\pi_{t+1}}} = \frac{\Lambda\sigma}{1 - \zeta\Lambda} + \int \frac{\Lambda^2 \tilde{\chi}_i \tau_{i\pi_{t+1}}}{(1 - \Lambda\rho)\bar{\tau}_{\pi_{t+1}}} di \quad (278)$$

We now turn to GE. Allowing  $r_t, \pi_t$ , and  $s_{it}(\pi_t)$  to respond to the shock implies that the unscaled causal effect is

$$\begin{aligned} \frac{dy_t}{d\varepsilon_{\pi_{t+1}}} = & -\frac{\Lambda\sigma}{1 - \zeta\Lambda}(\kappa\phi_\pi + \phi_y) \frac{dy_t}{d\varepsilon_{\pi_{t+1}}} + \frac{\Lambda\sigma}{1 - \zeta\Lambda} \int \left( \tau_{i\pi_{t+1}} + \rho\tau_{i\pi_t}(1 - \tau_{i\pi_{t+1}})\kappa \frac{dy_t}{d\varepsilon_{\pi_{t+1}}} \right) di \\ & + \int \frac{\Lambda^2 \tilde{\chi}_i}{(1 - \Lambda\rho)} \left( \tau_{i\pi_{t+1}} + \rho\tau_{i\pi_t}(1 - \tau_{i\pi_{t+1}})\kappa \frac{dy_t}{d\varepsilon_{\pi_{t+1}}} \right) di \end{aligned} \quad (279)$$

which rearranges to

$$\frac{dy_t}{d\varepsilon_{\pi_{t+1}}} = \frac{\bar{\tau}_{\pi_{t+1}}\Omega_y^{PE}(\pi_{t+1})}{D} \quad (280)$$

where  $D$  captures general equilibrium effects, and is defined as

$$D \equiv 1 + \frac{\Lambda\sigma}{1 - \zeta\Lambda}(\kappa\phi_\pi + \phi_y) - \frac{\Lambda\rho\kappa\sigma}{1 - \zeta\Lambda} \int \tau_{i\pi_t}(1 - \tau_{i\pi_{t+1}})di - \frac{\Lambda^2\rho\kappa}{1 - \Lambda\rho} \int \tilde{\chi}_i \tau_{i\pi_t}(1 - \tau_{i\pi_{t+1}})di \quad (281)$$

Given this, the effect of such an unscaled shock on  $\bar{E}_t\pi_{t+1}$  is

$$\frac{d\bar{E}_t\pi_{t+1}}{d\varepsilon_{\pi_{t+1}}} = \bar{\tau}_{\pi_{t+1}} + \rho\kappa \frac{dy_t}{d\varepsilon_{\pi_{t+1}}} \int \tau_{i\pi_t}(1 - \tau_{i\pi_{t+1}})di \quad (282)$$

$$= \bar{\tau}_{\pi_{t+1}} + \frac{\rho\kappa\bar{\tau}_{\pi_{t+1}}\Omega_y^{PE}(\pi_{t+1})}{D} \int \tau_{i\pi_t}(1 - \tau_{i\pi_{t+1}})di \quad (283)$$

---

<sup>53</sup>Note that, as in e.g. [Woodford \(2003\)](#) and [Angeletos and Lian \(2018\)](#) (among many others), I assume here that households do not invert equilibrium relationships to extract further information about inflation from other endogenous variables. This is consistent with the substantial dispersion in inflation perceptions in the IAS: if households could accurately invert equilibrium relationships and thus undo the information friction, they would all hold identical beliefs about  $\pi_t$ .

Dividing the unscaled effect in (279) by this gives the desired scaled causal effect

$$\Omega_y^{GE}(\pi_{t+1}) = \frac{\Omega_y^{PE}(\pi_{t+1})}{D + \rho\kappa\Omega_y^{PE}(\pi_{t+1}) \int \tau_{i\pi_t}(1 - \tau_{i\pi_{t+1}})di} \quad (284)$$

**Decomposition.** For all causal effects described above, the representative-agent component is found by taking the relevant equation (i.e.  $\Omega_y^{PE}(\pi_t)$ ,  $\Omega_y^{GE}(\pi_t)$ ,  $\Omega_y^{PE}(\pi_{t+1})$ , or  $\Omega_y^{GE}(\pi_{t+1})$ ) and replacing  $\tau_{i\pi_t} = \bar{\tau}_{\pi_t}$ ,  $\tau_{i\pi_{t+1}} = \bar{\tau}_{\pi_{t+1}}$ ,  $\chi_{iy\pi} = \bar{\chi}_{y\pi}$ . The narrative heterogeneity component is then the total effect minus the representative-agent component. For causal effects in PE, this yields a precise linear decomposition as in Proposition 6. In GE, the total effect is a nonlinear combination of representative-agent and narrative-heterogeneity terms, so the representative-agent component should be interpreted as the GE causal effect if there was no heterogeneity, but the remaining narrative heterogeneity component cannot be interpreted as a pure covariance as in Proposition 6.

**FIRE benchmarks.** In Table 2, I compute causal effects assuming FIRE. For these, take each causal effect definition above, and set  $\chi_{iy\pi} = \kappa^{-1}$ , and  $\tau_{i\pi_t}, \tau_{i\pi_{t+1}} \rightarrow 1$ . I take the limit here, rather than directly imposing  $\tau_{i\pi_t}, \tau_{i\pi_{t+1}} = 1$ , as with Bayesian updating  $\tau_{i\pi_t}, \tau_{i\pi_{t+1}} = 1$  would imply there is no noise in the signals at all, leaving no scope for the shocks to  $\varepsilon_{\pi_t}$  and  $\varepsilon_{\pi_{t+1}}$  that I use to compute causal effects.

## E.4 Calibration

Table 14 gives a full list of the calibrated parameters. Steady state  $A$  is omitted, as it is simply adjusted to generate the normalization  $Y = 1$ .

**Table 14:** Calibration for Section 6

| Parameter | Value | Source                              | Parameter  | Value  | Source                    |
|-----------|-------|-------------------------------------|------------|--------|---------------------------|
| $\beta$   | 0.905 | Target $R = 1.01$                   | $\phi_\pi$ | 1.5    | Standard                  |
| $\eta$    | 2.5   | Kaplan and Violante (2018)          | $\phi_y$   | 0.125  | Standard                  |
| $\lambda$ | 0.8   | Standard                            | $\chi_+$   | 7.245  | Inverse NKPC slope        |
| $\xi$     | 0.086 | Target $MPC = 0.35$                 | $\chi_-$   | -2.676 | Target $\tilde{\chi}_i^2$ |
| $\rho$    | 0.441 | AR(1) regression                    | $\psi$     | 1      | Standard                  |
| $\sigma$  | 1     | Standard                            | $p_+$      | 0.079  | IAS                       |
| $\tau_H$  | 0.276 | Target average Kalman gain estimate | $p_-$      | 0.588  | IAS                       |
| $\tau_L$  | 0     | Normalization                       |            |        |                           |

The Euler equation in steady state (238) implies a target for  $R$  is not sufficient to pin down  $\beta$  independently of  $\xi$ . I therefore also target the MPC out of a change in  $y_t$ , which is increasing in  $\xi$ . I target an impact MPC of 0.35, the mean of the empirical studies surveyed in Sokolova

(2023). These targets imply a substantially lower  $\beta$  than typically used in models without bonds in utility, consistent with the findings in e.g. Auclert et al. (2024). These parameters imply  $\mu = 0.353, \zeta = 0.589, \Lambda = 0.594$ .

For the slope of the Phillips curve, I make use of the well-known equivalence between Rotemberg- and Calvo-derived New Keynesian Phillips curves (NKPCs) up to first order (Ascari and Rossi, 2012). These two models give the same NKPC if

$$\frac{\varepsilon - 1}{\theta} = \frac{(1 - \lambda)(1 - \beta\lambda)}{\lambda} \quad (285)$$

where  $\lambda$ , as in Section 2, is the probability a firm cannot reset its price each period. I choose  $\lambda = 0.8$ , which is a standard value in the literature consistent with an average price duration of 5 quarters.

For  $\rho$ , I estimate an AR(1) model on quarterly UK CPI inflation data from 2001 Q2-2019 Q4. To do this, I take two approaches. In both, the raw data are monthly observations of the CPI All Items Index (not seasonally adjusted, base 2015 = 100; ONS series D7BT). I first aggregate to quarterly frequency by averaging the three monthly index values within each quarter. Quarterly log inflation is then  $\pi_t^q = \ln(\text{CPI}_t / \text{CPI}_{t-1})$ , where  $\text{CPI}_t$  is the quarterly average of the monthly index.

Since at the time of writing the UK does not publish officially seasonally adjusted CPI data (Dixon and Michail, 2025), I handle seasonality in two ways. Column 1 of Table 15 works with the unadjusted quarterly log inflation series and includes quarter-of-year fixed effects directly in the AR(1) regression. This absorbs the average seasonal pattern within the regression without pre-adjusting the data.

Column 2 instead applies a classical multiplicative decomposition before estimation. A  $2 \times 12$  centered moving average of the monthly index estimates the trend-cycle component; dividing through gives seasonal-irregular ratios. Monthly seasonal factors are computed as the mean ratio for each calendar month across all available years, normalized so the twelve factors average to 1. The seasonally adjusted monthly CPI is then the raw index divided by the corresponding factor. These adjusted monthly values are averaged within each quarter before computing log inflation and estimating the AR(1) regression without quarter fixed effects.

Both approaches yield an estimated persistence parameter of  $\rho = 0.44$ , as reported in Table 15. I use  $\rho = 0.441$  (Column 1) in the calibration.

The steady state values of  $p_+$  and  $p_-$  are taken from the average proportion of households responding “stronger” and “weaker” to Question 1 across all waves of the IAS.

With  $\chi_+$  set to the true inverse NKPC slope  $\kappa^{-1}$ , I set  $\chi_-$  such that  $\tilde{\chi}_i^2$  is the same in both cases. Formally, this implies

$$\chi_- = - \left( \chi_+ + \frac{2\sigma(\rho - \phi_\pi)}{\zeta - \sigma\phi_y} \right). \quad (286)$$

**Table 15:** AR(1) inflation regressions

|                 | (1)                 | (2)                 |
|-----------------|---------------------|---------------------|
|                 | CPI inflation       | CPI inflation       |
| L.CPI inflation | 0.441***<br>(0.128) | 0.440***<br>(0.127) |
| Quarter FE      | Yes                 | No                  |
| R-squared       | 0.549               | 0.197               |
| Observations    | 75                  | 75                  |

Robust standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

*Note:* The table reports estimates from regressing quarterly UK CPI log inflation on its own lag. Sample period: 2001 Q2–2019 Q4. In Column 1, the raw (unadjusted) quarterly series is used and quarter-of-year fixed effects absorb seasonal variation within the regression. In Column 2, the monthly CPI is seasonally adjusted prior to quarterly aggregation using a classical multiplicative decomposition (ratio-to-moving-average method with a  $2 \times 12$  centered moving average), and no quarter fixed effects are included.

Note that this implies a magnitude of  $\chi_-$  which is less than half that of  $\chi_+$ , which is itself derived from rational expectations.

Finally, with  $\tau_L$  set to 0, I set  $\tau_H$  such that

$$E_I[\tau_i] \equiv (p_+ + p_-)\tau_H + (1 - p_- - p_+)\tau_L = \bar{\tau}^* \quad (287)$$

where  $\bar{\tau}^*$  is an estimate of the average Kalman gain in inflation perceptions across all households and survey waves. To compute this, I take Equation (274) and average across households to give:

$$\bar{E}_t \pi_t = \bar{\tau}_{\pi_t} \pi_t + (1 - \bar{\tau}_{\pi_t}) \rho \bar{E}_{t-1} \pi_{t-1} + \bar{\tau}_{\pi_t} \varepsilon_{\pi t} \quad (288)$$

where I have used the fact that all households are calibrated to have the same  $\rho$ .

To take this to the survey data, I next accommodate a timing discrepancy: equation (288) describes quarterly inflation perceptions, but the IAS survey asks about perceived inflation over the past 12 months. To make the model and data consistent, I aggregate equation (288) over four quarters. Writing (288) at dates  $t, t-1, t-2, t-3$  and summing gives

$$\bar{E}_t^A \equiv \sum_{s=0}^3 \bar{E}_{t-s} \pi_{t-s} = \bar{\tau}_{\pi_t} \pi_t^A + (1 - \bar{\tau}_{\pi_t}) \rho \bar{E}_{t-1}^A + \bar{\tau}_{\pi_t} u_t \quad (289)$$

where  $\pi_t^A \equiv \sum_{s=0}^3 \pi_{t-s}$  is realized annual inflation,  $\bar{E}_t^A$  is annual perceived inflation, and  $u_t \equiv \sum_{s=0}^3 \varepsilon_{\pi_{t-s}}$  is the sum of public noise over the four quarters. We can therefore use average inflation perceptions in the IAS to estimate:

$$\bar{E}_t^A = \gamma_1 \pi_t + \gamma_2 \bar{E}_{t-1}^A + \nu_t, \quad (290)$$

restricting  $\gamma_2 = \rho(1 - \gamma_1)$ , where  $\rho$  is as in Table 14. The estimated  $\gamma_1$  gives an estimate of the average Kalman gain across the population.

However, in doing this, it is important to acknowledge that the composite error  $u_t$  contains  $\varepsilon_{\pi_{t-1}}, \varepsilon_{\pi_{t-2}},$  and  $\varepsilon_{\pi_{t-3}},$  each of which also enters  $\bar{E}_{t-1}^A$  through the quarterly perceptions that comprise it, and which affect  $\pi_t^A$  through the channels analyzed in Section 6. OLS estimation of (289) therefore suffers from an endogeneity problem.

To address this, I impose the constraint analytically and use instrumental variables. Rearranging (289) gives

$$\bar{E}_t^A - \rho \bar{E}_{t-1}^A = \gamma_1 (\pi_t^A - \rho \bar{E}_{t-1}^A) + \bar{\tau}_{\pi_t} u_t \quad (291)$$

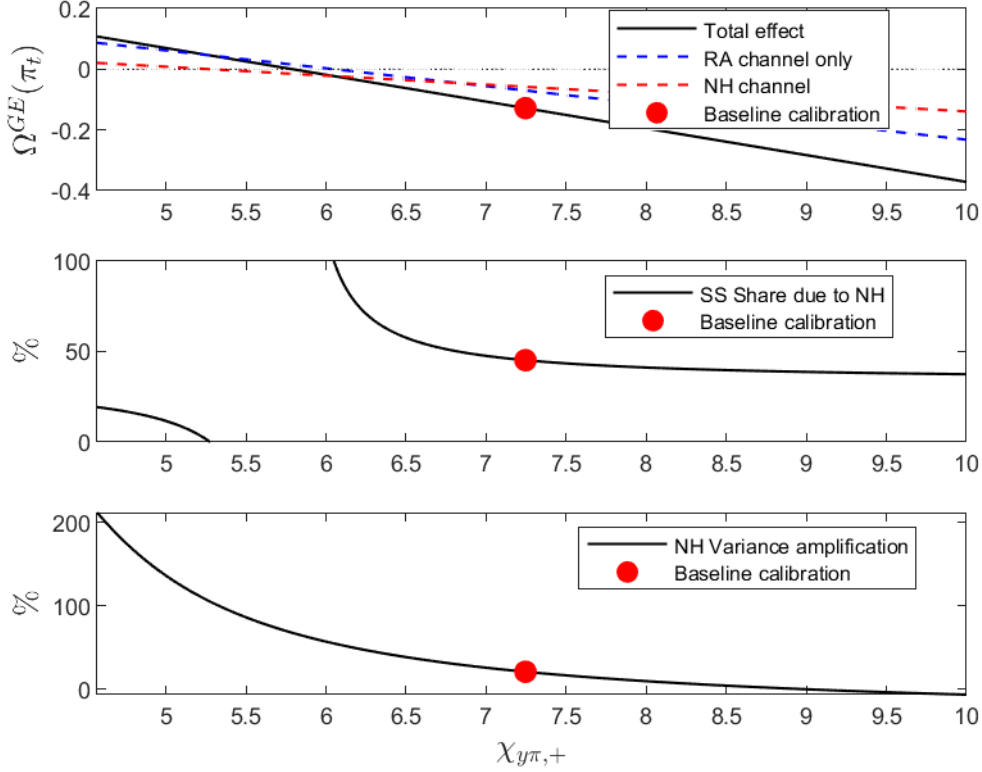
I then instrument  $\pi_t^A - \rho \bar{E}_{t-1}^A$  with  $\pi_{t-4}^A, \bar{E}_{t-5}^A,$  and  $\sum_{s=0}^3 \bar{E}_{t-5-s} \pi_{t-1-s}$ . The first two instruments are individual components of the regressor, lagged by 4 quarters. The final instrument is average annual one-year-ahead inflation expectations, also lagged to the same point as average inflation perceptions  $\bar{E}_{t-5}^A$ , and included to increase first-stage power. Since all three instruments are lagged at least a full year, they depend only on  $\varepsilon_{\pi_{t-4}}$  and earlier, which lie outside the support of  $u_t$ , so the exclusion restriction is satisfied by the timing structure. The first-stage is significant (Kleibergen-Paap  $F = 20.7$ ), consistent with other literature finding substantial persistence in household inflation expectations (Carroll, 2003). The estimated  $\hat{\gamma}_1$  implies a target of  $\bar{\tau}^* = 0.184$ .

## E.5 Robustness

**Alternative  $\chi_{iy\pi}$  calibrations.** Figure 9 shows how the key statistics computed in Sections 6.6 and 6.7 change with variation in the calibrated value of  $\chi_+$ . In all panels, I maintain the relationship between  $\chi_+$  and  $\chi_-$  from equation (286): a greater  $\chi_+$  therefore implies a correspondingly larger magnitude for  $\chi_-$ , in keeping with Empirical Result 1. I focus here on results for the causal effect of perceived  $\pi_t$ , but as discussed below the causal effects of expected  $\pi_{t+1}$  are proportional to those of perceived  $\pi_t$ . For any ratios (as in the middle and bottom panels of Figure 9) the results are exactly the same across the two cases. In each panel, I vary  $\chi_+$  from  $-\frac{2\sigma(\rho-\phi_\pi)}{\zeta-\sigma\phi_y}$ , the lower bound at which equation (286) implies  $\chi_- = 0$ , to 10.

The top panel shows how the total causal effect of perceived  $\pi_t$  in general equilibrium varies with  $\chi_+$ , along with the representative-agent and narrative-heterogeneity components, as in the first three columns of Table 2. As  $\chi_+$  rises, both the representative-agent and narrative-heterogeneity channels become more negative. This occurs because the largest groups of households are those on  $\chi_{iy\pi} = \chi_-$  and  $\chi_{iy\pi} = 0$ . The increase in the magnitude of  $\chi_i$  associated with an increase in  $\chi_+$  therefore pushes the average cross-learning parameter to be more negative (implying a more negative RA effect), and strengthens the negative covariance between cross-learning and  $\tau_{i\pi_t}$  (implying a more negative NH effect).

**Figure 9:** Robustness of narrative heterogeneity effects to the calibration of  $\chi_{y\pi,+}$



*Note:* Figure plots key statistics from the temporary-information equilibrium of the New Keynesian model as  $\chi_+$  varies, with  $\chi_-$  determined by the equal-squared-passthrough condition in equation (286). All other parameters are held at their baseline values (Section E.4). Top panel: the general equilibrium causal effect of aggregate inflation perceptions  $\bar{E}_t \pi_t$  on output  $y_t$ , decomposed into the representative-agent channel (dashed blue), the narrative heterogeneity channel (dashed red), and the total effect (solid black). The red circle marks the baseline calibration ( $\chi_+ = \kappa^{-1}$ ). Middle panel: the share of the total steady-state general equilibrium effect attributable to the narrative heterogeneity channel. Bottom panel: the percentage amplification of the variance of the time-varying general equilibrium causal effect due to narrative heterogeneity, computed using the time-varying proportions from the IAS as in Section 6.7. Results for the causal effects of  $\bar{E}_t \pi_{t+1}$  are proportional to those shown here; the ratios in the middle and bottom panels are identical across the two cases.

The converse of this is that if  $\chi_+$  becomes sufficiently small, then the substitution effect of greater perceived inflation implying lower perceived real interest rates starts to dominate, and the causal effect of inflation perceptions becomes positive. This occurs when  $\chi_+ \approx 5.8$ , approximately half-way between the baseline calibration ( $\chi_+ \approx 7.25$ ) and the minimum value such that  $\chi_- < 0$  ( $\chi_+ \approx 4.57$ ).

The middle panel shows the share of the total response that is due to the narrative heterogeneity channel ( $100 \times \Omega_y^{GE,NH}(\pi_t) / \Omega_y^{GE}(\pi_t)$ ). Away from the neighborhood where the total causal effect passes through 0, this share is relatively stable. It declines modestly as  $\chi_+$  rises, reaching 37.5% when  $\chi_+ = 10$  (relative to 45.2% in the baseline calibration). Thus the narrative heterogeneity channel is quantitatively substantial for a range of parametrizations.

Finally, the lower panel shows the amplification of the time-variation in the GE



causal effect due to the narrative heterogeneity channel  $(100 \times (Var(\Omega_y^{GE}(\pi_t)) - Var(\Omega_y^{GE,RA}(\pi_t)))/Var(\Omega_y^{GE,RA}(\pi_t)))$ . This is more sensitive to the value of  $\chi_+$  than the steady-state share due to narrative heterogeneity in the middle panel, though it again remains substantial for a wide range of parametrizations.

**Alternative  $\tau_L$  calibrations.** In Table 2, the steady state causal effects of inflation expectations are proportional to those of inflation perceptions. This is true for the total effects and their component parts (representative-agent and narrative-heterogeneity). This property arises due to the three-group structure for expectation-formation parameters assumed in (44), along with the assumption  $\tau_L = 0$ .

For partial equilibrium, the proportionality can be seen straightforwardly from equations (275) and (278), which together imply

$$\frac{\Omega_y^{PE}(\pi_{t+1})}{\Omega_y^{PE}(\pi_t)} = \frac{1}{\rho}. \quad (292)$$

Since this proportionality holds term-by-term, it is true of both the total effects and each component of the decomposition into representative-agent and narrative-heterogeneity channels.

For general equilibrium, the proportionality is less obvious from the general forms of the causal effects in Appendix E.3, as the formula for  $\Omega_y^{GE}(\pi_{t+1})$  (284) involves a different (and more complicated) denominator than that for  $\Omega_y^{GE}(\pi_t)$  (276). We show that the proportionality continues to hold with a guess-and-verify approach. Suppose that, as with partial equilibrium,

$$\frac{\Omega_y^{GE}(\pi_{t+1})}{\Omega_y^{GE}(\pi_t)} = \frac{1}{\rho}. \quad (293)$$

Combining this guess with equation (276) we have

$$\begin{aligned} \Omega_y^{GE}(\pi_{t+1}) &= \frac{\Omega_y^{GE}(\pi_t)}{\rho} = \frac{\Omega_y^{PE}(\pi_t)}{\rho} \cdot \frac{1 - \zeta\Lambda}{1 - \zeta\Lambda + \Lambda\sigma(\phi_\pi\kappa + \phi_y)} \\ &= \Omega_y^{PE}(\pi_{t+1}) \cdot \frac{1 - \zeta\Lambda}{1 - \zeta\Lambda + \Lambda\sigma(\phi_\pi\kappa + \phi_y)} \end{aligned} \quad (294)$$

Substituting this into equation (284) and rearranging, we obtain

$$D + \rho\kappa\Omega_y^{PE}(\pi_{t+1}) \int \tau_{i\pi_t}(1 - \tau_{i\pi_{t+1}})di = 1 + \frac{\Lambda\sigma(\phi_\pi\kappa + \phi_y)}{1 - \zeta\Lambda} \quad (295)$$

Substituting out for  $D$  using (281), several terms cancel, leaving

$$\int \tau_{i\pi_t}(1 - \tau_{i\pi_{t+1}})di \cdot \int \tilde{\chi}_i\tau_{i\pi_{t+1}}di = \bar{\tau}_{\pi_{t+1}} \int \tilde{\chi}_i\tau_{i\pi_t}(1 - \tau_{i\pi_{t+1}})di \quad (296)$$

Evaluating the integrals using the three-group structure in (44), and canceling more terms, yields

$$(\tilde{\chi}_{+p_{+}} + \tilde{\chi}_{-p_{-}})\tau_H\tau_L(\tau_H - \tau_L) = \tilde{\chi}_0(p_{+} + p_{-})\tau_H\tau_L(\tau_H - \tau_L) \quad (297)$$

This is trivially true when  $\tau_L = 0$ , thus verifying the guess in (293). However, when  $\tau_L \neq 0$ , it will not generally be true, and thus the guess (293) will be false.

Given this proportionality, we have that the shares of  $\Omega_y^{GE}(\pi_t)$  and  $\Omega_y^{GE}(\pi_{t+1})$  due to the narrative heterogeneity channel, in steady state or period-by-period, are the same as one another. However, this equivalence is not a general property, and would be broken if  $\tau_L > 0$ , although numerically the effect is small. Varying  $\tau_L$  between 0 (the baseline) and the calibration target for the average Kalman gain, and varying  $\tau_H$  accordingly to maintain that average, the maximum gap between the steady state narrative heterogeneity shares for  $\Omega_y^{GE}(\pi_t)$  and  $\Omega_y^{GE}(\pi_{t+1})$  is 0.006%. For the variance amplification, the maximum gap is 0.022%. These are orders of magnitude smaller than the levels of each ratio, as displayed in Sections 6.6 and 6.7.